

Tenable Vulnerability Management Report

Tenable Vulnerability Management

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Audits FAILED

UBTU-24-100450 - Ubuntu 24.04 LTS audit event multiplexor must be configured to offload audit logs onto a different system or storage media from the system being audited.

Info

Information stored in one location is vulnerable to accidental or incidental deletion or alteration. Offloading is a common process in information systems with limited audit storage capacity. Satisfies: SRG-OS-000342-GPOS-00133, SRG-OS-000479-GPOS-00224

Solution

Configure the audit event multiplexor to offload audit records to a different system or storage media from the system being audited. Install the audisp-remote plugin: `$ sudo apt install -y audispd-plugins` Set the audisp-remote plugin as active by editing the `/etc/audit/plugins.d/au-remote.conf` file: `$ sudo sed -i -E 's/active\s*=\s*no/active = yes/' /etc/audit/plugins.d/au-remote.conf` Set the address of the remote machine by editing the `/etc/audit/audisp-remote.conf` file: `$ sudo sed -i -E 's/(remote_server\s*=).*/\1 <remote addr>/' /etc/audit/audisp-remote.conf` where `<remote addr>` must be substituted by the address of the remote server receiving the audit log. Make the audit service reload its configuration files: `$ sudo systemctl restart auditd.service`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

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CAT	III
CCI	CCI-001851
CSF	PR.DS-4
CSF	PR.PT-1
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.6
ITSG-33	AU-4
NESA	T3.3.1
NESA	T3.6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270658r1067151_rule
STIG-ID	UBTU-24-100450
VULN-ID	V-270658

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
PASSED - audispd-plugins
The command '/bin/dpkg -s audispd-plugins | /bin/grep Status: 2>&1' returned :

Status: install ok installed

-----
FAILED - remote_server
Non-compliant file(s):
  /etc/audit/audisp-remote.conf - regex '^[\s]*remote_server[\s]*=[\s]*' found - expect
  '^[\s]*remote_server[\s]*=[\s]*10.0.1.2' not found in the following lines:
    6: remote_server =

-----
PASSED - active = yes
Compliant file(s):
  /etc/audit/plugins.d/au-remote.conf - regex '^[\s]*active[\s]*=' found - expect
  '^[\s]*active[\s]*=[\s]*yes[\s]*$' found in the following lines:
    6: active = yes
```

UBTU-24-100650 - Ubuntu 24.04 LTS must have the "SSSD" package installed.

Info

Without the use of multifactor authentication, the ease of access to privileged functions is greatly increased. Multifactor authentication requires using two or more factors to achieve authentication. Factors include: 1) Something a user knows (e.g., password/PIN); 2) Something a user has (e.g., cryptographic identification device, token); and 3) Something a user is (e.g., biometric). A privileged account is defined as an information system account with authorizations of a privileged user. Network access is defined as access to an information system by a user (or a process acting on behalf of a user) communicating through a network (e.g., local area network, wide area network, or the internet). The DOD common access card (CAC) with DOD-approved PKI is an example of multifactor authentication. Satisfies: SRG-OS-000705-GPOS-00150, SRG-OS-000105-GPOS-00052, SRG-OS-000106-GPOS-00053, SRG-OS-000107-GPOS-00054, SRG-OS-000108-GPOS-00055, SRG-OS-000375-GPOS-00160

Solution

Install the sssd.service and the required pam packages with the following commands: \$ sudo apt install -y sssd
\$ sudo apt install -y libpam-sss \$ sudo apt install -y libnss-sss

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.3
800-171R3	03.05.03
800-53	IA-2(1)
800-53	IA-2(2)
800-53R5	IA-2(1)
800-53R5	IA-2(2)
800-53R5	IA-2(6)(a)
800-53R5	IA-2(6)(b)
CAT	II
CCI	CCI-000765
CCI	CCI-000766
CCI	CCI-004046
CCI	CCI-004047
CN-L3	7.1.2.7(b)
CN-L3	7.1.3.1(a)
CN-L3	7.1.3.1(e)
CN-L3	8.1.4.1(a)
CN-L3	8.1.4.2(a)
CN-L3	8.5.4.1(a)
CSF	PR.AC-1
CSF2.0	PR.AA-01

CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ITSG-33	IA-2(1)
ITSG-33	IA-2(2)
NESA	T5.4.2
NIAV2	AM2
NIAV2	AM8
NIAV2	AM14b
NIAV2	AM36
NIAV2	VL3c
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270662r1067156_rule
STIG-ID	UBTU-24-100650
SWIFT-CSCV1	1.2
TBA-FIISB	35.1
TBA-FIISB	36.1
VULN-ID	V-270662

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
FAILED - sssd package is installed
The command '/bin/dpkg -s sssd 2>&1 | /bin/grep -E '(Status:|not installed)'' returned :

dpkg-query: package 'sssd' is not installed and no information is available

-----
FAILED - libpam-sss
The command '/bin/dpkg -s libpam-sss 2>&1 | /bin/grep -E '(Status:|not installed)'' returned :

dpkg-query: package 'libpam-sss' is not installed and no information is available

-----
FAILED - libnss-sss
The command '/bin/dpkg -s libnss-sss 2>&1 | /bin/grep -E '(Status:|not installed)'' returned :
```

dpkg-query: package 'libnss-sss' is not installed and no information is available

UBTU-24-100660 - Ubuntu 24.04 LTS must use the "SSSD" package for multifactor authentication services.

Info

Without the use of multifactor authentication, the ease of access to privileged functions is greatly increased. Multifactor authentication requires using two or more factors to achieve authentication. Factors include: 1) Something a user knows (e.g., password/PIN); 2) Something a user has (e.g., cryptographic identification device, token); and 3) Something a user is (e.g., biometric). A privileged account is defined as an information system account with authorizations of a privileged user. Network access is defined as access to an information system by a user (or a process acting on behalf of a user) communicating through a network (e.g., local area network, wide area network, or the internet). The DOD common access card (CAC) with DOD-approved PKI is an example of multifactor authentication. Satisfies: SRG-OS-000705-GPOS-00150, SRG-OS-000105-GPOS-00052, SRG-OS-000106-GPOS-00053, SRG-OS-000107-GPOS-00054, SRG-OS-000108-GPOS-00055, SRG-OS-000375-GPOS-00160

Solution

Enable the "sssd.service to start automatically on reboot with the following command: `$ sudo systemctl enable sssd.service` ensure the "sssd" service is running `$ sudo systemctl start sssd.service`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.3
800-171R3	03.05.03
800-53	IA-2(1)
800-53	IA-2(2)
800-53R5	IA-2(1)
800-53R5	IA-2(2)
800-53R5	IA-2(6)(a)
800-53R5	IA-2(6)(b)
CAT	II
CCI	CCI-000765
CCI	CCI-000766
CCI	CCI-004046
CCI	CCI-004047
CN-L3	7.1.2.7(b)
CN-L3	7.1.3.1(a)
CN-L3	7.1.3.1(e)
CN-L3	8.1.4.1(a)
CN-L3	8.1.4.2(a)
CN-L3	8.5.4.1(a)
CSF	PR.AC-1

CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ITSG-33	IA-2(1)
ITSG-33	IA-2(2)
NESA	T5.4.2
NIAV2	AM2
NIAV2	AM8
NIAV2	AM14b
NIAV2	AM36
NIAV2	VL3c
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270663r1066478_rule
STIG-ID	UBTU-24-100660
SWIFT-CSCV1	1.2
TBA-FIISB	35.1
TBA-FIISB	36.1
VULN-ID	V-270663

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```

-----
FAILED - is-active
The command '/bin/systemctl is-active sssd' returned :

inactive

-----
FAILED - is-enabled
The command '/bin/systemctl is-enabled sssd' returned :

not-found

```

UBTU-24-102000 - Ubuntu 24.04 LTS when booted must require authentication upon booting into single-user and maintenance modes.

Info

To mitigate the risk of unauthorized access to sensitive information by entities that have been issued certificates by DOD-approved PKIs, all DOD systems (e.g., web servers and web portals) must be properly configured to incorporate access control methods that do not rely solely on the possession of a certificate for access. Successful authentication must not automatically give an entity access to an asset or security boundary. Authorization procedures and controls must be implemented to ensure each authenticated entity also has a validated and current authorization.

Authorization is the process of determining whether an entity, once authenticated, is permitted to access a specific asset.

Information systems use access control policies and enforcement mechanisms to implement this requirement.

Access control policies include identity-based policies, role-based policies, and attribute-based policies.

Access enforcement mechanisms include access control lists, access control matrices, and cryptography.

These policies and mechanisms must be employed by the application to control access between users (or processes acting on behalf of users) and objects (e.g., devices, files, records, processes, programs, and domains) in the information system.

Solution

Configure the system to require a password for authentication upon booting into single-user and maintenance modes. Generate an encrypted (grub) password for root with the following command:

```
$ grub-mkpasswd-pbkdf2 Enter Password: Reenter Password: PBKDF2 hash of your password is grub.pbkdf2.sha512.10000.MFU48934NJD84NF8NSD39993JDHF84NG Using the hash from the output, modify the "/etc/grub.d/40_custom" file with the following command to add a boot password: $ sudo sed -i '$i set superusers="\root"\npbkdf2 root <hash>' /etc/grub.d/40_custom where <hash> is the hash generated by grub-mkpasswd-pbkdf2 command. Generate an updated "grub.conf" file with the new password by using the following command: $ sudo update-grub
```

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.1
800-171R3	03.01.02
800-53	AC-3
800-53R5	AC-3
CAT	I
CCI	CCI-000213
CN-L3	8.1.4.2(f)
CN-L3	8.1.4.11(b)
CN-L3	8.1.10.2(c)
CN-L3	8.5.3.1
CN-L3	8.5.4.1(a)
CSF	PR.AC-4
CSF	PR.PT-3
CSF2.0	PR.AA-05
CSF2.0	PR.DS-10

CSF2.0	PR.IR-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
ISO-27001-2022	A.5.15
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.3
ISO-27001-2022	A.8.18
ISO-27001-2022	A.8.20
ISO/IEC-27001	A.9.4.1
ISO/IEC-27001	A.9.4.5
ITSG-33	AC-3
NESA	T4.2.1
NESA	T5.4.4
NESA	T5.4.5
NESA	T5.5.4
NESA	T5.6.1
NESA	T7.5.2
NESA	T7.5.3
NIAV2	AM3
NIAV2	SS29
QCSC-V1	3.2
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270675r1137691_rule
STIG-ID	UBTU-24-102000
TBA-FIISB	31.1
VULN-ID	V-270675

Assets

strkbsubuntu9

The following file(s) do not contain "[\s]*password_pbkdf2[\s]":
/boot/grub/grub.cfg

UBTU-24-40020 - Ubuntu 24.04 LTS must implement smart card logins for multifactor authentication for local and network access to privileged and nonprivileged accounts.

Info

Without the use of multifactor authentication, the ease of access to privileged functions is greatly increased. Multifactor authentication requires using two or more factors to achieve authentication. Factors include: 1) Something a user knows (e.g., password/PIN); 2) Something a user has (e.g., cryptographic identification device, token); and 3) Something a user is (e.g., biometric). A privileged account is defined as an information system account with authorizations of a privileged user. Network access is defined as access to an information system by a user (or a process acting on behalf of a user) communicating through a network (e.g., local area network, wide area network, or the internet). The DOD common access card (CAC) with DOD-approved PKI is an example of multifactor authentication. Satisfies: SRG-OS-000105-GPOS-00052, SRG-OS-000107-GPOS-00054, SRG-OS-000106-GPOS-00053, SRG-OS-000108-GPOS-00055

Solution

Configure Ubuntu 24.04 LTS to use multifactor authentication for access to accounts. Add or update "pam_pkcs11.so" in "/etc/pam.d/common-auth" to match the following line: auth [success=2 default=ignore] pam_pkcs11.so

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.3
800-171R3	03.05.03
800-53	IA-2(1)
800-53	IA-2(2)
800-53R5	IA-2(1)
800-53R5	IA-2(2)
CAT	II
CCI	CCI-000765
CCI	CCI-000766
CN-L3	7.1.2.7(b)
CN-L3	7.1.3.1(a)
CN-L3	7.1.3.1(e)
CN-L3	8.1.4.1(a)
CN-L3	8.1.4.2(a)
CN-L3	8.5.4.1(a)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b

HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ITSG-33	IA-2(1)
ITSG-33	IA-2(2)
NESA	T5.4.2
NIAV2	AM2
NIAV2	AM8
NIAV2	AM14b
NIAV2	AM36
NIAV2	VL3c
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270721r1066652_rule
STIG-ID	UBTU-24-400020
SWIFT-CSCV1	1.2
TBA-FIISB	35.1
TBA-FIISB	36.1
VULN-ID	V-270721

Assets

strkbsubuntu9

The following file(s) do not contain "^[\s]*auth[\s]+([^\n\r]+)pam_pkcs11\.so[\s]*":
 /etc/pam.d/common-auth

UBTU-24-400030 - Ubuntu 24.04 LTS must implement smart card logins for multifactor authentication for local and network access to privileged and nonprivileged accounts over SSH.

Info

Without the use of multifactor authentication, the ease of access to privileged functions is greatly increased. Multifactor authentication requires using two or more factors to achieve authentication. Factors include: 1) Something a user knows (e.g., password/PIN); 2) Something a user has (e.g., cryptographic identification device, token); and 3) Something a user is (e.g., biometric). A privileged account is defined as an information system account with authorizations of a privileged user. Network access is defined as access to an information system by a user (or a process acting on behalf of a user) communicating through a network (e.g., local area network, wide area network, or the internet). The DOD common access card (CAC) with DOD-approved PKI is an example of multifactor authentication. Satisfies: SRG-OS-000105-GPOS-00052, SRG-OS-000107-GPOS-00054, SRG-OS-000106-GPOS-00053, SRG-OS-000108-GPOS-00055

Solution

Configure Ubuntu 24.04 LTS to use multifactor authentication for access to accounts. Set the sshd option "PubkeyAuthentication" to "yes" in the "/etc/ssh/sshd_config" file. PubkeyAuthentication yes

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.3
800-171R3	03.05.03
800-53	IA-2(1)
800-53	IA-2(2)
800-53R5	IA-2(1)
800-53R5	IA-2(2)
CAT	II
CCI	CCI-000765
CCI	CCI-000766
CN-L3	7.1.2.7(b)
CN-L3	7.1.3.1(a)
CN-L3	7.1.3.1(e)
CN-L3	8.1.4.1(a)
CN-L3	8.1.4.2(a)
CN-L3	8.5.4.1(a)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b

HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ITSG-33	IA-2(1)
ITSG-33	IA-2(2)
NESA	T5.4.2
NIAV2	AM2
NIAV2	AM8
NIAV2	AM14b
NIAV2	AM36
NIAV2	VL3c
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270722r1067130_rule
STIG-ID	UBTU-24-400030
SWIFT-CSCV1	1.2
TBA-FIISB	35.1
TBA-FIISB	36.1
VULN-ID	V-270722

Assets

strkbsubuntu9

No matching files were found
Less than 1 matches of regex found

UBTU-24-40060 - Ubuntu 24.04 LTS must electronically verify Personal Identity Verification (PIV) credentials.

Info

The use of PIV credentials facilitates standardization and reduces the risk of unauthorized access. DOD has mandated the use of the common access card (CAC) to support identity management and personal authentication for systems covered under Homeland Security Presidential Directive (HSPD) 12, as well as making the CAC a primary component of layered protection for national security systems.

Solution

Configure Ubuntu 24.04 LTS to do certificate status checking for multifactor authentication. Modify all of the "cert_policy" lines in "/etc/pam_pkcs11/pam_pkcs11.conf" to include "ocsp_on".

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.1
800-171R3	03.05.01a.
800-53	IA-2(12)
800-53R5	IA-2(12)
CAT	II
CCI	CCI-001954
CN-L3	7.1.3.1(a)
CN-L3	7.1.3.1(e)
CN-L3	8.1.4.1(a)
CN-L3	8.1.4.2(a)
CN-L3	8.5.4.1(a)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ITSG-33	IA-2
ITSG-33	IA-2a.
NESA	T2.3.8

NESA	T5.3.1
NESA	T5.4.2
NESA	T5.5.1
NESA	T5.5.2
NESA	T5.5.3
NIAV2	AM2
NIAV2	AM8
NIAV2	AM14b
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270723r1066658_rule
STIG-ID	UBTU-24-400060
TBA-FIISB	35.1
TBA-FIISB	36.1
VULN-ID	V-270723

Assets

strkbsubuntu9

The command `'/bin/grep use_pkcs11_module /etc/pam_pkcs11/pam_pkcs11.conf | /bin/awk '/pkcs11_module opensc {/,/}/' /etc/pam_pkcs11/pam_pkcs11.conf | /bin/grep -E '^s*cert_policy'` returned :

```
/bin/grep: /etc/pam_pkcs11/pam_pkcs11.conf: No such file or directory
awk: fatal: cannot open file `/etc/pam_pkcs11/pam_pkcs11.conf' for reading: No such file or directory
```

UBTU-24-400340 - Ubuntu 24.04 LTS must be configured such that Pluggable Authentication Module (PAM) prohibits the use of cached authentications after one day.

Info

If cached authentication information is out-of-date, the validity of the authentication information may be questionable.

Solution

Configure PAM to prohibit the use of cached authentications after one day.
Add or change the following line in "/etc/sss/sss.conf" just below the line "[pam]":
offline_credentials_expiration = 1 Note: It is valid for this configuration to be in a file with a name that ends with ".conf" and does not begin with a "." in the "/etc/sss/conf.d/" directory instead of the "/etc/sss/sss.conf" file.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.2
800-171R3	03.05.12
800-53	IA-5(13)
800-53R5	IA-5(13)
CAT	III
CCI	CCI-002007
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ITSG-33	IA-5
NESA	T5.2.3
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270734r1155240_rule
STIG-ID	UBTU-24-400340
VULN-ID	V-270734

Assets

strkbsubuntu9

No matching files were found
Less than 1 matches of regex found

UBTU-24-400360 - Ubuntu 24.04 LTS, for PKI-based authentication, SSSD must validate certificates by constructing a certification path (which includes status information) to an accepted trust anchor.

Info

Without path validation, an informed trust decision by the relying party cannot be made when presented with any certificate not already explicitly trusted. A trust anchor is an authoritative entity represented via a public key and associated data.

It is used in the context of public key infrastructures, X.509 digital certificates, and DNSSEC. When there is a chain of trust, usually the top entity to be trusted becomes the trust anchor; it can be, for example, a certification authority (CA).

A certification path starts with the subject certificate and proceeds through a number of intermediate certificates up to a trusted root certificate, typically issued by a trusted CA. This requirement verifies that a certification path to an accepted trust anchor is used for certificate validation and that the path includes status information.

Path validation is necessary for a relying party to make an informed trust decision when presented with any certificate not already explicitly trusted.

Status information for certification paths includes certificate revocation lists or online certificate status protocol responses.

Validation of the certificate status information is out of scope for this requirement. Satisfies: SRG-OS-000066-GPOS-00034, SRG-OS-000775-GPOS-00230

Solution

Configure Ubuntu 24.04 LTS, for PKI-based authentication, to validate certificates by constructing a certification path to an accepted trust anchor. Add or update the /etc/sss/sss.conf so that the following entries are in the correct sections of the file: `$ sudo vi /etc/sss/sss.conf [sss] services = nss,pam,ssh config_file_version = 2 [pam] pam_cert_auth = True [domain/example.com] ldap_user_certificate = usercertificate;binary certificate_verification = ca_cert,ocsp ca_cert = /etc/ssl/certs/ca-certificates.crt`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.2
800-171R3	03.05.12
800-53	IA-5(2)(a)
800-53R5	IA-5(2)(b)(1)
800-53R5	SC-17b.
CAT	II
CCI	CCI-000185
CCI	CCI-004909
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)

ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ITSG-33	IA-5(2)(a)
NESA	T5.2.3
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270735r1066694_rule
STIG-ID	UBTU-24-400360
VULN-ID	V-270735

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
FAILED - sssd.conf services
No files found: /etc/sssds/sssds.conf
```

```
-----
FAILED - sssd.conf pam
No files found: /etc/sssds/sssds.conf
```

```
-----
FAILED - sssd.conf certificate_verification
No files found: /etc/sssds/sssds.conf
```

UBTU-24-400370 - Ubuntu 24.04 LTS must map the authenticated identity to the user or group account for PKI-based authentication.

Info

Without mapping the certificate used to authenticate to the user account, the ability to determine the identity of the individual user or group will not be available for forensic analysis.

Solution

Configure sssd to map authenticated certificates to the appropriate user group by adding the following line to the "/etc/sss/sss.conf" file: ldap_user_certificate=userCertificate;binary

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.2
800-171R3	03.05.12
800-53	IA-5(2)(c)
800-53R5	IA-5(2)(a)(2)
CAT	I
CCI	CCI-000187
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ITSG-33	IA-5(2)(c)
NESA	T5.2.3
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270736r1066697_rule
STIG-ID	UBTU-24-400370
VULN-ID	V-270736

Assets

strkbsubuntu9

No files found: /etc/sss/sss.conf

UBTU-24-400375 - Ubuntu 24.04 LTS, for PKI-based authentication, Privileged Access Management (PAM) must validate certificates by constructing a certification path (which includes status information) to an accepted trust anchor.

Info

Without path validation, an informed trust decision by the relying party cannot be made when presented with any certificate not already explicitly trusted. A trust anchor is an authoritative entity represented via a public key and associated data.

It is used in the context of public key infrastructures, X.509 digital certificates, and DNSSEC. When there is a chain of trust, usually the top entity to be trusted becomes the trust anchor; it can be, for example, a certification authority (CA).

A certification path starts with the subject certificate and proceeds through a number of intermediate certificates up to a trusted root certificate, typically issued by a trusted CA. This requirement verifies that a certification path to an accepted trust anchor is used for certificate validation and that the path includes status information.

Path validation is necessary for a relying party to make an informed trust decision when presented with any certificate not already explicitly trusted.

Status information for certification paths includes certificate revocation lists or online certificate status protocol responses.

Validation of the certificate status information is out of scope for this requirement. Satisfies: SRG-OS-000066-GPOS-00034, SRG-OS-000775-GPOS-00230

Solution

Configure Ubuntu 24.04 LTS, for PKI-based authentication, to validate certificates by constructing a certification path to an accepted trust anchor. Determine which pkcs11 module is being used via the "use_pkcs11_module" in "/etc/pam_pkcs11/pam_pkcs11.conf" and ensure "ca" is enabled in "cert_policy". Add or update the "cert_policy" to ensure "ca" is enabled: cert_policy = ca,signature,ocsp_on; If the system is missing an "/etc/pam_pkcs11/" directory and an "/etc/pam_pkcs11/pam_pkcs11.conf", find an example to copy into place and modify accordingly at "https://manpages.ubuntu.com/manpages/xenial/man8/pam_pkcs11.8.html".

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.2
800-171R3	03.05.12
800-53	IA-5(2)(a)
800-53R5	IA-5(2)(b)(1)
800-53R5	SC-17b.
CAT	II
CCI	CCI-000185
CCI	CCI-004909
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)

HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ITSG-33	IA-5(2)(a)
NESA	T5.2.3
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270737r1067178_rule
STIG-ID	UBTU-24-400375
VULN-ID	V-270737

Assets

strkbsubuntu9

The command `/bin/grep use_pkcs11_module /etc/pam_pkcs11/pam_pkcs11.conf | /bin/awk '/pkcs11_module opensc {/,/}/' /etc/pam_pkcs11/pam_pkcs11.conf | /bin/grep -E '^s*cert_policy'` returned :

```
/bin/grep: /etc/pam_pkcs11/pam_pkcs11.conf: No such file or directory
awk: fatal: cannot open file `/etc/pam_pkcs11/pam_pkcs11.conf' for reading: No such file or directory
```

UBTU-24-400380 - Ubuntu 24.04 LTS for PKI-based authentication, must implement a local cache of revocation data in case of the inability to access revocation information via the network.

Info

Without configuring a local cache of revocation data, there is the potential to allow access to users who are no longer authorized (users with revoked certificates).

Solution

Configure Ubuntu 24.04 LTS, for PKI-based authentication, to use local revocation data when unable to access the network to obtain it remotely. Add or update the "cert_policy" option in "/etc/pam_pkcs11/pam_pkcs11.conf" to include "crl_auto" or "crl_offline". cert_policy = ca,signature,ocsp_on, crl_auto; If the system is missing an "/etc/pam_pkcs11/" directory and an "/etc/pam_pkcs11/pam_pkcs11.conf", find an example to copy into place and modify accordingly at "/usr/share/doc/libpam-pkcs11/examples/pam_pkcs11.conf.example.gz".

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.2
800-171R3	03.05.12
800-53	IA-5(2)(d)
800-53R5	IA-5(2)(b)(2)
CAT	II
CCI	CCI-004068
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ITSG-33	IA-5(2)
NESA	T5.2.3
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270738r1066703_rule
STIG-ID	UBTU-24-400380

VULN-ID

V-270738

Assets

strkbsubuntu9

No files found: /etc/pam_pkcs11/pam_pkcs11.conf

UBTU-24-60030 - Ubuntu 24.04 LTS must implement NIST FIPS-validated cryptography to protect classified information and for the following: To provision digital signatures, to generate cryptographic hashes, and to protect unclassified information requiring confidentiality and cryptographic protection in accordance with applicable federal laws, Executive Orders, directives, policies, regulations, and standards.

Info

Use of weak or untested encryption algorithms undermines the purposes of utilizing encryption to protect data.

Ubuntu 24.04 LTS must implement cryptographic modules adhering to the higher standards approved by the federal government since this provides assurance they have been tested and validated. Satisfies: SRG-OS-000478-GPOS-00223, SRG-OS-000396-GPOS-00176

Solution

Configure the system to run in FIPS mode.

Add "fips=1" to the kernel parameter during Ubuntu 24.04 LTSs install. Enabling a FIPS mode on a pre-existing system involves a number of modifications to Ubuntu 24.04 LTS.

Refer to the Ubuntu Pro security certification documentation for instructions. A subscription to the "Ubuntu Pro" plan is required to obtain the FIPS Kernel cryptographic modules and enable FIPS. Note: Ubuntu Pro security certification instructions can be found at: <https://ubuntu.com/security/certifications/docs/fips-enablement> The basic steps use the following commands: \$ sudo pro attach <token> \$ sudo pro enable fips-updates

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.13.11
800-171R3	03.13.11
800-53	SC-13
800-53R5	SC-13b.
CAT	I
CCI	CCI-002450
CSF	PR.DS-5
CSF2.0	PR.DS-01
CSF2.0	PR.DS-02
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.a
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(iv)
HIPAA	164.312(e)(2)(ii)
ISO-27001-2022	A.8.24
ISO/IEC-27001	A.10.1.1

ITSG-33	SC-13
ITSG-33	SC-13a.
NESA	M5.2.6
NESA	T7.4.1
NIAV2	CY3
NIAV2	CY4
NIAV2	CY5b
NIAV2	CY5c
NIAV2	CY5d
NIAV2	CY7
NIAV2	NS5e
QCSC-V1	6.2
RULE-ID	SV-270744r1137699_rule
STIG-ID	UBTU-24-600030
VULN-ID	V-270744

Assets

strkbsubuntu9

No files found: /proc/sys/crypto/fips_enabled

UBTU-24-600070 - Ubuntu 24.04 LTS must disable kernel core dumps.

Info

Kernel core dumps may contain the full contents of system memory at the time of the crash. Kernel core dumps may consume a considerable amount of disk space and may result in denial of service by exhausting the available space on the target file system partition.

Solution

If kernel core dumps are not required, disable the "kdump-tools" service with the following command: `$ sudo systemctl disable kdump-tools.service` If kernel core dumps are required, document the need with the ISSO.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SC-24
800-53R5	SC-24
CAT	II
CCI	CCI-001190
CSF2.0	PR.DS-10
CSF2.0	PR.IR-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SC-24
ITSG-33	SC-24a.
QCSC-V1	5.2.1
RULE-ID	SV-270746r1155242_rule
STIG-ID	UBTU-24-600070
VULN-ID	V-270746

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
PASSED - inactive
The command '/bin/systemctl is-active kdump-tools.service' returned :

inactive

-----
FAILED - masked
The command '/bin/systemctl is-enabled kdump-tools.service' returned :

not-found
```

UBTU-24-600160 - Ubuntu 24.04 LTS must compare internal information system clocks at least every 24 hours with an authoritative time server.

Info

Inaccurate time stamps make it more difficult to correlate events and can lead to an inaccurate analysis. Determining the correct time a particular event occurred on a system is critical when conducting forensic analysis and investigating system events.

Sources outside the configured acceptable allowance (drift) may be inaccurate. Synchronizing internal information system clocks provides uniformity of time stamps for information systems with multiple system clocks and systems connected over a network. Organizations must consider endpoints that may not have regular access to the authoritative time server (e.g., mobile, teleworking, and tactical endpoints). Satisfies: SRG-OS-000785-GPOS-00250, SRG-OS-000355-GPOS-00143

Solution

To configure the system clock to compare to the authoritative time source at least every 24 hours, edit the "/etc/chrony/chrony.conf" file.

Add or correct the following lines, by replacing "[source]" in the following line with an authoritative time source: server [source] iburst maxpoll 16 If the "chrony" service was running and the value of "maxpoll" or "server" was updated, the service must be restarted using the following command: \$ sudo systemctl restart chrony.service

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.7
800-171R3	03.03.07
800-53	AU-8(1)
800-53R5	SC-45
800-53R5	SC-45(1)(a)
CAT	III
CCI	CCI-004922
CCI	CCI-004923
CN-L3	8.1.4.3(b)
CSF	PR.PT-1
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.17
ISO/IEC-27001	A.12.4.4
ITSG-33	AU-8(1)
NESA	T3.6.7
NIAV2	NS44
NIAV2	NS45

NIAV2	NS46
NIAV2	NS47
PCI-DSSV3.2.1	10.4
PCI-DSSV3.2.1	10.4.1
PCI-DSSV3.2.1	10.4.3
PCI-DSSV4.0	10.6
PCI-DSSV4.0	10.6.1
PCI-DSSV4.0	10.6.2
PCI-DSSV4.0	10.6.3
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270751r1066742_rule
STIG-ID	UBTU-24-600160
TBA-FIISB	37.4
VULN-ID	V-270751

Assets

strkbsubuntu9

No matching files were found
Less than 1 matches of regex found

UBTU-24-700020 - Ubuntu 24.04 LTS must generate system journal entries without revealing information that could be exploited by adversaries.

Info

Any operating system providing too much information in error messages risks compromising the data and security of the structure, and content of error messages must be carefully considered by the organization. Organizations carefully consider the structure/content of error messages.

The extent to which information systems are able to identify and handle error conditions is guided by organizational policy and operational requirements.

Information that could be exploited by adversaries includes, for example, erroneous logon attempts with passwords entered by mistake as the username, mission/business information that can be derived from (if not stated explicitly by) information recorded, and personal information, such as account numbers, social security numbers, and credit card numbers.

Solution

Configure the system to set the appropriate permissions to the files and directories used by the systemd journal. Create a drop-in file if it does not already exist with the following command: `$ sudo vi /etc/tmpfiles.d/zzz-systemd-stig.conf` Add or modify the following lines in the `"/usr/lib/tmpfiles.d/zzz-systemd-stig.conf"` file:
`z /run/log/journal 0640 root systemd-journal - - Z /run/log/journal/%m ~0640 root systemd-journal - - z /var/log/journal 0640 root systemd-journal - - z /var/log/journal/%m 0640 root systemd-journal - - z /var/log/journal/%m/system.journal 0640 root systemd-journal - -` Note: Restart the system for these settings to take effect.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11a.
800-53R5	SI-11a.
CAT	II
CCI	CCI-001312
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11b.
RULE-ID	SV-270757r1184072_rule
STIG-ID	UBTU-24-700020
VULN-ID	V-270757

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----  
PASSED - files  
No results found.  
  
-----  
FAILED - directories  
/run/log/journal  
* /run/log/journal  
/var/log/journal  
* /var/log/journal
```

Audits SKIPPED

Audits PASSED

DISA_STIG_Canonical_Ubuntu_24.04_LTS_v1r5.audit from DISA Canonical Ubuntu 24.04 LTS STIG v1r5

Info

Solution

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - Check distribution release

Compliant file(s):

```
/etc/lsb-release - regex '^[\s]*DISTRIB_RELEASE[\s]*=' found - expect  
'^[\s]*DISTRIB_RELEASE[\s]*=[\s]*24.04[\s]*$' found in the following lines:  
2: DISTRIB_RELEASE=24.04
```

UBTU-24-100010 - Ubuntu 24.04 LTS must not have the "systemd-timesyncd" package installed.

Info

Inaccurate time stamps make it more difficult to correlate events and can lead to an inaccurate analysis. Determining the correct time a particular event occurred on a system is critical when conducting forensic analysis and investigating system events. Sources outside the configured acceptable allowance (drift) may be inaccurate. Organizations must consider endpoints that may not have regular access to the authoritative time server (e.g., mobile, teleworking, and tactical endpoints).

Solution

The "systemd-timesyncd" package will be uninstalled as part of the "chrony" package install. Purge the remaining configuration files for "systemd-timesyncd" from Ubuntu 24.04 LTS: `$ sudo apt-get purge systemd-timesyncd`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	III
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270645r1068357_rule
STIG-ID	UBTU-24-100010
SWIFT-CSCV1	2.3
VULN-ID	V-270645

Assets

strkbsubuntu9

The command `'/bin/dpkg -s systemd-timesyncd 2>&1 | /bin/grep -E '^(^Status:|not installed)''` returned :

dpkg-query: package 'systemd-timesyncd' is not installed and no information is available

UBTU-24-100020 - Ubuntu 24.04 LTS must not have the "ntp" package installed.

Info

Inaccurate time stamps make it more difficult to correlate events and can lead to an inaccurate analysis. Determining the correct time a particular event occurred on a system is critical when conducting forensic analysis and investigating system events. Sources outside the configured acceptable allowance (drift) may be inaccurate. Organizations must consider endpoints that may not have regular access to the authoritative time server (e.g., mobile, teleworking, and tactical endpoints).

Solution

Uninstall the "ntp" package using the following command: `$ sudo apt remove ntp` If there are additional configuration files on the system that must be removed, the following command can be used instead: `$ sudo apt-get purge ntp`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	III
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270646r1068358_rule
STIG-ID	UBTU-24-100020
SWIFT-CSCV1	2.3
VULN-ID	V-270646

Assets

strkbsubuntu9

The command `'/bin/dpkg -s ntp 2>&1 | /bin/grep -E '^(Status:|not installed)''` returned :

dpkg-query: package 'ntp' is not installed and no information is available

UBTU-24-100030 - Ubuntu 24.04 LTS must not have the telnet package installed.

Info

Remote access services, such as those providing remote access to network devices and information systems, which lack automated control capabilities, increase risk and make remote user access management difficult at best. Remote access is access to DOD nonpublic information systems by an authorized user (or an information system) communicating through an external, nonorganization-controlled network.

Remote access methods include, for example, dial-up, broadband, and wireless. Ubuntu 24.04 LTS functionality (e.g., RDP) must be capable of taking enforcement action if the audit reveals unauthorized activity.

Automated control of remote access sessions allows organizations to ensure ongoing compliance with remote access policies by enforcing connection rules of remote access applications on a variety of information system components (e.g., servers, workstations, notebook computers, smartphones, and tablets).

Solution

Remove the telnet package from Ubuntu 24.04 LTS with the following command: `$ sudo apt remove telnetd`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.10
800-171R3	03.05.07c.
800-53	IA-5(1)(c)
800-53R5	IA-5(1)(c)
CAT	I
CCI	CCI-000197
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ITSG-33	IA-5(1)(c)
NESA	T5.2.3
NIAV2	CY6
QCSC-V1	5.2.2
QCSC-V1	13.2

RULE-ID	SV-270647r1066430_rule
STIG-ID	UBTU-24-100030
SWIFT-CSCV1	4.1
TBA-FIISB	26.1
VULN-ID	V-270647

Assets

strkbsubuntu9

The command `'/bin/dpkg -s telnetd 2>&1 | /bin/grep -E '^(Status:|not installed)''` returned :

`dpkg-query: package 'telnetd' is not installed and no information is available`

UBTU-24-100040 - Ubuntu 24.04 LTS must not have the rsh-server package installed.

Info

It is detrimental for operating systems to provide, or install by default, functionality exceeding requirements or mission objectives.

These unnecessary capabilities or services are often overlooked and therefore, may remain unsecured.

They increase the risk to the platform by providing additional attack vectors. Operating systems are capable of providing a wide variety of functions and services.

Some of the functions and services, provided by default, may not be necessary to support essential organizational operations (e.g., key missions, functions). Examples of nonessential capabilities include, but are not limited to, games, software packages, tools, and demonstration software, not related to requirements or providing a wide array of functionality not required for every mission, but which cannot be disabled.

Solution

Configure Ubuntu 24.04 LTS to disable nonessential capabilities by removing the rsh-server package from the system with the following command: `$ sudo apt remove rsh-server`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.6
800-171	3.4.7
800-171R3	03.04.06a.
800-53	CM-7a.
800-53R5	CM-7a.
CAT	I
CCI	CCI-000381
CN-L3	7.1.3.5(c)
CN-L3	8.1.4.4(a)
CSF	PR.IP-1
CSF	PR.PT-3
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	CM-7a.
NIAV2	SS15a
PCI-DSSV3.2.1	2.2.1
PCI-DSSV4.0	2.2.3
QCSC-V1	3.2
RULE-ID	SV-270648r1066433_rule

STIG-ID UBTU-24-100040

SWIFT-CSCV1 2.3

VULN-ID V-270648

Assets

strkbsubuntu9

The command `'/bin/dpkg -s rsh-server 2>&1 | /bin/grep -E '^(Status:|not installed)''` returned :

`dpkg-query: package 'rsh-server' is not installed and no information is available`

UBTU-24-100050 - Ubuntu 24.04 LTS must not have the nfs-kernel-server package installed.

Info

It is detrimental for operating systems to provide, or install by default, functionality exceeding requirements or mission objectives.

These unnecessary capabilities or services are often overlooked and therefore may remain unsecured.

They increase the risk to the platform by providing additional attack vectors. Operating systems are capable of providing a wide variety of functions and services.

Some of the functions and services, provided by default, may not be necessary to support essential organizational operations (e.g., key missions, functions). Examples of non-essential capabilities include, but are not limited to, games, software packages, tools, and demonstration software, not related to requirements or providing a wide array of functionality not required for every mission, but which cannot be disabled.

Solution

Configure Ubuntu 24.04 LTS to disable non-essential capabilities by removing the nfs-common and nfs-kernel-server packages from the system with the following commands: Remove packages if present: `$ sudo apt purge --yes nfs-common nfs-kernel-server` Remove any unneeded dependencies `$ sudo apt autoremove --yes` Verify NFS services are gone: `$ sudo systemctl list-units --type=service | grep nfs`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.6
800-171	3.4.7
800-171R3	03.04.06a.
800-53	CM-7a.
800-53R5	CM-7a.
CAT	I
CCI	CCI-000381
CN-L3	7.1.3.5(c)
CN-L3	8.1.4.4(a)
CSF	PR.IP-1
CSF	PR.PT-3
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	CM-7a.
NIAV2	SS15a
PCI-DSSV3.2.1	2.2.1
PCI-DSSV4.0	2.2.3
QCSC-V1	3.2

RULE-ID	SV-279938r1156367_rule
STIG-ID	UBTU-24-100050
SWIFT-CSCV1	2.3
VULN-ID	V-279938

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - nfs-common not installed

The command '/bin/dpkg -s nfs-common 2>&1 | /bin/grep -E '^(Status:|not installed)'' returned :

dpkg-query: package 'nfs-common' is not installed and no information is available

PASSED - nfs-kernel-server not installed

The command '/bin/dpkg -s nfs-kernel-server 2>&1 | /bin/grep -E '^(Status:|not installed)'' returned :

dpkg-query: package 'nfs-kernel-server' is not installed and no information is available

UBTU-24-100100 - Ubuntu 24.04 LTS must use a file integrity tool to verify correct operation of all security functions.

Info

Without verification, security functions may not operate correctly and the failure may go unnoticed. Security function is defined as the hardware, software, and/or firmware of the information system responsible for enforcing the system security policy and supporting the isolation of code and data on which the protection is based.

Security functionality includes, but is not limited to, establishing system accounts, configuring access authorizations (i.e., permissions, privileges), setting events to be audited, and setting intrusion detection parameters. This requirement applies to Ubuntu 24.04 LTS performing security function verification/testing and/or systems and environments that require this functionality.

Solution

Install the "AIDE" file integrity package: `$ sudo apt install -y aide`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-6a.
800-53R5	SI-6a.
CAT	II
CCI	CCI-002696
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-6a.
RULE-ID	SV-270649r1067136_rule
STIG-ID	UBTU-24-100100
VULN-ID	V-270649

Assets

strkbsubuntu9

The command `'/bin/dpkg -s aide 2>&1 | /bin/grep -E '(Status:|not installed)''` returned :

Status: install ok installed

UBTU-24-100130 - Ubuntu 24.04 LTS must notify designated personnel if baseline configurations are changed in an unauthorized manner. The file integrity tool must notify the system administrator (SA) when changes to the baseline configuration or anomalies in the operation of any security functions are discovered.

Info

Unauthorized changes to the baseline configuration could make the system vulnerable to various attacks or allow unauthorized access to Ubuntu 24.04 LTS.

Changes to Ubuntu 24.04 LTS configurations can have unintended side effects, some of which may be relevant to security. Detecting such changes and providing an automated response can help avoid unintended, negative consequences that could ultimately affect the security state of Ubuntu 24.04 LTS.

Ubuntu 24.04 LTS' IMO/information system security officer (ISSO) and SAs must be notified via email and/or monitoring system trap when there is an unauthorized modification of a configuration item. Satisfies: SRG-OS-000447-GPOS-00201, SRG-OS-000363-GPOS-00150

Solution

Configure Ubuntu 24.04 LTS to notify designated personnel if baseline configurations are changed in an unauthorized manner. Modify the "SILENTREPORTS" parameter in the "/etc/default/aide" file with a value of "no" if it does not already exist as follows: SILENTREPORTS=no

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.3
800-171R3	03.04.03
800-53	CM-3(5)
800-53	SI-6d.
800-53R5	CM-3(5)
800-53R5	SI-6d.
CAT	II
CCI	CCI-001744
CCI	CCI-002702
CN-L3	8.1.10.6(g)
CSF	DE.CM-1
CSF	DE.CM-7
CSF	PR.IP-1
CSF	PR.IP-3
CSF2.0	DE.CM-01
CSF2.0	DE.CM-09
CSF2.0	ID.RA-07
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
GDPR	32.4
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ISO-27001-2022	A.8.32
ISO-27001-2022	8.1
ISO-27001-2022	9.3.3
ISO/IEC-27001	A.12.1.2
ITSG-33	CM-3
ITSG-33	SI-6a.
NESA	T3.2.3
NESA	T3.3.2
NESA	T7.5.1
NESA	T7.6.1
NESA	T7.6.2
NESA	T7.6.3
NIAV2	CM1
NIAV2	CM1a
NIAV2	CM1b
NIAV2	CM1c
PCI-DSSV3.2.1	11.5
PCI-DSSV4.0	11.5.2
QCSC-V1	3.2
QCSC-V1	5.2.1
QCSC-V1	6.2
QCSC-V1	7.2
QCSC-V1	8.2.1
RULE-ID	SV-270652r1067138_rule
STIG-ID	UBTU-24-100130
VULN-ID	V-270652

Assets

strkbsubuntu9

Compliant file(s):

```
/etc/default/aide - regex '^[\s]*SILENTREPORTS[\s]*=' found - expect  
'^[\s]*SILENTREPORTS[\s]*=[\s]*no[\s]*$' found in the following lines:  
94: SILENTREPORTS=no
```

UBTU-24-100200 - Ubuntu 24.04 LTS must be configured to preserve log records from failure events.

Info

Failure to a known state can address safety or security in accordance with the mission/business needs of the organization.

Failure to a known secure state helps prevent a loss of confidentiality, integrity, or availability in the event of a failure of the information system or a component of the system. Preserving operating system state information helps to facilitate operating system restart and return to the operational mode of the organization with least disruption to mission/business processes.

Solution

Configure the log service to collect failure events. Install the log service (if the log service is not already installed) with the following command: `$ sudo apt install -y rsyslog` Enable the log service with the following command: `$ sudo systemctl enable --now rsyslog`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SC-24
800-53R5	SC-24
CAT	II
CCI	CCI-001665
CSF2.0	PR.DS-10
CSF2.0	PR.IR-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SC-24
ITSG-33	SC-24a.
QCSC-V1	5.2.1
RULE-ID	SV-270653r1067141_rule
STIG-ID	UBTU-24-100200
VULN-ID	V-270653

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
PASSED - rsyslog
The command '/bin/dpkg -s rsyslog | /bin/grep Status: 2>&1' returned :

Status: install ok installed

-----
PASSED - rsyslog enabled
The command '/bin/systemctl is-enabled rsyslog' returned :

enabled
```

```
-----  
PASSED - rsyslog check - active  
The command '/bin/systemctl is-active rsyslog' returned :  
active
```

UBTU-24-100300 - Ubuntu 24.04 LTS must have an application firewall installed in order to control remote access methods.

Info

Remote access services, such as those providing remote access to network devices and information systems, which lack automated control capabilities, increase risk, and make remote user access management difficult at best. Remote access is access to DOD nonpublic information systems by an authorized user (or an information system) communicating through an external, nonorganization-controlled network.

Remote access methods include, for example, dial-up, broadband, and wireless. Ubuntu 24.04 LTS functionality (e.g., RDP) must be capable of taking enforcement action if the audit reveals unauthorized activity.

Automated control of remote access sessions allows organizations to ensure ongoing compliance with remote access policies by enforcing connection rules of remote access applications on a variety of information system components (e.g., servers, workstations, notebook computers, smartphones, and tablets).

Solution

Install the ufw by using the following command: `$ sudo apt install -y ufw`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.12
800-171R3	03.01.12
800-53	AC-17(1)
800-53R5	AC-17(1)
CAT	II
CCI	CCI-002314
CN-L3	8.1.4.4(c)
CN-L3	8.1.10.6(i)
CSF	PR.AC-3
CSF	PR.PT-4
CSF2.0	PR.AA-05
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
ISO-27001-2022	A.8.16
ISO/IEC-27001	A.6.2.2
ITSG-33	AC-17(1)
NESA	T5.4.4
QCSC-V1	3.2
QCSC-V1	5.2.1

QCSC-V1	5.2.2
RULE-ID	SV-270654r1067143_rule
STIG-ID	UBTU-24-100300
SWIFT-CSCV1	2.6
VULN-ID	V-270654

Assets

strkbsubuntu9

The command '/bin/dpkg -s ufw 2>&1 | /bin/grep -E '(Status:|not installed)'' returned :

Status: install ok installed

UBTU-24-100310 - Ubuntu 24.04 LTS must enable and run the Uncomplicated Firewall (ufw).

Info

Remote access services, such as those providing remote access to network devices and information systems, which lack automated control capabilities, increase risk, and make remote user access management difficult at best. Remote access is access to DOD nonpublic information systems by an authorized user (or an information system) communicating through an external, nonorganization-controlled network.

Remote access methods include, for example, dial-up, broadband, and wireless. Ubuntu 24.04 LTS functionality (e.g., RDP) must be capable of taking enforcement action if the audit reveals unauthorized activity.

Automated control of remote access sessions allows organizations to ensure ongoing compliance with remote access policies by enforcing connection rules of remote access applications on a variety of information system components (e.g., servers, workstations, notebook computers, smartphones, and tablets). Satisfies: SRG-OS-000297-GPOS-00115, SRG-OS-000480-GPOS-00232

Solution

Enable the ufw by using the following command: `$ sudo ufw enable` Note: Enabling the firewall will potentially disrupt ssh sessions.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.12
800-171R3	03.01.12
800-53	AC-17(1)
800-53R5	AC-17(1)
CAT	II
CCI	CCI-002314
CN-L3	8.1.4.4(c)
CN-L3	8.1.10.6(i)
CSF	PR.AC-3
CSF	PR.PT-4
CSF2.0	PR.AA-05
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
ISO-27001-2022	A.8.16
ISO/IEC-27001	A.6.2.2
ITSG-33	AC-17(1)
NESA	T5.4.4
QCSC-V1	3.2
QCSC-V1	5.2.1

QCSC-V1	5.2.2
RULE-ID	SV-270655r1067145_rule
STIG-ID	UBTU-24-100310
SWIFT-CSCV1	2.6
VULN-ID	V-270655

Assets

strkbsubuntu9

The command '/sbin/ufw status' returned :

Status: active

To	Action	From
--	-----	----
22/tcp	ALLOW	Anywhere
22/tcp (v6)	ALLOW	Anywhere (v6)

UBTU-24-100400 - Ubuntu 24.04 LTS must have the "auditd" package installed.

Info

Without establishing the when, where, type, source, and outcome of events that occurred, it would be difficult to establish, correlate, and investigate the events leading up to an outage or attack. Without the capability to generate audit records, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit record content that may be necessary to satisfy this requirement includes, for example, time stamps, source and destination addresses, user/process identifiers, event descriptions, success/fail indications, filenames involved, and access control or flow control rules invoked. Reconstruction of harmful events or forensic analysis is not possible if audit records do not contain enough information. Successful incident response and auditing relies on timely, accurate system information and analysis to allow the organization to identify and respond to potential incidents in a proficient manner. If Ubuntu 24.04 LTS does not provide the ability to centrally review Ubuntu 24.04 LTS logs, forensic analysis is negatively impacted. Associating event types with detected events in Ubuntu 24.04 LTS audit logs provides a means of investigating an attack; recognizing resource utilization or capacity thresholds; or identifying an improperly configured operating system. Satisfies: SRG-OS-000365-GPOS-00152, SRG-OS-000337-GPOS-00129, SRG-OS-000062-GPOS-00031, SRG-OS-000475-GPOS-00220, SRG-OS-000042-GPOS-00020, SRG-OS-000042-GPOS-00021, SRG-OS-000037-GPOS-00015, SRG-OS-000038-GPOS-00016, SRG-OS-000039-GPOS-00017, SRG-OS-000040-GPOS-00018, SRG-OS-000041-GPOS-00019, SRG-OS-000051-GPOS-00024, SRG-OS-000054-GPOS-00025, SRG-OS-000122-GPOS-00063, SRG-OS-000348-GPOS-00136, SRG-OS-000349-GPOS-00137, SRG-OS-000350-GPOS-00138, SRG-OS-000351-GPOS-00139, SRG-OS-000352-GPOS-00140, SRG-OS-000353-GPOS-00141, SRG-OS-000354-GPOS-00142

Solution

Configure the audit service to produce audit records containing the information needed to establish when (date and time) an event occurred. Install the audit service (if the audit service is not already installed) with the following command: `$ sudo apt install -y auditd`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171	3.3.6
800-171R3	03.03.02a.
800-171R3	03.03.02b.
800-171R3	03.03.03
800-171R3	03.03.03a.
800-171R3	03.03.05
800-171R3	03.03.06a.
800-171R3	03.03.06b.
800-53	AU-3
800-53	AU-3(1)
800-53	AU-6(4)
800-53	AU-7a.
800-53	AU-7b.
800-53	AU-7(1)

800-53	AU-12a.
800-53	AU-12c.
800-53	AU-12(3)
800-53R5	AU-3a.
800-53R5	AU-3b.
800-53R5	AU-3c.
800-53R5	AU-3d.
800-53R5	AU-3e.
800-53R5	AU-3(1)
800-53R5	AU-6(4)
800-53R5	AU-7a.
800-53R5	AU-7b.
800-53R5	AU-7(1)
800-53R5	AU-12a.
800-53R5	AU-12c.
800-53R5	AU-12(3)
800-53R5	CM-5(1)(b)
CAT	II
CCI	CCI-000130
CCI	CCI-000131
CCI	CCI-000132
CCI	CCI-000133
CCI	CCI-000134
CCI	CCI-000135
CCI	CCI-000154
CCI	CCI-000158
CCI	CCI-000169
CCI	CCI-000172
CCI	CCI-001875
CCI	CCI-001876
CCI	CCI-001877

CCI	CCI-001878
CCI	CCI-001879
CCI	CCI-001880
CCI	CCI-001881
CCI	CCI-001882
CCI	CCI-001914
CCI	CCI-003938
CN-L3	7.1.2.3(a)
CN-L3	7.1.2.3(b)
CN-L3	7.1.2.3(c)
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	7.1.3.3(d)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CN-L3	8.1.4.3(b)
CSF	DE.AE-2
CSF	DE.AE-3
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	DE.DP-4
CSF	PR.PT-1
CSF	RS.AN-1
CSF	RS.AN-3
CSF	RS.CO-2
CSF2.0	DE.AE-02
CSF2.0	DE.AE-03
CSF2.0	DE.CM-01

CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
CSF2.0	RS.AN-03
CSF2.0	RS.AN-06
CSF2.0	RS.AN-07
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.25
ISO-27001-2022	A.5.28
ISO-27001-2022	A.6.8
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-3
ITSG-33	AU-3(1)
ITSG-33	AU-6(4)
ITSG-33	AU-7
ITSG-33	AU-7(1)
ITSG-33	AU-12
ITSG-33	AU-12a.
ITSG-33	AU-12c.
NESA	M5.2.5
NESA	T3.6.2
NESA	T3.6.3
NESA	T3.6.5
NESA	T3.6.6
NIAV2	AM34a
NIAV2	AM34b
NIAV2	AM34c

NIAV2	AM34d
NIAV2	AM34e
NIAV2	AM34f
NIAV2	AM34g
NIAV2	SM8
PCI-DSSV3.2.1	10.1
PCI-DSSV3.2.1	10.3
PCI-DSSV3.2.1	10.3.1
PCI-DSSV3.2.1	10.3.2
PCI-DSSV3.2.1	10.3.3
PCI-DSSV3.2.1	10.3.4
PCI-DSSV3.2.1	10.3.5
PCI-DSSV3.2.1	10.3.6
PCI-DSSV4.0	10.2.2
QCSC-V1	3.2
QCSC-V1	5.2.3
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	10.2.1
QCSC-V1	11.2
QCSC-V1	13.2
RULE-ID	SV-270656r1067148_rule
STIG-ID	UBTU-24-100400
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270656

Assets

strkbsubuntu9

The command '/bin/dpkg -s auditd | /bin/grep Status: 2>&1' returned :

Status: install ok installed

UBTU-24-100410 - Ubuntu 24.04 LTS must produce audit records and reports containing information to establish when, where, what type, the source, and the outcome for all DOD-defined auditable events and actions in near real time.

Info

Without establishing the when, where, type, source, and outcome of events that occurred, it would be difficult to establish, correlate, and investigate the events leading up to an outage or attack. Without the capability to generate audit records, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit record content that may be necessary to satisfy this requirement includes, for example, time stamps, source and destination addresses, user/process identifiers, event descriptions, success/fail indications, filenames involved, and access control or flow control rules invoked. Reconstruction of harmful events or forensic analysis is not possible if audit records do not contain enough information. Successful incident response and auditing relies on timely, accurate system information and analysis to allow the organization to identify and respond to potential incidents in a proficient manner. If Ubuntu 24.04 LTS does not provide the ability to centrally review Ubuntu 24.04 LTS logs, forensic analysis is negatively impacted. Associating event types with detected events in Ubuntu 24.04 LTS audit logs provides a means of investigating an attack; recognizing resource utilization or capacity thresholds; or identifying an improperly configured operating system. Satisfies: SRG-OS-000365-GPOS-00152, SRG-OS-000337-GPOS-00129, SRG-OS-000062-GPOS-00031, SRG-OS-000475-GPOS-00220, SRG-OS-000042-GPOS-00020, SRG-OS-000042-GPOS-00021, SRG-OS-000037-GPOS-00015, SRG-OS-000038-GPOS-00016, SRG-OS-000039-GPOS-00017, SRG-OS-000040-GPOS-00018, SRG-OS-000041-GPOS-00019, SRG-OS-000051-GPOS-00024, SRG-OS-000054-GPOS-00025, SRG-OS-000122-GPOS-00063, SRG-OS-000348-GPOS-00136, SRG-OS-000349-GPOS-00137, SRG-OS-000350-GPOS-00138, SRG-OS-000351-GPOS-00139, SRG-OS-000352-GPOS-00140, SRG-OS-000353-GPOS-00141, SRG-OS-000354-GPOS-00142

Solution

Configure the audit service to produce audit records containing the information needed to establish when (date and time) an event occurred. Enable the audit service with the following command: `$ sudo systemctl enable auditd.service` To reload the rules file, issue the following command: `$ sudo augenrules --load`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171	3.3.6
800-171R3	03.03.02a.
800-171R3	03.03.02b.
800-171R3	03.03.03
800-171R3	03.03.03a.
800-171R3	03.03.05
800-171R3	03.03.06a.
800-171R3	03.03.06b.
800-53	AU-3
800-53	AU-3(1)
800-53	AU-6(4)
800-53	AU-7a.
800-53	AU-7b.

800-53	AU-7(1)
800-53	AU-12a.
800-53	AU-12c.
800-53	AU-12(3)
800-53R5	AU-3a.
800-53R5	AU-3b.
800-53R5	AU-3c.
800-53R5	AU-3d.
800-53R5	AU-3e.
800-53R5	AU-3(1)
800-53R5	AU-6(4)
800-53R5	AU-7a.
800-53R5	AU-7b.
800-53R5	AU-7(1)
800-53R5	AU-12a.
800-53R5	AU-12c.
800-53R5	AU-12(3)
800-53R5	CM-5(1)(b)
CAT	II
CCI	CCI-000130
CCI	CCI-000131
CCI	CCI-000132
CCI	CCI-000133
CCI	CCI-000134
CCI	CCI-000135
CCI	CCI-000154
CCI	CCI-000158
CCI	CCI-000169
CCI	CCI-000172
CCI	CCI-001875
CCI	CCI-001876

CCI	CCI-001877
CCI	CCI-001878
CCI	CCI-001879
CCI	CCI-001880
CCI	CCI-001881
CCI	CCI-001882
CCI	CCI-001914
CCI	CCI-003938
CN-L3	7.1.2.3(a)
CN-L3	7.1.2.3(b)
CN-L3	7.1.2.3(c)
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	7.1.3.3(d)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CN-L3	8.1.4.3(b)
CSF	DE.AE-2
CSF	DE.AE-3
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	DE.DP-4
CSF	PR.PT-1
CSF	RS.AN-1
CSF	RS.AN-3
CSF	RS.CO-2
CSF2.0	DE.AE-02
CSF2.0	DE.AE-03

CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
CSF2.0	RS.AN-03
CSF2.0	RS.AN-06
CSF2.0	RS.AN-07
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.25
ISO-27001-2022	A.5.28
ISO-27001-2022	A.6.8
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-3
ITSG-33	AU-3(1)
ITSG-33	AU-6(4)
ITSG-33	AU-7
ITSG-33	AU-7(1)
ITSG-33	AU-12
ITSG-33	AU-12a.
ITSG-33	AU-12c.
NESA	M5.2.5
NESA	T3.6.2
NESA	T3.6.3
NESA	T3.6.5
NESA	T3.6.6
NIAV2	AM34a
NIAV2	AM34b

NIAV2	AM34c
NIAV2	AM34d
NIAV2	AM34e
NIAV2	AM34f
NIAV2	AM34g
NIAV2	SM8
PCI-DSSV3.2.1	10.1
PCI-DSSV3.2.1	10.3
PCI-DSSV3.2.1	10.3.1
PCI-DSSV3.2.1	10.3.2
PCI-DSSV3.2.1	10.3.3
PCI-DSSV3.2.1	10.3.4
PCI-DSSV3.2.1	10.3.5
PCI-DSSV3.2.1	10.3.6
PCI-DSSV4.0	10.2.2
QCSC-V1	3.2
QCSC-V1	5.2.3
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	10.2.1
QCSC-V1	11.2
QCSC-V1	13.2
RULE-ID	SV-270657r1066460_rule
STIG-ID	UBTU-24-100410
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270657

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - active

The command '/bin/systemctl is-active auditd' returned :

active

```
-----  
PASSED - enabled  
The command '/bin/systemctl is-enabled auditd' returned :  
enabled
```

UBTU-24-100500 - Ubuntu 24.04 LTS must have AppArmor installed.

Info

Control of program execution is a mechanism used to prevent execution of unauthorized programs. Some operating systems may provide a capability that runs counter to the mission or provides users with functionality that exceeds mission requirements.

This includes functions and services installed at Ubuntu 24.04 LTS-level. Some of the programs, installed by default, may be harmful or may not be necessary to support essential organizational operations (e.g., key missions, functions).

Removal of executable programs is not always possible; therefore, establishing a method of preventing program execution is critical to maintaining a secure system baseline. Methods for complying with this requirement include restricting execution of programs in certain environments, while preventing execution in other environments; or limiting execution of certain program functionality based on organization-defined criteria (e.g., privileges, subnets, sandboxed environments, or roles). Satisfies: SRG-OS-000312-GPOS-00124, SRG-OS-000368-GPOS-00154, SRG-OS-000370-GPOS-00155

Solution

Install "AppArmor" with the following command: `$ sudo apt install apparmor` Note: AppArmor must have properly configured profiles for applications and home directories.

All configurations will be based on the actual system setup and organization and normally are on a per role basis.

Refer to the AppArmor documentation for more information on configuring profiles.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.1
800-171	3.4.7
800-171	3.4.8
800-171R3	03.01.02
800-171R3	03.04.06
800-171R3	03.04.08b.
800-53	AC-3(4)
800-53	CM-7(2)
800-53	CM-7(5)(b)
800-53R5	AC-3(4)
800-53R5	CM-7(2)
800-53R5	CM-7(5)(b)
CAT	II
CCI	CCI-001764
CCI	CCI-001774
CCI	CCI-002165
CN-L3	8.1.4.2(f)
CN-L3	8.1.4.11(b)

CN-L3	8.1.10.2(c)
CN-L3	8.5.3.1
CN-L3	8.5.4.1(a)
CSF	PR.AC-4
CSF	PR.IP-1
CSF	PR.PT-3
CSF2.0	PR.AA-05
CSF2.0	PR.DS-10
CSF2.0	PR.IR-01
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
ISO-27001-2022	A.5.15
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.3
ISO-27001-2022	A.8.18
ISO-27001-2022	A.8.19
ISO-27001-2022	A.8.20
ISO/IEC-27001	A.9.4.1
ISO/IEC-27001	A.9.4.5
ISO/IEC-27001	A.12.5.1
ISO/IEC-27001	A.12.6.2
ITSG-33	AC-3(4)
ITSG-33	CM-7
ITSG-33	CM-7(2)
NESA	T4.2.1
NESA	T5.4.4
NESA	T5.4.5
NESA	T5.5.4

NESA	T5.6.1
NESA	T7.5.2
NESA	T7.5.3
NIAV2	AM3
NIAV2	SS15a
NIAV2	SS29
PCI-DSSV3.2.1	2.2.2
QCSC-V1	3.2
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270659r1066466_rule
STIG-ID	UBTU-24-100500
SWIFT-CSCV1	2.3
TBA-FIISB	31.1
TBA-FIISB	44.2.2
TBA-FIISB	49.2.3
VULN-ID	V-270659

Assets

strkbsubuntu9

The command '/bin/dpkg -s apparmor | /bin/grep Status: 2>&1' returned :

Status: install ok installed

UBTU-24-100510 - Ubuntu 24.04 LTS must be configured to use AppArmor.

Info

Control of program execution is a mechanism used to prevent execution of unauthorized programs. Some operating systems may provide a capability that runs counter to the mission or provides users with functionality that exceeds mission requirements.

This includes functions and services installed at Ubuntu 24.04 LTS-level. Some of the programs, installed by default, may be harmful or may not be necessary to support essential organizational operations (e.g., key missions, functions).

Removal of executable programs is not always possible; therefore, establishing a method of preventing program execution is critical to maintaining a secure system baseline. Methods for complying with this requirement include restricting execution of programs in certain environments, while preventing execution in other environments; or limiting execution of certain program functionality based on organization-defined criteria (e.g., privileges, subnets, sandboxed environments, or roles). Satisfies: SRG-OS-000368-GPOS-00154, SRG-OS-000324-GPOS-00125, SRG-OS-000370-GPOS-00155

Solution

Enable "apparmor" with the following command: `$ sudo systemctl enable apparmor.service` Start "apparmor" with the following command: `$ sudo systemctl start apparmor.service` Note: AppArmor must have properly configured profiles for applications and home directories.

All configurations will be based on the actual system setup and organization and normally are on a per role basis.

Refer to the AppArmor documentation for more information on configuring profiles.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.7
800-171	3.4.7
800-171	3.4.8
800-171R3	03.01.07a.
800-171R3	03.04.06
800-171R3	03.04.08b.
800-53	AC-6(10)
800-53	CM-7(2)
800-53	CM-7(5)(b)
800-53R5	AC-6(10)
800-53R5	CM-7(2)
800-53R5	CM-7(5)(b)
CAT	II
CCI	CCI-001764
CCI	CCI-001774
CCI	CCI-002235
CN-L3	7.1.3.2(b)
CN-L3	7.1.3.2(g)

CN-L3	8.1.4.2(d)
CN-L3	8.1.10.6(a)
CSF	PR.AC-4
CSF	PR.IP-1
CSF	PR.PT-3
CSF2.0	PR.AA-05
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
ISO-27001-2022	A.5.15
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.18
ISO-27001-2022	A.8.19
ISO/IEC-27001	A.12.5.1
ISO/IEC-27001	A.12.6.2
ITSG-33	AC-6
ITSG-33	CM-7
ITSG-33	CM-7(2)
NESA	T5.1.1
NESA	T5.2.2
NESA	T5.4.1
NESA	T5.4.4
NESA	T5.4.5
NESA	T5.5.4
NESA	T5.6.1
NESA	T7.5.3
NIAV2	AM1
NIAV2	AM23f
NIAV2	SS13c

NIAV2	SS15a
NIAV2	SS15c
PCI-DSSV3.2.1	2.2.2
PCI-DSSV3.2.1	7.1.2
PCI-DSSV4.0	7.2.1
PCI-DSSV4.0	7.2.2
QCSC-V1	3.2
QCSC-V1	5.2.2
QCSC-V1	6.2
RULE-ID	SV-270660r1066469_rule
STIG-ID	UBTU-24-100510
SWIFT-CSCV1	2.3
SWIFT-CSCV1	5.1
TBA-FIISB	31.4.2
TBA-FIISB	31.4.3
TBA-FIISB	44.2.2
TBA-FIISB	49.2.3
VULN-ID	V-270660

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - apparmor.service active

The command '/bin/systemctl is-active apparmor.service' returned :

active

PASSED - apparmor.service enabled

The command '/bin/systemctl is-enabled apparmor.service' returned :

enabled

UBTU-24-100600 - Ubuntu 24.04 LTS must have the "libpam-pwquality" package installed.

Info

Use of a complex password helps to increase the time and resources required to compromise the password. Password complexity, or strength, is a measure of the effectiveness of a password in resisting attempts at guessing and brute-force attacks.

"pwquality" enforces complex password construction configuration and has the ability to limit brute-force attacks on the system.

Solution

Install the "pam_pwquality" package by using the following command: `$ sudo apt install -y libpam-pwquality`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	II
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270661r1067175_rule
STIG-ID	UBTU-24-100600
SWIFT-CSCV1	2.3
VULN-ID	V-270661

Assets

strkbsubuntu9

The command `'/bin/dpkg -s libpam-pwquality 2>&1 | /bin/grep -E '(Status:|not installed)''` returned :

Status: install ok installed

UBTU-24-100700 - Ubuntu 24.04 LTS must have the "chrony" package installed.

Info

Inaccurate time stamps make it more difficult to correlate events and can lead to an inaccurate analysis. Determining the correct time a particular event occurred on a system is critical when conducting forensic analysis and investigating system events. Sources outside the configured acceptable allowance (drift) may be inaccurate. Organizations must consider endpoints that may not have regular access to the authoritative time server (e.g., mobile, teleworking, and tactical endpoints).

Solution

Install the "chrony" network time protocol package using the following command: `$ sudo apt install -y chrony`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	III
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270664r1068359_rule
STIG-ID	UBTU-24-100700
SWIFT-CSCV1	2.3
VULN-ID	V-270664

Assets

strkbsubuntu9

The command `'/bin/dpkg -s chrony 2>&1 | /bin/grep -E '(Status:|not installed)''` returned :

Status: install ok installed

UBTU-24-100800 - Ubuntu 24.04 LTS must have SSH installed.

Info

Without protection of the transmitted information, confidentiality and integrity may be compromised because unprotected communications can be intercepted and either read or altered. This requirement applies to both internal and external networks and all types of information system components from which information can be transmitted (e.g., servers, mobile devices, notebook computers, printers, copiers, scanners, and facsimile machines).

Communication paths outside the physical protection of a controlled boundary are exposed to the possibility of interception and modification. Protecting the confidentiality and integrity of organizational information can be accomplished by physical means (e.g., employing physical distribution systems) or by logical means (e.g., employing cryptographic techniques).

If physical means of protection are employed, then logical means (cryptography) do not have to be employed, and vice versa. Satisfies: SRG-OS-000423-GPOS-00187, SRG-OS-000425-GPOS-00189, SRG-OS-000426-GPOS-00190

Solution

Install the "ssh" meta-package on the system with the following command: `$ sudo apt install -y ssh`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.13.8
800-171R3	03.13.08
800-53	SC-8
800-53	SC-8(2)
800-53R5	SC-8
800-53R5	SC-8(2)
CAT	I
CCI	CCI-002418
CCI	CCI-002420
CCI	CCI-002422
CN-L3	8.1.2.2(a)
CN-L3	8.1.2.2(b)
CN-L3	8.1.4.7(a)
CN-L3	8.1.4.8(a)
CN-L3	8.2.4.5(c)
CN-L3	8.2.4.5(d)
CN-L3	8.5.2.2
CSF	PR.DS-2
CSF	PR.DS-5
CSF2.0	PR.DS-02

DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.a
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(e)(1)
HIPAA	164.312(e)(2)(i)
ISO-27001-2022	A.5.10
ISO-27001-2022	A.5.14
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.20
ITSG-33	SC-8
ITSG-33	SC-8a.
ITSG-33	SC-8(2)
ITSG-33	SC-9(2)
NESA	T4.3.1
NESA	T4.3.2
NESA	T4.5.1
NESA	T4.5.2
NESA	T7.3.3
NESA	T7.4.1
NIAV2	IE8
NIAV2	IE9
NIAV2	IE12
NIAV2	NS29
NIAV2	SS24
PCI-DSSV3.2.1	2.3
PCI-DSSV3.2.1	4.1
PCI-DSSV4.0	2.2.7
PCI-DSSV4.0	4.2.1
QCSC-V1	5.2.2
QCSC-V1	6.2

RULE-ID	SV-270665r1067133_rule
STIG-ID	UBTU-24-100800
SWIFT-CSCV1	2.1
VULN-ID	V-270665

Assets

strkbsubuntu9

The command '/bin/dpkg -s openssh-server 2>&1 | /bin/grep -E '(Status:|not installed)''
returned :

Status: install ok installed

UBTU-24-100810 - Ubuntu 24.04 LTS must use SSH to protect the confidentiality and integrity of transmitted information.

Info

Without protection of the transmitted information, confidentiality and integrity may be compromised because unprotected communications can be intercepted and either read or altered. This requirement applies to both internal and external networks and all types of information system components from which information can be transmitted (e.g., servers, mobile devices, notebook computers, printers, copiers, scanners, and facsimile machines).

Communication paths outside the physical protection of a controlled boundary are exposed to the possibility of interception and modification. Protecting the confidentiality and integrity of organizational information can be accomplished by physical means (e.g., employing physical distribution systems) or by logical means (e.g., employing cryptographic techniques).

If physical means of protection are employed, then logical means (cryptography) do not have to be employed, and vice versa. Satisfies: SRG-OS-000423-GPOS-00187, SRG-OS-000425-GPOS-00189, SRG-OS-000426-GPOS-00190

Solution

Enable the "ssh" service to start automatically on reboot with the following command: `$ sudo systemctl enable ssh.service` ensure the "ssh" service is running `$ sudo systemctl start ssh.service`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.13.8
800-171R3	03.13.08
800-53	SC-8
800-53	SC-8(2)
800-53R5	SC-8
800-53R5	SC-8(2)
CAT	I
CCI	CCI-002418
CCI	CCI-002420
CCI	CCI-002422
CN-L3	8.1.2.2(a)
CN-L3	8.1.2.2(b)
CN-L3	8.1.4.7(a)
CN-L3	8.1.4.8(a)
CN-L3	8.2.4.5(c)
CN-L3	8.2.4.5(d)
CN-L3	8.5.2.2
CSF	PR.DS-2
CSF	PR.DS-5

CSF2.0	PR.DS-02
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.a
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(e)(1)
HIPAA	164.312(e)(2)(i)
ISO-27001-2022	A.5.10
ISO-27001-2022	A.5.14
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.20
ITSG-33	SC-8
ITSG-33	SC-8a.
ITSG-33	SC-8(2)
ITSG-33	SC-9(2)
NESA	T4.3.1
NESA	T4.3.2
NESA	T4.5.1
NESA	T4.5.2
NESA	T7.3.3
NESA	T7.4.1
NIAV2	IE8
NIAV2	IE9
NIAV2	IE12
NIAV2	NS29
NIAV2	SS24
PCI-DSSV3.2.1	2.3
PCI-DSSV3.2.1	4.1
PCI-DSSV4.0	2.2.7
PCI-DSSV4.0	4.2.1
QCSC-V1	5.2.2

QCSC-V1	6.2
RULE-ID	SV-270666r1066487_rule
STIG-ID	UBTU-24-100810
SWIFT-CSCV1	2.1
VULN-ID	V-270666

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
PASSED - sshd.service
The command '/bin/systemctl is-active ssh' returned :

active

-----
PASSED - openssh-server
The command '/bin/systemctl is-enabled ssh' returned :

enabled
```

UBTU-24-100820 - Ubuntu 24.04 LTS must configure the SSH daemon to use FIPS 140-3 approved ciphers to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.

Info

Without cryptographic integrity protections, information can be altered by unauthorized users without detection. Remote access (e.g., RDP) is access to DOD nonpublic information systems by an authorized user (or an information system) communicating through an external, nonorganization-controlled network.

Remote access methods include, for example, dial-up, broadband, and wireless. Nonlocal maintenance and diagnostic activities are those activities conducted by individuals communicating through a network, either an external network (e.g., the internet) or an internal network. Local maintenance and diagnostic activities are those activities carried out by individuals physically present at the information system or information system component and not communicating across a network connection. Encrypting information for transmission protects information from unauthorized disclosure and modification.

Cryptographic mechanisms implemented to protect information integrity include, for example, cryptographic hash functions that have common application in digital signatures, checksums, and message authentication codes. Satisfies: SRG-OS-000033-GPOS-00014, SRG-OS-000394-GPOS-00174, SRG-OS-000424-GPOS-00188

Solution

Configure Ubuntu 24.04 LTS to allow the SSH daemon to only implement FIPS-approved algorithms. Add the following line (or modify the line to have the required value) to the "/etc/ssh/sshd_config" file (this file may be named differently or be in a different location if using a version of SSH that is provided by a third-party vendor): Ciphers aes256-gcm@openssh.com,aes128-gcm@openssh.com,aes256-ctr,aes128-ctr Restart the "sshd" service for changes to take effect: \$ sudo systemctl restart sshd

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.13
800-171	3.7.5
800-171	3.13.8
800-171R3	03.07.05
800-171R3	03.13.08
800-53	AC-17(2)
800-53	MA-4(6)
800-53	SC-8(1)
800-53R5	AC-17(2)
800-53R5	MA-4(6)
800-53R5	SC-8(1)
CAT	II
CCI	CCI-000068
CCI	CCI-002421
CCI	CCI-003123
CN-L3	7.1.2.7(g)
CN-L3	7.1.3.1(d)

CN-L3	8.1.2.2(a)
CN-L3	8.1.2.2(b)
CN-L3	8.1.4.1(c)
CN-L3	8.1.4.7(a)
CN-L3	8.1.4.8(a)
CN-L3	8.2.4.5(c)
CN-L3	8.2.4.5(d)
CN-L3	8.5.2.2
CSF	PR.AC-3
CSF	PR.DS-2
CSF	PR.DS-5
CSF	PR.MA-2
CSF	PR.PT-4
CSF2.0	PR.AA-05
CSF2.0	PR.DS-02
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.a
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
HIPAA	164.312(e)(1)
HIPAA	164.312(e)(2)(i)
ISO-27001-2022	A.5.10
ISO-27001-2022	A.5.14
ISO-27001-2022	A.5.33
ISO-27001-2022	A.6.7
ISO-27001-2022	A.8.20
ISO/IEC-27001	A.6.2.2
ISO/IEC-27001	A.10.1.1
ISO/IEC-27001	A.13.2.3
ITSG-33	AC-17(2)

ITSG-33	MA-4(6)
ITSG-33	SC-8(1)
NESA	T2.3.4
NESA	T5.4.2
NESA	T5.4.4
NESA	T7.4.1
NIAV2	AM37
NIAV2	NS5d
NIAV2	NS6b
PCI-DSSV3.2.1	2.3
PCI-DSSV3.2.1	4.1
PCI-DSSV4.0	2.2.7
PCI-DSSV4.0	4.2.1
QCSC-V1	3.2
QCSC-V1	5.2.1
QCSC-V1	5.2.2
QCSC-V1	6.2
RULE-ID	SV-270667r1067107_rule
STIG-ID	UBTU-24-100820
SWIFT-CSCV1	2.1
SWIFT-CSCV1	2.6
TBA-FIISB	29.1
TBA-FIISB	45.2.3
VULN-ID	V-270667

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/ssh/sshd_config - regex '(?i)^\s*Ciphers\s+' found - expect '(?i)^\s*Ciphers\s+"?aes256-gcm@openssh.com,\s*aes128-gcm@openssh.com,\s*aes256-ctr,\s*aes128-ctr"?[\s]*$' found in the following lines:
    132: Ciphers aes256-gcm@openssh.com,aes128-gcm@openssh.com,aes256-ctr,aes128-ctr
/etc/ssh/sshd_config.d/50-cloud-init.conf - regex not found
/etc/ssh/sshd_config.d/50-cloudimg-settings.conf - regex not found
/etc/ssh/sshd_config.d/60-cloudimg-settings.conf - regex not found
```

UBTU-24-100830 - Ubuntu 24.04 LTS must configure the SSH daemon to use Message Authentication Codes (MACs) employing FIPS 140-3 approved cryptographic hashes to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.

Info

Without cryptographic integrity protections, information can be altered by unauthorized users without detection. Remote access (e.g., RDP) is access to DOD nonpublic information systems by an authorized user (or an information system) communicating through an external, nonorganization-controlled network.

Remote access methods include, for example, dial-up, broadband, and wireless.

Nonlocal maintenance and diagnostic activities are those activities conducted by individuals communicating through a network, either an external network (e.g., the internet) or an internal network. Local maintenance and diagnostic activities are those activities carried out by individuals physically present at the information system or information system component and not communicating across a network connection. Encrypting information for transmission protects information from unauthorized disclosure and modification.

Cryptographic mechanisms implemented to protect information integrity include, for example, cryptographic hash functions that have common application in digital signatures, checksums, and message authentication codes. Satisfies: SRG-OS-000250-GPOS-00093, SRG-OS-000393-GPOS-00173, SRG-OS-000424-GPOS-00188

Solution

Configure Ubuntu 24.04 LTS to allow the SSH daemon to only use MACs that employ FIPS 140-3 approved ciphers. Add the following line (or modify the line to have the required value) to the "/etc/ssh/sshd_config" file (this file may be named differently or be in a different location if using a version of SSH that is provided by a third-party vendor): MACs hmac-sha2-512-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512,hmac-sha2-256 Restart the "sshd" service for changes to take effect: \$ sudo systemctl restart sshd

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.13
800-171	3.7.5
800-171	3.13.8
800-171R3	03.07.05
800-171R3	03.13.08
800-53	AC-17(2)
800-53	MA-4(6)
800-53	SC-8(1)
800-53R5	AC-17(2)
800-53R5	MA-4(6)
800-53R5	SC-8(1)
CAT	II
CCI	CCI-001453
CCI	CCI-002421
CCI	CCI-002890
CN-L3	7.1.2.7(g)
CN-L3	7.1.3.1(d)

CN-L3	8.1.2.2(a)
CN-L3	8.1.2.2(b)
CN-L3	8.1.4.1(c)
CN-L3	8.1.4.7(a)
CN-L3	8.1.4.8(a)
CN-L3	8.2.4.5(c)
CN-L3	8.2.4.5(d)
CN-L3	8.5.2.2
CSF	PR.AC-3
CSF	PR.DS-2
CSF	PR.DS-5
CSF	PR.MA-2
CSF	PR.PT-4
CSF2.0	PR.AA-05
CSF2.0	PR.DS-02
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.a
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
HIPAA	164.312(e)(1)
HIPAA	164.312(e)(2)(i)
ISO-27001-2022	A.5.10
ISO-27001-2022	A.5.14
ISO-27001-2022	A.5.33
ISO-27001-2022	A.6.7
ISO-27001-2022	A.8.20
ISO/IEC-27001	A.6.2.2
ISO/IEC-27001	A.10.1.1
ISO/IEC-27001	A.13.2.3
ITSG-33	AC-17(2)

ITSG-33	MA-4(6)
ITSG-33	SC-8(1)
NESA	T2.3.4
NESA	T5.4.2
NESA	T5.4.4
NESA	T7.4.1
NIAV2	AM37
NIAV2	NS5d
NIAV2	NS6b
PCI-DSSV3.2.1	2.3
PCI-DSSV3.2.1	4.1
PCI-DSSV4.0	2.2.7
PCI-DSSV4.0	4.2.1
QCSC-V1	3.2
QCSC-V1	5.2.1
QCSC-V1	5.2.2
QCSC-V1	6.2
RULE-ID	SV-270668r1067110_rule
STIG-ID	UBTU-24-100830
SWIFT-CSCV1	2.1
SWIFT-CSCV1	2.6
TBA-FIISB	29.1
TBA-FIISB	45.2.3
VULN-ID	V-270668

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/ssh/sshd_config - regex '(?i)^\s*MACs\s+' found - expect '(?i)^\s*MACs\s+("?
hmac-sha2-512-etm@openssh.com,\s*hmac-sha2-256-etm@openssh.com,\s*hmac-sha2-512,\s*hmac-
sha2-256"?[\s]*$' found in the following lines:
133: MACs hmac-sha2-512-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-
sha2-512,hmac-sha2-256
/etc/ssh/sshd_config.d/50-cloud-init.conf - regex not found
/etc/ssh/sshd_config.d/50-cloudimg-settings.conf - regex not found
/etc/ssh/sshd_config.d/60-cloudimg-settings.conf - regex not found
```


UBTU-24-100840 - Ubuntu 24.04 LTS SSH server must be configured to use only FIPS 140-3 validated key exchange algorithms.

Info

Without cryptographic integrity protections provided by FIPS-validated cryptographic algorithms, information can be viewed and altered by unauthorized users without detection. The system will attempt to use the first algorithm presented by the client that matches the server list.

Solution

Configure the SSH daemon to use only FIPS-validated key exchange algorithms by adding or modifying the following line in `/etc/ssh/sshd_config`: `KexAlgorithms ecdh-sha2-nistp521,ecdh-sha2-nistp384,ecdh-sha2-nistp256,diffie-hellman-group-exchange-sha256,diffie-hellman-group16-sha512,diffie-hellman-group14-sha256`
Restart the "sshd" service for changes to take effect: `$ sudo systemctl restart sshd`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.13
800-171R3	03.13.08
800-53	AC-17(2)
800-53R5	AC-17(2)
CAT	II
CCI	CCI-000068
CN-L3	7.1.2.7(g)
CN-L3	7.1.3.1(d)
CN-L3	8.1.4.1(c)
CSF	PR.AC-3
CSF	PR.PT-4
CSF2.0	PR.AA-05
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
ISO-27001-2022	A.5.14
ISO-27001-2022	A.6.7
ISO/IEC-27001	A.6.2.2
ITSG-33	AC-17(2)
NESA	T5.4.2
NIAV2	AM37

PCI-DSSV3.2.1	2.3
PCI-DSSV4.0	2.2.7
QCSC-V1	3.2
QCSC-V1	5.2.1
QCSC-V1	5.2.2
RULE-ID	SV-270669r1134804_rule
STIG-ID	UBTU-24-100840
SWIFT-CSCV1	2.6
VULN-ID	V-270669

Assets

strkbsubuntu9

Compliant file(s):

```
/etc/ssh/sshd_config - regex '(?i)^\s*KexAlgorithms\s+' found - expect '(?i)^\s*KexAlgorithms\s+((ecdh-sha2-nistp521,?|ecdh-sha2-nistp384,?|ecdh-sha2-nistp256,?|diffie-hellman-group-exchange-sha256,?|diffie-hellman-group16-sha512,?|diffie-hellman-group14-sha256,?)+)\s*$' found in the following lines:
134: KexAlgorithms ecdh-sha2-nistp521,ecdh-sha2-nistp384,ecdh-sha2-nistp256,diffie-hellman-group-exchange-sha256,diffie-hellman-group16-sha512,diffie-hellman-group14-sha256
/etc/ssh/sshd_config.d/50-cloud-init.conf - regex not found
/etc/ssh/sshd_config.d/50-cloudimg-settings.conf - regex not found
/etc/ssh/sshd_config.d/60-cloudimg-settings.conf - regex not found
```

UBTU-24-100850 - Ubuntu 24.04 LTS must configure the SSH client to use FIPS 140-3 approved ciphers to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.

Info

Without cryptographic integrity protections, information can be altered by unauthorized users without detection. Remote access (e.g., RDP) is access to DOD nonpublic information systems by an authorized user (or an information system) communicating through an external, nonorganization-controlled network.

Remote access methods include, for example, dial-up, broadband, and wireless. Nonlocal maintenance and diagnostic activities are those activities conducted by individuals communicating through a network, either an external network (e.g., the internet) or an internal network. Local maintenance and diagnostic activities are those activities carried out by individuals physically present at the information system or information system component and not communicating across a network connection. Encrypting information for transmission protects information from unauthorized disclosure and modification.

Cryptographic mechanisms implemented to protect information integrity include, for example, cryptographic hash functions that have common application in digital signatures, checksums, and message authentication codes. By specifying a cipher list with the order of ciphers being in a "strongest to weakest" orientation, the system will automatically attempt to use the strongest cipher for securing SSH connections.

Solution

Configure the Ubuntu 24.04 LTS SSH client to use only ciphers employing FIPS 140-3 approved algorithms by updating the "/etc/ssh/ssh_config" file with the following line: Ciphers aes256-gcm@openssh.com,aes128-gcm@openssh.com,aes256-ctr,aes128-ctr Restart the "ssh" service for changes to take effect: \$ sudo systemctl restart ssh

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.13
800-171R3	03.13.08
800-53	AC-17(2)
800-53R5	AC-17(2)
CAT	II
CCI	CCI-001453
CN-L3	7.1.2.7(g)
CN-L3	7.1.3.1(d)
CN-L3	8.1.4.1(c)
CSF	PR.AC-3
CSF	PR.PT-4
CSF2.0	PR.AA-05
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
ISO-27001-2022	A.5.14

ISO-27001-2022	A.6.7
ISO/IEC-27001	A.6.2.2
ITSG-33	AC-17(2)
NESA	T5.4.2
NIAV2	AM37
PCI-DSSV3.2.1	2.3
PCI-DSSV4.0	2.2.7
QCSC-V1	3.2
QCSC-V1	5.2.1
QCSC-V1	5.2.2
RULE-ID	SV-270670r1067115_rule
STIG-ID	UBTU-24-100850
SWIFT-CSCV1	2.6
VULN-ID	V-270670

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/ssh/ssh_config - regex '(?i)^\s*Ciphers\s+' found - expect '(?i)^\s*Ciphers\s+"?aes256-gcm@openssh.com,\s*aes128-gcm@openssh.com,\s*aes256-ctr,\s*aes128-ctr"?[\s]*$' found in the following lines:
54:      Ciphers aes256-gcm@openssh.com,aes128-gcm@openssh.com,aes256-ctr,aes128-ctr
```

UBTU-24-100860 - Ubuntu 24.04 LTS SSH client must be configured to use only Message Authentication Codes (MACs) employing FIPS 140-3 validated cryptographic hash algorithms.

Info

Without cryptographic integrity protections, information can be altered by unauthorized users without detection. Remote access (e.g., RDP) is access to DOD nonpublic information systems by an authorized user (or an information system) communicating through an external, nonorganization-controlled network.

Remote access methods include, for example, dial-up, broadband, and wireless.

Nonlocal maintenance and diagnostic activities are those activities conducted by individuals communicating through a network, either an external network (e.g., the internet) or an internal network. Local maintenance and diagnostic activities are those activities carried out by individuals physically present at the information system or information system component and not communicating across a network connection. Encrypting information for transmission protects information from unauthorized disclosure and modification.

Cryptographic mechanisms implemented to protect information integrity include, for example, cryptographic hash functions that have common application in digital signatures, checksums, and message authentication codes.

Solution

Configure the Ubuntu 24.04 LTS SSH client to use only MACs employing FIPS 140-3 approved algorithms by updating the "/etc/ssh/ssh_config" file with the following line: MACs hmac-sha2-512-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512,hmac-sha2-256 Restart the "ssh" service for changes to take effect: \$ sudo systemctl restart ssh

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.13
800-171R3	03.13.08
800-53	AC-17(2)
800-53R5	AC-17(2)
CAT	II
CCI	CCI-001453
CN-L3	7.1.2.7(g)
CN-L3	7.1.3.1(d)
CN-L3	8.1.4.1(c)
CSF	PR.AC-3
CSF	PR.PT-4
CSF2.0	PR.AA-05
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
ISO-27001-2022	A.5.14
ISO-27001-2022	A.6.7

ISO/IEC-27001	A.6.2.2
ITSG-33	AC-17(2)
NESA	T5.4.2
NIAV2	AM37
PCI-DSSV3.2.1	2.3
PCI-DSSV4.0	2.2.7
QCSC-V1	3.2
QCSC-V1	5.2.1
QCSC-V1	5.2.2
RULE-ID	SV-270671r1155244_rule
STIG-ID	UBTU-24-100860
SWIFT-CSCV1	2.6
VULN-ID	V-270671

Assets

strkbsubuntu9

```
Compliant file(s):
    /etc/ssh/ssh_config - regex '(?i)^[\\s]*MACs[\\s]+' found - expect '(?i)^[\\s]*MACs[\\s]+'?
hmac-sha2-512-etm@openssh.com, [\\s]*hmac-sha2-256-etm@openssh.com, [\\s]*hmac-sha2-512, [\\s]*hmac-
sha2-256"?[\\s]*$' found in the following lines:
    55:      MACs hmac-sha2-512-etm@openssh.com, hmac-sha2-256-etm@openssh.com, hmac-
sha2-512, hmac-sha2-256
```

UBTU-24-100900 - Ubuntu 24.04 LTS must accept Personal Identity Verification (PIV) credentials.

Info

The use of PIV credentials facilitates standardization and reduces the risk of unauthorized access. DOD has mandated the use of the common access card (CAC) to support identity management and personal authentication for systems covered under Homeland Security Presidential Directive (HSPD) 12, as well as making the CAC a primary component of layered protection for national security systems.

Solution

Configure Ubuntu 24.04 LTS to accept PIV credentials. Install the "opensc-pkcs11" package using the following command: `$ sudo apt install -y opensc-pkcs11`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.1
800-171R3	03.05.01a.
800-53	IA-2(12)
800-53R5	IA-2(12)
CAT	II
CCI	CCI-001953
CN-L3	7.1.3.1(a)
CN-L3	7.1.3.1(e)
CN-L3	8.1.4.1(a)
CN-L3	8.1.4.2(a)
CN-L3	8.5.4.1(a)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ITSG-33	IA-2
ITSG-33	IA-2a.
NESA	T2.3.8

NESA	T5.3.1
NESA	T5.4.2
NESA	T5.5.1
NESA	T5.5.2
NESA	T5.5.3
NIAV2	AM2
NIAV2	AM8
NIAV2	AM14b
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270672r1067161_rule
STIG-ID	UBTU-24-100900
TBA-FIISB	35.1
TBA-FIISB	36.1
VULN-ID	V-270672

Assets

strkbsubuntu9

The command `'/bin/dpkg -s opensc-pkcs11 2>&1 | /bin/grep -E '(Status:|not installed)''` returned :

Status: install ok installed

UBTU-24-100910 - Ubuntu 24.04 LTS must accept Personal Identity Verification (PIV) credentials managed through the Privileged Access Management (PAM) framework.

Info

The use of PIV credentials facilitates standardization and reduces the risk of unauthorized access. DOD has mandated the use of the common access card (CAC) to support identity management and personal authentication for systems covered under Homeland Security Presidential Directive (HSPD) 12, as well as making the CAC a primary component of layered protection for national security systems.

Solution

Configure Ubuntu 24.04 LTS to accept PIV credentials that are managed through the PAM framework. Install the "libpam-pkcs11" package using the following command: `$ sudo apt install -y libpam-pkcs11`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.1
800-171R3	03.05.01a.
800-53	IA-2(12)
800-53R5	IA-2(12)
CAT	II
CCI	CCI-001953
CN-L3	7.1.3.1(a)
CN-L3	7.1.3.1(e)
CN-L3	8.1.4.1(a)
CN-L3	8.1.4.2(a)
CN-L3	8.5.4.1(a)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ITSG-33	IA-2
ITSG-33	IA-2a.
NESA	T2.3.8

NESA	T5.3.1
NESA	T5.4.2
NESA	T5.5.1
NESA	T5.5.2
NESA	T5.5.3
NIAV2	AM2
NIAV2	AM8
NIAV2	AM14b
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270673r1067164_rule
STIG-ID	UBTU-24-100910
TBA-FIISB	35.1
TBA-FIISB	36.1
VULN-ID	V-270673

Assets

strkbsubuntu9

The command `'/bin/dpkg -s libpam-pkcs11 2>&1 | /bin/grep -E '(Status:|not installed)''` returned :

Status: install ok installed

UBTU-24-101000 - Ubuntu 24.04 LTS must allow users to directly initiate a session lock for all connection types.

Info

A session lock is a temporary action taken when a user stops work and moves away from the immediate physical vicinity of the information system but does not want to log out because of the temporary nature of the absence. The session lock is implemented at the point where session activity can be determined. Rather than be forced to wait for a period of time to expire before the user session can be locked, Ubuntu 24.04 LTSs need to provide users with the ability to manually invoke a session lock so users may secure their session if they need to temporarily vacate the immediate physical vicinity. Satisfies: SRG-OS-000031-GPOS-00012, SRG-OS-000030-GPOS-00011

Solution

Install the "vlock" package (if it is not already installed) by running the following command: `$ sudo apt install -y vlock`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.10
800-171R3	03.01.10
800-171R3	03.01.10a.
800-53	AC-11a.
800-53	AC-11(1)
800-53R5	AC-11a.
800-53R5	AC-11(1)
CAT	II
CCI	CCI-000057
CCI	CCI-000060
CN-L3	8.1.4.1(b)
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(iii)
ISO-27001-2022	A.7.7
ISO-27001-2022	A.8.1
ISO/IEC-27001	A.11.2.8
ITSG-33	AC-11a.
ITSG-33	AC-11(1)
NESA	T2.3.8

NESA	T2.3.9
NIAV2	AM23a
NIAV2	AM23b
NIAV2	AM23c
NIAV2	AM23d
PCI-DSSV3.2.1	8.1.8
PCI-DSSV4.0	8.2.8
RULE-ID	SV-270674r1067167_rule
STIG-ID	UBTU-24-101000
VULN-ID	V-270674

Assets

strkbsubuntu9

The command '/bin/dpkg -s vlock | /bin/grep Status: 2>&1' returned :

Status: install ok installed

UBTU-24-102010 - Ubuntu 24.04 LTS must initiate session audits at system startup.

Info

If auditing is enabled late in the startup process, the actions of some startup processes may not be audited. Some audit systems also maintain state information only available if auditing is enabled before a given process is created.

Solution

Configure Ubuntu 24.04 LTS to produce audit records at system startup. Edit the "/etc/default/grub" file and add "audit=1" to the "GRUB_CMDLINE_LINUX" option and to the "GRUB_CMDLINE_LINUX_DEFAULT" option. GRUB_CMDLINE_LINUX_DEFAULT="audit=1" GRUB_CMDLINE_LINUX="audit=1" To update the grub config file, run: \$ sudo update-grub

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	AU-14(1)
800-53R5	AU-14(1)
CAT	II
CCI	CCI-001464
CSF	PR.PT-1
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ITSG-33	AU-14
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270676r1155245_rule
STIG-ID	UBTU-24-102010
SWIFT-CSCV1	6.4
VULN-ID	V-270676

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
PASSED - GRUB_CMDLINE_LINUX
Compliant file(s):
    /etc/default/grub - regex '^[\\s]*GRUB_CMDLINE_LINUX[\\s]*=' found - expect
    '^[\\s]*GRUB_CMDLINE_LINUX[\\s]*=[^#]*audit=1\\b' found in the following lines:
        11: GRUB_CMDLINE_LINUX="audit=1 "
```

```
-----
PASSED - GRUB_CMDLINE_LINUX_DEFAULT
Compliant file(s):
```

```
/etc/default/grub - regex '^[\\s]*GRUB_CMDLINE_LINUX_DEFAULT[\\s]*=' found - expect
'^[\\s]*GRUB_CMDLINE_LINUX_DEFAULT[\\s]*=[^#]*audit=1\\b' found in the following lines:
10: GRUB_CMDLINE_LINUX_DEFAULT="quiet splash audit=1"
```

```
-----
PASSED - linux bootloader
Compliant file(s):
/boot/grub/grub.cfg - regex '^[\\s]*linux[\\s]+' found - expect
'^[\\s]*linux[\\s]+[\\^#]*audit=1\\b' found in the following lines:
161: linux /vmlinuz-6.17.0-1022-azure root=PARTUUID=7766a12d-b53b-453f-a51f-
d354e4fa13b1 ro audit=1 console=tty1 console=ttyS0 earlyprintk=ttyS0 nvme_core.io_timeout=240
165: linux /vmlinuz-6.17.0-1022-azure root=PARTUUID=7766a12d-b53b-453f-a51f-
d354e4fa13b1 ro audit=1 console=tty1 console=ttyS0 earlyprintk=ttyS0 nvme_core.io_timeout=240
panic=-1
182: linux /vmlinuz-6.17.0-1022-azure root=PARTUUID=7766a12d-b53b-453f-a51f-
d354e4fa13b1 ro audit=1 console=tty1 console=ttyS0 earlyprintk=ttyS0 nvme_core.io_timeout=240
187: linux /vmlinuz-6.17.0-1022-azure root=PARTUUID=7766a12d-b53b-453f-a51f-
d354e4fa13b1 ro audit=1 console=tty1 console=ttyS0 earlyprintk=ttyS0 nvme_core.io_timeout=240
panic=-1
202: linux /vmlinuz-6.17.0-1022-azure root=PARTUUID=7766a12d-b53b-453f-a51f-
d354e4fa13b1 ro recovery nomodeset dis_ucode_ldr audit=1 console=tty1 console=ttyS0
earlyprintk=ttyS0 nvme_core.io_timeout=240
207: linux /vmlinuz-6.17.0-1022-azure root=PARTUUID=7766a12d-b53b-453f-a51f-
d354e4fa13b1 ro recovery nomodeset dis_ucode_ldr audit=1 console=tty1 console=ttyS0
earlyprintk=ttyS0 nvme_core.io_timeout=240 panic=-1
223: linux /vmlinuz-6.8.0-1007-azure root=PARTUUID=7766a12d-b53b-453f-a51f-
d354e4fa13b1 ro audit=1 console=tty1 console=ttyS0 earlyprintk=ttyS0 nvme_core.io_timeout=240
228: linux /vmlinuz-6.8.0-1007-azure root=PARTUUID=7766a12d-b53b-453f-a51f-
d354e4fa13b1 ro audit=1 console=tty1 console=ttyS0 earlyprintk=ttyS0 nvme_core.io_timeout=240
panic=-1
243: linux /vmlinuz-6.8.0-1007-azure root=PARTUUID=7766a12d-b53b-453f-a51f-
d354e4fa13b1 ro recovery nomodeset dis_ucode_ldr audit=1 console=tty1 console=ttyS0
earlyprintk=ttyS0 nvme_core.io_timeout=240
248: linux /vmlinuz-6.8.0-1007-azure root=PARTUUID=7766a12d-b53b-453f-a51f-
d354e4fa13b1 ro recovery nomodeset dis_ucode_ldr audit=1 console=tty1 console=ttyS0
earlyprintk=ttyS0 nvme_core.io_timeout=240 panic=-1
```

UBTU-24-200000 - Ubuntu 24.04 LTS must limit the number of concurrent sessions to 10 for all accounts and/or account types.

Info

Ubuntu 24.04 LTS management includes the ability to control the number of users and user sessions that utilize an operating system.

Limiting the number of allowed users and sessions per user is helpful in reducing the risks related to denial-of-service (DoS) attacks. This requirement addresses concurrent sessions for information system accounts and does not address concurrent sessions by single users via multiple system accounts.

The maximum number of concurrent sessions must be defined based upon mission needs and the operational environment for each system.

Solution

Configure Ubuntu 24.04 LTS to limit the number of concurrent sessions to 10 for all accounts and/or account types. Add the following line to the top of the /etc/security/limits.conf or in a ".conf" file defined in /etc/security/limits.d/: * hard maxlogins 10

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	AC-10
800-53R5	AC-10
CAT	III
CCI	CCI-000054
CSF2.0	PR.AA-05
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	AC-10
NESA	T5.5.1
QCSC-V1	5.2.1
QCSC-V1	5.2.2
RULE-ID	SV-270677r1101774_rule
STIG-ID	UBTU-24-200000
VULN-ID	V-270677

Assets

strkbsubuntu9

Compliant file(s):
/etc/security/limits.conf - regex '^[\s]*\s*hard[\s]+maxlogins[\s]+' found - expect
'^[\s]*\s*hard[\s]+maxlogins[\s]+([1-9]|10)[\s]*\$' found in the following lines:
68: * hard maxlogins 10

UBTU-24-200020 - Ubuntu 24.04 LTS must initiate a graphical session lock after 10 minutes of inactivity.

Info

A session lock is a temporary action taken when a user stops work and moves away from the immediate physical vicinity of the information system but does not want to log out because of the temporary nature of the absence. The session lock is implemented at the point where session activity can be determined. Regardless of where the session lock is determined and implemented, once invoked, a session lock of Ubuntu 24.04 LTS must remain in place until the user reauthenticates.

No other activity aside from reauthentication must unlock the system.

Solution

Configure Ubuntu 24.04 LTS to lock the current graphical user interface session after 10 minutes of inactivity. Create or edit a file named `/etc/dconf/db/local.d/00-screensaver` with the following contents: `[org/gnome/desktop/session] idle-delay=uint32 900 [org/gnome/desktop/screensaver] lock-enabled=true lock-delay=uint32 600` Update the dconf settings: `$ sudo dconf update`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.10
800-171R3	03.01.10a.
800-53	AC-11a.
800-53R5	AC-11a.
CAT	II
CCI	CCI-000057
CN-L3	8.1.4.1(b)
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(iii)
ISO-27001-2022	A.7.7
ISO-27001-2022	A.8.1
ISO/IEC-27001	A.11.2.8
ITSG-33	AC-11a.
NESA	T2.3.8
NESA	T2.3.9
NIAV2	AM23a
NIAV2	AM23b
PCI-DSSV3.2.1	8.1.8
PCI-DSSV4.0	8.2.8

RULE-ID SV-270678r1101791_rule

STIG-ID UBTU-24-200020

VULN-ID V-270678

Assets

strkbsubuntu9

UBTU-24-200040 - Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user interface automount function.

Info

A nonprivileged account is any operating system account with authorizations of a nonprivileged user. Satisfies: SRG-OS-000114-GPOS-00059, SRG-OS-000378-GPOS-00163, SRG-OS-000480-GPOS-00227

Solution

Configure Ubuntu 24.04 LTS so the GNOME desktop does not allow a user to change the setting that disables automated mounting of removable media. Add the following line to "/etc/dconf/db/local.d/locks/00-security-settings-lock" to prevent user modification: /org/gnome/desktop/media-handling/automount-open Update the dconf system databases: \$ sudo dconf update

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.10
800-171R3	03.01.10b.
800-53	AC-11b.
800-53R5	AC-11b.
CAT	II
CCI	CCI-000056
CN-L3	8.1.4.1(b)
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(iii)
ISO-27001-2022	A.7.7
ISO-27001-2022	A.8.1
ISO/IEC-27001	A.11.2.8
ITSG-33	AC-11b.
NIAV2	AM23e
PCI-DSSV3.2.1	8.1.8
PCI-DSSV4.0	8.2.8
RULE-ID	SV-270679r1107295_rule
STIG-ID	UBTU-24-200040
VULN-ID	V-270679

Assets

strkbsubuntu9

UBTU-24-200041 - Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user interface autorun function.

Info

Techniques used to address this include protocols using nonces (e.g., numbers generated for a specific one-time use) or challenges (e.g., TLS, WS_Security).

Additional techniques include time-synchronous or challenge-response one-time authenticators. Satisfies: SRG-OS-000114-GPOS-00059, SRG-OS-000378-GPOS-00163, SRG-OS-000480-GPOS-00227

Solution

Configure the Ubuntu 24.04 LTS GNOME desktop to not allow a user to change the setting that disables autorun on removable media. Add the following line to "/etc/dconf/db/local.d/locks/00-security-settings-lock" to prevent user modification: /org/gnome/desktop/media-handling/autorun-never Update the dconf system databases: \$ sudo dconf update

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171R3	03.05.02
800-53	IA-3
800-53R5	IA-3
CAT	II
CCI	CCI-000778
CCI	CCI-001958
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ITSG-33	IA-3
ITSG-33	IA-3a.
NESA	T5.4.3
QCSC-V1	13.2
RULE-ID	SV-274872r1107297_rule
STIG-ID	UBTU-24-200041
TBA-FIISB	27.1
VULN-ID	V-274872

UBTU-24-200042 - Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user smart card removal action.

Info

A session lock is a temporary action taken when a user stops work and moves away from the immediate physical vicinity of the information system but does not want to log out because of the temporary nature of the absence. The session lock is implemented at the point where session activity can be determined. Rather than be forced to wait for a period of time to expire before the user session can be locked, Ubuntu 24.04 LTS must provide users with the ability to manually invoke a session lock so users can secure their session if it is necessary to temporarily vacate the immediate physical vicinity. Satisfies: SRG-OS-000028-GPOS-00009, SRG-OS-000030-GPOS-00011

Solution

Configure Ubuntu 24.04 LTS to prevent a user from overriding the disabling of the graphical user smart card removal action. Add the following line to `/etc/dconf/db/local.d/locks/00-security-settings-lock`: `/org/gnome/settings-daemon/peripherals/smartcard/removal-action` Update the dconf system databases: `$ sudo dconf update`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.10
800-171R3	03.01.10a.
800-171R3	03.01.10b.
800-53	AC-11a.
800-53	AC-11b.
800-53R5	AC-11a.
800-53R5	AC-11b.
CAT	II
CCI	CCI-000056
CCI	CCI-000057
CCI	CCI-000058
CN-L3	8.1.4.1(b)
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(iii)
ISO-27001-2022	A.7.7
ISO-27001-2022	A.8.1
ISO/IEC-27001	A.11.2.8
ITSG-33	AC-11a.

ITSG-33	AC-11b.
NESA	T2.3.8
NESA	T2.3.9
NIAV2	AM23a
NIAV2	AM23b
NIAV2	AM23e
PCI-DSSV3.2.1	8.1.8
PCI-DSSV4.0	8.2.8
RULE-ID	SV-274873r1107300_rule
STIG-ID	UBTU-24-200042
VULN-ID	V-274873

Assets

strkbsubuntu9

UBTU-24-200043 - Ubuntu 24.04 LTS must conceal, via the session lock, information previously visible on the display with a publicly viewable image.

Info

Setting the screensaver mode to blank-only conceals the contents of the display from passersby.

Solution

Configure Ubuntu 24.04 LTS to prevent a user from overriding the picture-uri setting for graphical user interfaces. In the file "/etc/dconf/db/local.d/00-security-settings", add or update the following lines: [org/gnome/desktop/screensaver] picture-uri=" Prevent user modification by adding the following line to "/etc/dconf/db/local.d/locks/00-security-settings-lock": /org/gnome/desktop/screensaver/picture-uri Update the dconf system databases: \$ sudo dconf update

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.10
800-171R3	03.01.10
800-53	AC-11(1)
800-53R5	AC-11(1)
CAT	II
CCI	CCI-000060
CN-L3	8.1.4.1(b)
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(iii)
ISO-27001-2022	A.7.7
ISO-27001-2022	A.8.1
ISO/IEC-27001	A.11.2.8
ITSG-33	AC-11(1)
NIAV2	AM23c
NIAV2	AM23d
PCI-DSSV3.2.1	8.1.8
PCI-DSSV4.0	8.2.8
RULE-ID	SV-274871r1107302_rule
STIG-ID	UBTU-24-200043
VULN-ID	V-274871

Assets

strkbsubuntu9

UBTU-24-200060 - Ubuntu 24.04 LTS must automatically terminate a user session after inactivity timeouts have expired.

Info

Automatic session termination addresses the termination of user-initiated logical sessions in contrast to the termination of network connections associated with communications sessions (i.e., network disconnect). A logical session (for local, network, and remote access) is initiated whenever a user (or process acting on behalf of a user) accesses an organizational information system.

Such user sessions can be terminated (and thus terminate user access) without terminating network sessions. Session termination terminates all processes associated with a user's logical session except those processes specifically created by the user (i.e., session owner) to continue after the session is terminated. Conditions or trigger events requiring automatic session termination can include, for example, organization-defined periods of user inactivity, targeted responses to certain types of incidents, and time-of-day restrictions on information system use. This capability is typically reserved for specific operating system functionality where the system owner, data owner, or organization requires additional assurance.

Solution

Configure Ubuntu 24.04 LTS to automatically terminate a user session after inactivity timeouts have expired or at shutdown. Create the file `/etc/profile.d/99-terminal_tmout.sh` if it does not exist. Modify or append the following line in the `/etc/profile.d/99-terminal_tmout.sh` file: `TMOUT=600` This will set a timeout value of 10 minutes for all future sessions. To set the timeout for the current sessions, execute the following command over the terminal session: `$ export TMOUT=600`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.11
800-171R3	03.01.11
800-53	AC-12
800-53R5	AC-12
CAT	II
CCI	CCI-002361
CN-L3	7.1.2.2(d)
CN-L3	7.1.3.7(b)
CN-L3	8.1.4.1(b)
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(iii)
ITSG-33	AC-12
NIAV2	NS49
RULE-ID	SV-270680r1066529_rule
STIG-ID	UBTU-24-200060

Assets**strkbsubuntu9**

```
Compliant file(s):
/etc/bash.bashrc - regex not found
/etc/profile.d/01-locale-fix.sh - regex not found
/etc/profile.d/99-terminal_tmout.sh - regex '^[\s]*TMOUT[\s]*=' found - expect
'^[\s]*TMOUT[\s]*=[\s]*(?:[1-9]|[1-9][0-9]|[1-5][0-9]{2}|600)$' found in the following lines:
7: TMOUT=600
/etc/profile.d/Z97-byobu.sh - regex not found
/etc/profile.d/Z99-cloud-locale-test.sh - regex not found
/etc/profile.d/Z99-cloudinit-warnings.sh - regex not found
/etc/profile.d/apps-bin-path.sh - regex not found
/etc/profile.d/bash_completion.sh - regex not found
/etc/profile.d/gawk.csh - regex not found
/etc/profile.d/gawk.sh - regex not found
/etc/profile.d/linux-base-sgx.sh - regex not found
```

UBTU-24-200090 - Ubuntu 24.04 LTS must monitor remote access methods.

Info

Remote access services, such as those providing remote access to network devices and information systems, which lack automated monitoring capabilities, increase risk, and make remote user access management difficult at best. Remote access is access to DOD nonpublic information systems by an authorized user (or an information system) communicating through an external, nonorganization-controlled network.

Remote access methods include, for example, dial-up, broadband, and wireless. Automated monitoring of remote access sessions allows organizations to detect cyberattacks and also ensure ongoing compliance with remote access policies by auditing connection activities of remote access capabilities, such as Remote Desktop Protocol (RDP), on a variety of information system components (e.g., servers, workstations, notebook computers, smartphones, and tablets).

Solution

Configure Ubuntu 24.04 LTS to monitor all remote access methods by adding the following lines to the "/etc/rsyslog.d/50-default.conf" file: auth.*,authpriv.* /var/log/secure daemon.* /var/log/messages For the changes to take effect, restart the "rsyslog" service with the following command: \$ sudo systemctl restart rsyslog.service

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.12
800-171R3	03.01.12
800-53	AC-17(1)
800-53R5	AC-17(1)
CAT	II
CCI	CCI-000067
CN-L3	8.1.4.4(c)
CN-L3	8.1.10.6(i)
CSF	PR.AC-3
CSF	PR.PT-4
CSF2.0	PR.AA-05
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
ISO-27001-2022	A.8.16
ISO/IEC-27001	A.6.2.2
ITSG-33	AC-17(1)
NESA	T5.4.4
QCSC-V1	3.2
QCSC-V1	5.2.1

QCSC-V1	5.2.2
RULE-ID	SV-270681r1134806_rule
STIG-ID	UBTU-24-200090
SWIFT-CSCV1	2.6
VULN-ID	V-270681

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - authpriv

Compliant file(s):

/etc/rsyslog.conf - regex not found

/etc/rsyslog.d/20-ufw.conf - regex not found

/etc/rsyslog.d/21-cloudinit.conf - regex not found

/etc/rsyslog.d/50-default.conf - regex '^([#])*auth,authpriv\.*' found - expect

'^([#])*auth,authpriv\.*' found in the following lines:

5: auth,authpriv.* /var/log/secure

/etc/rsyslog.d/postfix.conf - regex not found

PASSED - daemon

Compliant file(s):

/etc/rsyslog.conf - regex not found

/etc/rsyslog.d/20-ufw.conf - regex not found

/etc/rsyslog.d/21-cloudinit.conf - regex not found

/etc/rsyslog.d/50-default.conf - regex '^([#])*daemon\.*' found - expect '^([#])*daemon\.*'

found in the following lines:

6: daemon.* /var/log/messages

/etc/rsyslog.d/postfix.conf - regex not found

UBTU-24-200260 - Ubuntu 24.04 LTS must disable account identifiers (individuals, groups, roles, and devices) after 35 days of inactivity.

Info

Inactive identifiers pose a risk to systems and applications because attackers may exploit an inactive identifier and potentially obtain undetected access to the system.

Owners of inactive accounts will not notice if unauthorized access to their user account has been obtained.

Operating systems need to track periods of inactivity and disable application identifiers after 35 days of inactivity. Satisfies: SRG-OS-000118-GPOS-00060, SRG-OS-000590-GPOS-00110

Solution

Configure Ubuntu 24.04 LTS to disable account identifiers after 35 days of inactivity after the password expiration. Run the following command to change the configuration for adduser: `$ sudo useradd -D -f 35` Note: DOD recommendation is 35 days, but a lower value is acceptable.

The value "0" will disable the account immediately after the password expires.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.1
800-171R3	03.01.01f.
800-53	AC-2(3)
800-53R5	AC-2(3)(a)
800-53R5	AC-2(3)(b)
CAT	II
CCI	CCI-003627
CCI	CCI-003628
CN-L3	7.1.3.2(e)
CN-L3	8.1.4.2(c)
CSF	PR.AC-1
CSF	PR.AC-4
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	PR.AA-01
CSF2.0	PR.AA-05
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)

ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.18
ISO-27001-2022	A.8.2
ISO/IEC-27001	A.9.2.1
ISO/IEC-27001	A.9.2.6
ITSG-33	AC-2(3)
NIAV2	AM26
QCSC-V1	5.2.2
QCSC-V1	8.2.1
QCSC-V1	13.2
QCSC-V1	15.2
RULE-ID	SV-270683r1066538_rule
STIG-ID	UBTU-24-200260
TBA-FIISB	36.2.2
VULN-ID	V-270683

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/default/useradd - regex '^[\s]*INACTIVE[\s]*=' found - expect
'^[\s]*INACTIVE[\s]*=[\s]*([1-2]?[0-9]|3[0-5])[\s]*$' found in the following lines:
38: INACTIVE=35
```

UBTU-24-200270 - Ubuntu 24.04 LTS must audit any script or executable called by cron as root or by any privileged user.

Info

Any script or executable called by cron as root or by any privileged user must be owned by that user, must have the permissions 755 or more restrictive, and should have no extended rights that allow any nonprivileged user to modify the script or executable.

Solution

Configure Ubuntu 24.04 LTS to audit the execution of any system call made by cron as root or as any privileged user. Add or update the following file system rules to "/etc/audit/rules.d/audit.rules": -w /etc/cron.d/ -p wa -k cronjobs -w /var/spool/cron/ -p wa -k cronjobs To load the rules to the kernel immediately, use the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-274870r1155243_rule
STIG-ID	UBTU-24-200270
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-274870

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
PASSED - auditctl /etc/cron.d
The command '/sbin/auditctl -l | /bin/grep /etc/cron.d' returned :

-w /etc/cron.d -p wa -k cronjobs

-----
PASSED - auditctl /var/spool/cron
The command '/sbin/auditctl -l | /bin/grep /var/spool/cron' returned :

-w /var/spool/cron -p wa -k cronjobs
```

UBTU-24-200280 - Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/passwd.

Info

Once an attacker establishes access to a system, the attacker often attempts to create a persistent method of reestablishing access.

One way to accomplish this is for the attacker to create an account.

Auditing account creation actions provides logging that can be used for forensic purposes. To address access requirements, many operating systems may be integrated with enterprise-level authentication/access/auditing mechanisms that meet or exceed access control policy requirements. Satisfies: SRG-OS-000004-GPOS-00004, SRG-OS-000239-GPOS-00089, SRG-OS-000240-GPOS-00090, SRG-OS-000241-GPOS-00091, SRG-OS-000303-GPOS-00120, SRG-OS-000458-GPOS-00203, SRG-OS-000463-GPOS-00207, SRG-OS-000476-GPOS-00221

Solution

Configure Ubuntu 24.04 LTS to generate audit records for all account creations, modifications, disabling, and termination events that affect "/etc/passwd". Add or update the following rule to "/etc/audit/rules.d/stig.rules":
-w /etc/passwd -p wa -k usergroup_modification To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.1
800-171	3.3.1
800-171	3.3.2
800-171R3	03.01.01
800-171R3	03.03.03a.
800-53	AC-2(4)
800-53	AU-12c.
800-53R5	AC-2(4)
800-53R5	AU-12c.
CAT	II
CCI	CCI-000018
CCI	CCI-000172
CCI	CCI-001403
CCI	CCI-001404
CCI	CCI-001405
CCI	CCI-002130
CN-L3	7.1.3.2(d)
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)

CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.AC-1
CSF	PR.AC-4
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.AA-01
CSF2.0	PR.AA-05
CSF2.0	PR.DS-10
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.18
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.9.2.1
ISO/IEC-27001	A.12.4.1
ITSG-33	AC-2(4)
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5

NESA	T3.6.6
NESA	T5.2.2
NIAV2	AM9a
NIAV2	AM9b
NIAV2	AM9c
NIAV2	AM9d
NIAV2	AM9e
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	5.2.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
QCSC-V1	15.2
RULE-ID	SV-270684r1066541_rule
STIG-ID	UBTU-24-200280
SWIFT-CSCV1	6.4
TBA-FIISB	36.2.3
TBA-FIISB	45.1.1
VULN-ID	V-270684

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep passwd' returned :

```
-w /etc/passwd -p wa -k usergroup_modification
-w /etc/security/opasswd -p wa -k usergroup_modification
-a always,exit -S all -F path=/usr/bin/passwd -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-passwd
-a always,exit -S all -F path=/usr/bin/gpasswd -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-gpasswd
```

UBTU-24-200290 - Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/group.

Info

Once an attacker establishes access to a system, the attacker often attempts to create a persistent method of reestablishing access.

One way to accomplish this is for the attacker to create an account.

Auditing account creation actions provides logging that can be used for forensic purposes. To address access requirements, many operating systems may be integrated with enterprise-level authentication/access/auditing mechanisms that meet or exceed access control policy requirements. Satisfies: SRG-OS-000004-GPOS-00004, SRG-OS-000239-GPOS-00089, SRG-OS-000240-GPOS-00090, SRG-OS-000241-GPOS-00091, SRG-OS-000303-GPOS-00120, SRG-OS-000458-GPOS-00203, SRG-OS-000463-GPOS-00207, SRG-OS-000476-GPOS-00221

Solution

Configure Ubuntu 24.04 LTS to generate audit records for all account creations, modifications, disabling, and termination events that affect "/etc/group". Add or update the following rule to "/etc/audit/rules.d/stig.rules":
-w /etc/group -p wa -k usergroup_modification To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.1
800-171	3.3.1
800-171	3.3.2
800-171R3	03.01.01
800-171R3	03.03.03a.
800-53	AC-2(4)
800-53	AU-12c.
800-53R5	AC-2(4)
800-53R5	AU-12c.
CAT	II
CCI	CCI-000018
CCI	CCI-000172
CCI	CCI-001403
CCI	CCI-001404
CCI	CCI-001405
CCI	CCI-002130
CN-L3	7.1.3.2(d)
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)

CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.AC-1
CSF	PR.AC-4
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.AA-01
CSF2.0	PR.AA-05
CSF2.0	PR.DS-10
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.18
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.9.2.1
ISO/IEC-27001	A.12.4.1
ITSG-33	AC-2(4)
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5

NESA	T3.6.6
NESA	T5.2.2
NIAV2	AM9a
NIAV2	AM9b
NIAV2	AM9c
NIAV2	AM9d
NIAV2	AM9e
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	5.2.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
QCSC-V1	15.2
RULE-ID	SV-270685r1066544_rule
STIG-ID	UBTU-24-200290
SWIFT-CSCV1	6.4
TBA-FIISB	36.2.3
TBA-FIISB	45.1.1
VULN-ID	V-270685

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep group' returned :

```
-w /etc/passwd -p wa -k usergroup_modification
-w /etc/group -p wa -k usergroup_modification
-w /etc/shadow -p wa -k usergroup_modification
-w /etc/gshadow -p wa -k usergroup_modification
-w /etc/security/opasswd -p wa -k usergroup_modification
```

UBTU-24-200300 - Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/shadow.

Info

Once an attacker establishes access to a system, the attacker often attempts to create a persistent method of reestablishing access.

One way to accomplish this is for the attacker to create an account.

Auditing account creation actions provides logging that can be used for forensic purposes. To address access requirements, many operating systems may be integrated with enterprise-level authentication/access/auditing mechanisms that meet or exceed access control policy requirements. Satisfies: SRG-OS-000004-GPOS-00004, SRG-OS-000239-GPOS-00089, SRG-OS-000240-GPOS-00090, SRG-OS-000241-GPOS-00091, SRG-OS-000303-GPOS-00120, SRG-OS-000458-GPOS-00203, SRG-OS-000463-GPOS-00207, SRG-OS-000476-GPOS-00221

Solution

Configure Ubuntu 24.04 LTS to generate audit records for all account creations, modifications, disabling, and termination events that affect "/etc/shadow". Add or update the following rule to "/etc/audit/rules.d/stig.rules":
-w /etc/shadow -p wa -k usergroup_modification To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.1
800-171	3.3.1
800-171	3.3.2
800-171R3	03.01.01
800-171R3	03.03.03a.
800-53	AC-2(4)
800-53	AU-12c.
800-53R5	AC-2(4)
800-53R5	AU-12c.
CAT	II
CCI	CCI-000018
CCI	CCI-000172
CCI	CCI-001403
CCI	CCI-001404
CCI	CCI-001405
CCI	CCI-002130
CN-L3	7.1.3.2(d)
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)

CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.AC-1
CSF	PR.AC-4
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.AA-01
CSF2.0	PR.AA-05
CSF2.0	PR.DS-10
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.18
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.9.2.1
ISO/IEC-27001	A.12.4.1
ITSG-33	AC-2(4)
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5

NESA	T3.6.6
NESA	T5.2.2
NIAV2	AM9a
NIAV2	AM9b
NIAV2	AM9c
NIAV2	AM9d
NIAV2	AM9e
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	5.2.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
QCSC-V1	15.2
RULE-ID	SV-270686r1066547_rule
STIG-ID	UBTU-24-200300
SWIFT-CSCV1	6.4
TBA-FIISB	36.2.3
TBA-FIISB	45.1.1
VULN-ID	V-270686

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep shadow' returned :

```
-w /etc/shadow -p wa -k usergroup_modification
-w /etc/gshadow -p wa -k usergroup_modification
```


UBTU-24-200310 - Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/gshadow.

Info

Once an attacker establishes access to a system, the attacker often attempts to create a persistent method of reestablishing access.

One way to accomplish this is for the attacker to create an account.

Auditing account creation actions provides logging that can be used for forensic purposes. To address access requirements, many operating systems may be integrated with enterprise-level authentication/access/auditing mechanisms that meet or exceed access control policy requirements. Satisfies: SRG-OS-000004-GPOS-00004, SRG-OS-000239-GPOS-00089, SRG-OS-000240-GPOS-00090, SRG-OS-000241-GPOS-00091, SRG-OS-000303-GPOS-00120, SRG-OS-000458-GPOS-00203, SRG-OS-000463-GPOS-00207, SRG-OS-000476-GPOS-00221

Solution

Configure Ubuntu 24.04 LTS to generate audit records for all account creations, modifications, disabling, and termination events that affect "/etc/gshadow". Add or update the following rule to "/etc/audit/rules.d/stig.rules": -w /etc/gshadow -p wa -k usergroup_modification To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.1
800-171	3.3.1
800-171	3.3.2
800-171R3	03.01.01
800-171R3	03.03.03a.
800-53	AC-2(4)
800-53	AU-12c.
800-53R5	AC-2(4)
800-53R5	AU-12c.
CAT	II
CCI	CCI-000018
CCI	CCI-000172
CCI	CCI-001403
CCI	CCI-001404
CCI	CCI-001405
CCI	CCI-002130
CN-L3	7.1.3.2(d)
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)

CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.AC-1
CSF	PR.AC-4
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.AA-01
CSF2.0	PR.AA-05
CSF2.0	PR.DS-10
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.18
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.9.2.1
ISO/IEC-27001	A.12.4.1
ITSG-33	AC-2(4)
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5

NESA	T3.6.6
NESA	T5.2.2
NIAV2	AM9a
NIAV2	AM9b
NIAV2	AM9c
NIAV2	AM9d
NIAV2	AM9e
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	5.2.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
QCSC-V1	15.2
RULE-ID	SV-270687r1066550_rule
STIG-ID	UBTU-24-200310
SWIFT-CSCV1	6.4
TBA-FIISB	36.2.3
TBA-FIISB	45.1.1
VULN-ID	V-270687

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep gshadow' returned :

```
-w /etc/gshadow -p wa -k usergroup_modification
```

UBTU-24-200320 - Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/passwd.

Info

Once an attacker establishes access to a system, the attacker often attempts to create a persistent method of reestablishing access.

One way to accomplish this is for the attacker to create an account.

Auditing account creation actions provides logging that can be used for forensic purposes. To address access requirements, many operating systems may be integrated with enterprise-level authentication/access/auditing mechanisms that meet or exceed access control policy requirements. Satisfies: SRG-OS-000004-GPOS-00004, SRG-OS-000239-GPOS-00089, SRG-OS-000240-GPOS-00090, SRG-OS-000241-GPOS-00091, SRG-OS-000303-GPOS-00120, SRG-OS-000458-GPOS-00203, SRG-OS-000463-GPOS-00207, SRG-OS-000476-GPOS-00221

Solution

Configure Ubuntu 24.04 LTS to generate audit records for all account creations, modifications, disabling, and termination events that affect "/etc/security/passwd". Add or update the following rule to "/etc/audit/rules.d/stig.rules": -w /etc/security/passwd -p wa -k usergroup_modification To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.1
800-171	3.3.1
800-171	3.3.2
800-171R3	03.01.01
800-171R3	03.03.03a.
800-53	AC-2(4)
800-53	AU-12c.
800-53R5	AC-2(4)
800-53R5	AU-12c.
CAT	II
CCI	CCI-000018
CCI	CCI-000172
CCI	CCI-001403
CCI	CCI-001404
CCI	CCI-001405
CCI	CCI-002130
CN-L3	7.1.3.2(d)
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)

CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.AC-1
CSF	PR.AC-4
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.AA-01
CSF2.0	PR.AA-05
CSF2.0	PR.DS-10
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.18
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.9.2.1
ISO/IEC-27001	A.12.4.1
ITSG-33	AC-2(4)
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5

NESA	T3.6.6
NESA	T5.2.2
NIAV2	AM9a
NIAV2	AM9b
NIAV2	AM9c
NIAV2	AM9d
NIAV2	AM9e
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	5.2.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
QCSC-V1	15.2
RULE-ID	SV-270688r1066553_rule
STIG-ID	UBTU-24-200320
SWIFT-CSCV1	6.4
TBA-FIISB	36.2.3
TBA-FIISB	45.1.1
VULN-ID	V-270688

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep opasswd' returned :

```
-w /etc/security/opasswd -p wa -k usergroup_modification
```

UBTU-24-200580 - Ubuntu 24.04 LTS must prevent all software from executing at higher privilege levels than users executing the software and the audit system must be configured to audit the execution of privileged functions.

Info

In certain situations, software applications/programs need to execute with elevated privileges to perform required functions.

However, if the privileges required for execution are at a higher level than the privileges assigned to organizational users invoking such applications/programs, those users are indirectly provided with greater privileges than assigned by the organizations. Some programs and processes are required to operate at a higher privilege level and therefore, must be excluded from the organization-defined software list after review. Satisfies: SRG-OS-000326-GPOS-00126, SRG-OS-000327-GPOS-00127, SRG-OS-000755-GPOS-00220

Solution

Configure Ubuntu 24.04 LTS to audit the execution of all privileged functions. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F arch=b64 -S execve -C uid!=euid -F euid=0 -F key=execpriv -a always,exit -F arch=b64 -S execve -C gid!=egid -F egid=0 -F key=execpriv -a always,exit -F arch=b32 -S execve -C uid!=euid -F euid=0 -F key=execpriv -a always,exit -F arch=b32 -S execve -C gid!=egid -F egid=0 -F key=execpriv Notes: For 32-bit architectures, only the 32-bit specific entries are required. To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.5
800-171	3.1.7
800-171R3	03.01.05a.
800-171R3	03.01.07b.
800-53	AC-6(8)
800-53	AC-6(9)
800-53R5	AC-6(8)
800-53R5	AC-6(9)
800-53R5	MA-3(5)
CAT	II
CCI	CCI-002233
CCI	CCI-002234
CCI	CCI-004188
CN-L3	7.1.3.2(b)
CN-L3	7.1.3.2(g)
CN-L3	8.1.4.2(d)
CN-L3	8.1.10.6(a)
CSF	PR.AC-4
CSF2.0	PR.AA-05

DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
ISO-27001-2022	A.5.15
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.18
ISO/IEC-27001	A.12.4.3
ITSG-33	AC-6
NESA	T5.1.1
NESA	T5.2.2
NESA	T5.4.1
NESA	T5.4.4
NESA	T5.4.5
NESA	T5.5.4
NESA	T5.6.1
NESA	T7.5.3
NIAV2	AM1
NIAV2	AM23f
NIAV2	SS13c
NIAV2	SS15c
PCI-DSSV3.2.1	7.1.2
PCI-DSSV4.0	7.2.1
PCI-DSSV4.0	7.2.2
QCSC-V1	5.2.2
QCSC-V1	6.2
RULE-ID	SV-270689r1066556_rule
STIG-ID	UBTU-24-200580
SWIFT-CSCV1	5.1
TBA-FIISB	31.4.2
TBA-FIISB	31.4.3

Assets**strkbsubuntu9**

All of the following must pass to satisfy this requirement:

```
-----
PASSED - b32 uid
The command '/sbin/auditctl -l | /bin/grep execve' returned :

-a always,exit -F arch=b64 -S execve -C uid!=euid -F euid=0 -F key=execpriv
-a always,exit -F arch=b64 -S execve -C gid!=egid -F egid=0 -F key=execpriv
-a always,exit -F arch=b32 -S execve -C uid!=euid -F euid=0 -F key=execpriv
-a always,exit -F arch=b32 -S execve -C gid!=egid -F egid=0 -F key=execpriv

-----
PASSED - b64 uid
The command '/sbin/auditctl -l | /bin/grep execve' returned :

-a always,exit -F arch=b64 -S execve -C uid!=euid -F euid=0 -F key=execpriv
-a always,exit -F arch=b64 -S execve -C gid!=egid -F egid=0 -F key=execpriv
-a always,exit -F arch=b32 -S execve -C uid!=euid -F euid=0 -F key=execpriv
-a always,exit -F arch=b32 -S execve -C gid!=egid -F egid=0 -F key=execpriv

-----
PASSED - b64 gid
The command '/sbin/auditctl -l | /bin/grep execve' returned :

-a always,exit -F arch=b64 -S execve -C uid!=euid -F euid=0 -F key=execpriv
-a always,exit -F arch=b64 -S execve -C gid!=egid -F egid=0 -F key=execpriv
-a always,exit -F arch=b32 -S execve -C uid!=euid -F euid=0 -F key=execpriv
-a always,exit -F arch=b32 -S execve -C gid!=egid -F egid=0 -F key=execpriv

-----
PASSED - b32 gid
The command '/sbin/auditctl -l | /bin/grep execve' returned :

-a always,exit -F arch=b64 -S execve -C uid!=euid -F euid=0 -F key=execpriv
-a always,exit -F arch=b64 -S execve -C gid!=egid -F egid=0 -F key=execpriv
-a always,exit -F arch=b32 -S execve -C uid!=euid -F euid=0 -F key=execpriv
-a always,exit -F arch=b32 -S execve -C gid!=egid -F egid=0 -F key=execpriv
```

UBTU-24-200610 - Ubuntu 24.04 LTS must automatically lock an account until the locked account is released by an administrator when three unsuccessful logon attempts have been made.

Info

By limiting the number of failed logon attempts, the risk of unauthorized system access via user password guessing, otherwise known as brute-forcing, is reduced.

Limits are imposed by locking the account. Satisfies: SRG-OS-000021-GPOS-00005, SRG-OS-000329-GPOS-00128

Solution

Configure Ubuntu 24.04 LTS to utilize the "pam_faillock" module. Edit the /etc/pam.d/common-auth file to add the following lines below the "auth" definition for pam_unix.so: auth [default=die] pam_faillock.so authfail auth sufficient pam_faillock.so authsucc Configure the "pam_faillock" module to use the following options: Edit the /etc/security/faillock.conf file and add/update the following keywords and values: audit silent deny = 3 fail_interval = 900 unlock_time = 0

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.8
800-171R3	03.01.08a.
800-171R3	03.01.08b.
800-53	AC-7a.
800-53	AC-7b.
800-53R5	AC-7a.
800-53R5	AC-7b.
CAT	III
CCI	CCI-000044
CCI	CCI-002238
CN-L3	7.1.2.7(f)
CN-L3	7.1.3.1(c)
CN-L3	8.1.4.1(b)
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.5
ITSG-33	AC-7a.
ITSG-33	AC-7b.
NESA	T5.5.1
NIAV2	AM24

PCI-DSSV3.2.1	8.1.6
PCI-DSSV3.2.1	8.1.7
PCI-DSSV4.0	8.3.4
RULE-ID	SV-270690r1067126_rule
STIG-ID	UBTU-24-200610
TBA-FIISB	36.2.4
TBA-FIISB	45.1.2
TBA-FIISB	45.2.1
TBA-FIISB	45.2.2
VULN-ID	V-270690

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - pam_faillock.so module is present in /etc/pam.d/common-auth file

Compliant file(s):

```
/etc/pam.d/common-auth - regex '^[\\s]*auth([\\#\\n\\r]+)pam_faillock.so[\\s]+' found - expect
'^[\\s]*auth([\\#\\n\\r]+)pam_faillock.so[\\s]+' found in the following lines:
    20: auth    [default=die]                pam_faillock.so authfail
    21: auth    sufficient                    pam_faillock.so authsucc
```

PASSED - pam_faillock.so is used with audit

Compliant file(s):

```
/etc/security/faillock.conf - regex '^[\\s]*audit[\\s]*$' found - expect '^[\\s]*audit[\\s]*$'
found in the following lines:
    10: audit
```

PASSED - pam_faillock.so is used with silent

Compliant file(s):

```
/etc/security/faillock.conf - regex '^[\\s]*silent[\\s]*$' found - expect '^[\\s]*silent[\\s]*$'
found in the following lines:
    14: silent
```

PASSED - pam_faillock.so is used with deny set to 3 or less

Compliant file(s):

```
/etc/security/faillock.conf - regex '^[\\s]*deny[\\s]*=' found - expect
'^[\\s]*deny[\\s]*=[\\s]*[1-3][\\s]*$' found in the following lines:
    32: deny = 3
```

PASSED - pam_faillock.so is used with fail_interval set to 900 or less

Compliant file(s):

```
/etc/security/faillock.conf - regex '^[\\s]*fail_interval[\\s]*=' found - expect
'^[\\s]*fail_interval[\\s]*=[\\s]*([1-9]|[1-9][0-9]|[1-8][0-9][0-9]|900)[\\s]*$' found in the
following lines:
    38: fail_interval = 900
```

PASSED - pam_faillock.so is used with unlock_time set to 0

Compliant file(s):

```
/etc/security/faillock.conf - regex '^[\\s]*unlock_time[\\s]*=' found - expect
'^[\\s]*unlock_time[\\s]*=[\\s]*0[\\s]*$' found in the following lines:
    45: unlock_time = 0
```

UBTU-24-200640 - Ubuntu 24.04 LTS must display the Standard Mandatory DOD Notice and Consent Banner before granting access to via an SSH login.

Info

Display of a standardized and approved use notification before granting access to the publicly accessible operating system ensures privacy and security notification verbiage used is consistent with applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance. System use notifications are required only for access via logon interfaces with human users and are not required when such human interfaces do not exist. The banner must be formatted in accordance with applicable DOD policy.

Use the following verbiage for operating systems that can accommodate banners of 1300 characters: "You are accessing a U.S.

Government (USG) Information System (IS) that is provided for USG-authorized use only. By using this IS (which includes any device attached to this IS), you consent to the following conditions: -The USG routinely intercepts and monitors communications on this IS for purposes including, but not limited to, penetration testing, COMSEC monitoring, network operations and defense, personnel misconduct (PM), law enforcement (LE), and counterintelligence (CI) investigations. -At any time, the USG may inspect and seize data stored on this IS. -Communications using, or data stored on, this IS are not private, are subject to routine monitoring, interception, and search, and may be disclosed or used for any USG-authorized purpose. -This IS includes security measures (e.g., authentication and access controls) to protect USG interests--not for your personal benefit or privacy. -Notwithstanding the above, using this IS does not constitute consent to PM, LE or CI investigative searching or monitoring of the content of privileged communications, or work product, related to personal representation or services by attorneys, psychotherapists, or clergy, and their assistants. Such communications and work product are private and confidential.

See User Agreement for details." Use the following verbiage for operating systems that have severe limitations on the number of characters that can be displayed in the banner: "I've read (literal ampersand) consent to terms in IS user agreem't." Satisfies: SRG-OS-000023-GPOS-00006, SRG-OS-000228-GPOS-00088

Solution

Configure Ubuntu 24.04 LTS to display the Standard Mandatory DOD Notice and Consent Banner before granting access via an SSH login. Set the parameter Banner in "/etc/ssh/sshd_config" to point to the "/etc/issue.net" file: `$ sudo sed -i '/^Banner/d' /etc/ssh/sshd_config $ sudo sed -i '$aBanner /etc/issue.net' /etc/ssh/sshd_config` Either create the file containing the banner or replace the text in the file with the Standard Mandatory DOD Notice and Consent Banner.

The DOD required text is: "You are accessing a U.S.

Government (USG) Information System (IS) that is provided for USG-authorized use only. By using this IS (which includes any device attached to this IS), you consent to the following conditions: -The USG routinely intercepts and monitors communications on this IS for purposes including, but not limited to, penetration testing, COMSEC monitoring, network operations and defense, personnel misconduct (PM), law enforcement (LE), and counterintelligence (CI) investigations. -At any time, the USG may inspect and seize data stored on this IS. -Communications using, or data stored on, this IS are not private, are subject to routine monitoring, interception, and search, and may be disclosed or used for any USG-authorized purpose. -This IS includes security measures (e.g., authentication and access controls) to protect USG interests--not for your personal benefit or privacy. -Notwithstanding the above, using this IS does not constitute consent to PM, LE or CI investigative searching or monitoring of the content of privileged communications, or work product, related to personal representation or services by attorneys, psychotherapists, or clergy, and their assistants. Such communications and work product are private and confidential.

See User Agreement for details." Restart the SSH daemon for the changes to take effect and then signal the SSH server to reload the configuration file: `$ sudo systemctl -s SIGHUP kill sshd`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.9
800-171R3	03.01.09
800-53	AC-8a.
800-53	AC-8c.1.
800-53	AC-8c.2.
800-53	AC-8c.3.

800-53R5	AC-8a.
800-53R5	AC-8c.1.
800-53R5	AC-8c.2.
800-53R5	AC-8c.3.
CAT	II
CCI	CCI-000048
CCI	CCI-001384
CCI	CCI-001385
CCI	CCI-001386
CCI	CCI-001387
CCI	CCI-001388
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.5
ITSG-33	AC-8a.
ITSG-33	AC-8c.a.
ITSG-33	AC-8c.b.
ITSG-33	AC-8c.c.
NESA	M5.2.5
NESA	T5.5.1
NIAV2	AM10a
NIAV2	AM10b
NIAV2	AM10c
NIAV2	AM10d
NIAV2	AM10e
RULE-ID	SV-270691r1066562_rule
STIG-ID	UBTU-24-200640
TBA-FIISB	45.2.4
VULN-ID	V-270691

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
PASSED - sshd_config
Compliant file(s):
  /etc/ssh/sshd_config - regex '(?i)^\s*Banner\s+' found - expect '(?
i)^\s*Banner\s+"?/etc/issue.net"' found in the following lines:
    118: Banner /etc/issue.net
  /etc/ssh/sshd_config.d/50-cloud-init.conf - regex not found
  /etc/ssh/sshd_config.d/50-cloudimg-settings.conf - regex not found
  /etc/ssh/sshd_config.d/60-cloudimg-settings.conf - regex not found
-----
```

```
PASSED - banner text
You are accessing a U.S. Government (USG) Information System (IS) that is provided for USG-
authorized use only.
```

By using this IS (which includes any device attached to this IS), you consent to the following conditions:

- The USG routinely intercepts and monitors communications on this IS for purposes including, but not limited to, penetration testing, COMSEC monitoring, network operations and defense, personnel misconduct (PM), law enforcement (LE), and counterintelligence (CI) investigations.
- At any time, the USG may inspect and seize data stored on this IS.
- Communications using, or data stored on, this IS are not private, are subject to routine monitoring, interception, and search, and may be disclosed or used for any USG-authorized purpose.
- This IS includes security measures (e.g., authentication and access controls) to protect USG interests--not for your personal benefit or privacy.
- Notwithstanding the above, using this IS does not constitute consent to PM, LE or CI investigative searching or monitoring of the content of privileged communications, or work product, related to personal representation or services by attorneys, psychotherapists, or clergy, and their assistants. Such communications and work product are private and confidential. See User Agreement for details.

UBTU-24-200650 - Ubuntu 24.04 LTS must enable the graphical user logon banner to display the Standard Mandatory DOD Notice and Consent Banner before granting local access to the system via a graphical user logon.

Info

Display of a standardized and approved use notification before granting access to Ubuntu 24.04 LTS ensures privacy and security notification verbiage used is consistent with applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance. System use notifications are required only for access via logon interfaces with human users and are not required when such human interfaces do not exist. The banner must be formatted in accordance with applicable DOD policy.

Use the following verbiage for operating systems that can accommodate banners of 1300 characters: "You are accessing a U.S.

Government (USG) Information System (IS) that is provided for USG-authorized use only. By using this IS (which includes any device attached to this IS), you consent to the following conditions: -The USG routinely intercepts and monitors communications on this IS for purposes including, but not limited to, penetration testing, COMSEC monitoring, network operations and defense, personnel misconduct (PM), law enforcement (LE), and counterintelligence (CI) investigations. -At any time, the USG may inspect and seize data stored on this IS. -Communications using, or data stored on, this IS are not private, are subject to routine monitoring, interception, and search, and may be disclosed or used for any USG-authorized purpose. -This IS includes security measures (e.g., authentication and access controls) to protect USG interests--not for your personal benefit or privacy. -Notwithstanding the above, using this IS does not constitute consent to PM, LE or CI investigative searching or monitoring of the content of privileged communications, or work product, related to personal representation or services by attorneys, psychotherapists, or clergy, and their assistants. Such communications and work product are private and confidential.

See User Agreement for details." Use the following verbiage for operating systems that have severe limitations on the number of characters that can be displayed in the banner: "I've read (literal ampersand) consent to terms in IS user agreem't."

Solution

Configure Ubuntu 24.04 LTS to display the Standard Mandatory DOD Notice and Consent Banner before granting access to Ubuntu 24.04 LTS via a graphical user logon. Edit the `/etc/gdm3/greeter.dconf-defaults` file. Look for the `"banner-message-enable"` parameter under the `"[org/gnome/login-screen]"` section and uncomment it (remove the leading `"#"` characters): `[org/gnome/login-screen] banner-message-enable=true`
Update the GDM with the new configuration: `$ sudo dconf update $ sudo systemctl restart gdm3`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.9
800-171R3	03.01.09
800-53	AC-8a.
800-53R5	AC-8a.
CAT	II
CCI	CCI-000048
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.5
ITSG-33	AC-8a.
NESA	M5.2.5

NESA	T5.5.1
NIAV2	AM10a
NIAV2	AM10b
NIAV2	AM10c
NIAV2	AM10d
NIAV2	AM10e
RULE-ID	SV-270692r1066565_rule
STIG-ID	UBTU-24-200650
TBA-FIISB	45.2.4
VULN-ID	V-270692

Assets

strkbsubuntu9

UBTU-24-200660 - Ubuntu 24.04 LTS must display the Standard Mandatory DOD Notice and Consent Banner before granting local access to the system via a graphical user logon.

Info

Display of a standardized and approved use notification before granting access to Ubuntu 24.04 LTS ensures privacy and security notification verbiage used is consistent with applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance. System use notifications are required only for access via logon interfaces with human users and are not required when such human interfaces do not exist. The banner must be formatted in accordance with applicable DOD policy.

Use the following verbiage for operating systems that can accommodate banners of 1300 characters: "You are accessing a U.S.

Government (USG) Information System (IS) that is provided for USG-authorized use only. By using this IS (which includes any device attached to this IS), you consent to the following conditions: -The USG routinely intercepts and monitors communications on this IS for purposes including, but not limited to, penetration testing, COMSEC monitoring, network operations and defense, personnel misconduct (PM), law enforcement (LE), and counterintelligence (CI) investigations. -At any time, the USG may inspect and seize data stored on this IS. -Communications using, or data stored on, this IS are not private, are subject to routine monitoring, interception, and search, and may be disclosed or used for any USG-authorized purpose. -This IS includes security measures (e.g., authentication and access controls) to protect USG interests--not for your personal benefit or privacy. -Notwithstanding the above, using this IS does not constitute consent to PM, LE or CI investigative searching or monitoring of the content of privileged communications, or work product, related to personal representation or services by attorneys, psychotherapists, or clergy, and their assistants.

Such communications and work product are private and confidential.

See User Agreement for details." Use the following verbiage for operating systems that have severe limitations on the number of characters that can be displayed in the banner: "I've read (literal ampersand) consent to terms in IS user agreem't."

Solution

Configure Ubuntu 24.04 LTS displays the Standard Mandatory DOD Notice and Consent Banner before granting access to Ubuntu 24.04 LTS via a graphical user logon. Edit the "/etc/gdm3/greeter.dconf-defaults" file. Set the "banner-message-text" line to contain the appropriate banner message text as shown below: banner-message-text="You are accessing a U.S.

Government (USG) Information System (IS) that is provided for USG-authorized use only.\n\nBy using this IS (which includes any device attached to this IS), you consent to the following conditions:\n\n-The USG routinely intercepts and monitors communications on this IS for purposes including, but not limited to, penetration testing, COMSEC monitoring, network operations and defense, personnel misconduct (PM), law enforcement (LE), and counterintelligence (CI) investigations.\n\n-At any time, the USG may inspect and seize data stored on this IS.\n\n-Communications using, or data stored on, this IS are not private, are subject to routine monitoring, interception, and search, and may be disclosed or used for any USG-authorized purpose.\n\n-This IS includes security measures (e.g., authentication and access controls) to protect USG interests--not for your personal benefit or privacy.\n\n-Notwithstanding the above, using this IS does not constitute consent to PM, LE or CI investigative searching or monitoring of the content of privileged communications, or work product, related to personal representation or services by attorneys, psychotherapists, or clergy, and their assistants.

Such communications and work product are private and confidential.

See User Agreement for details.' Update the GDM with the new configuration: \$ sudo dconf update \$ sudo systemctl restart gdm3

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.9
800-171R3	03.01.09
800-53	AC-8a.
800-53R5	AC-8a.
CAT	II
CCI	CCI-000048
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.5
ITSG-33	AC-8a.
NESA	M5.2.5
NESA	T5.5.1
NIAV2	AM10a
NIAV2	AM10b
NIAV2	AM10c
NIAV2	AM10d
NIAV2	AM10e
RULE-ID	SV-270693r1066568_rule
STIG-ID	UBTU-24-200660
TBA-FIISB	45.2.4
VULN-ID	V-270693

Assets
strkbsubuntu9

UBTU-24-300001 - Ubuntu 24.04 LTS Advance Package Tool (APT) must be configured to prevent the installation of patches, service packs, device drivers, or Ubuntu 24.04 LTS components without verification they have been digitally signed using a certificate that is recognized and approved by the organization.

Info

Changes to any software components can have significant effects on the overall security of Ubuntu 24.04 LTS. This requirement ensures the software has not been tampered with and that it has been provided by a trusted vendor. Accordingly, patches, service packs, device drivers, or operating system components must be signed with a certificate recognized and approved by the organization. Verifying the authenticity of the software prior to installation validates the integrity of the patch or upgrade received from a vendor.

This ensures the software has not been tampered with and that it has been provided by a trusted vendor.

Self-signed certificates are disallowed by this requirement.

Ubuntu 24.04 LTS will not have to verify the software again.

This requirement does not mandate DOD certificates for this purpose; however, the certificate used to verify the software must be from an approved CA.

Solution

Configure APT to prevent the installation of patches, service packs, device drivers, or Ubuntu 24.04 LTS components without verification they have been digitally signed using a certificate recognized and approved by the organization. Remove/update any APT configuration files that contain the variable "AllowUnauthenticated" to "false" or remove "AllowUnauthenticated" entirely from each file.

Below is an example of setting the "AllowUnauthenticated" variable to "false": `APT::Get::AllowUnauthenticated "false";`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.5
800-171R3	03.04.05
800-53	CM-5(3)
800-53R5	CM-14
CAT	III
CCI	CCI-003992
CSF	PR.IP-1
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.4
ISO-27001-2022	A.8.9
ISO-27001-2022	A.8.19
ISO-27001-2022	A.8.31
ISO-27001-2022	A.8.32

ITSG-33	CM-5(3)
NESA	T3.2.3
NESA	T5.1.1
NESA	T5.6.1
NESA	T7.5.1
NESA	T7.5.3
NESA	T7.6.3
QCSC-V1	7.2
RULE-ID	SV-270695r1066574_rule
STIG-ID	UBTU-24-300001
VULN-ID	V-270695

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/apt/apt.conf.d/01-vendor-ubuntu - regex '^[\\s]*APT::Get::AllowUnauthenticated[\\s]+'
found - expect '^[\\s]*APT::Get::AllowUnauthenticated[\\s]+["']*true["']*\\s*;' not found in the
following lines:
    3: APT::Get::AllowUnauthenticated "false";
/etc/apt/apt.conf.d/01autoremove - regex not found
/etc/apt/apt.conf.d/10periodic - regex not found
/etc/apt/apt.conf.d/15update-stamp - regex not found
/etc/apt/apt.conf.d/20apt-esm-hook.conf - regex not found
/etc/apt/apt.conf.d/20archive - regex not found
/etc/apt/apt.conf.d/20auto-upgrades - regex not found
/etc/apt/apt.conf.d/20packagekit - regex not found
/etc/apt/apt.conf.d/20snapd.conf - regex not found
/etc/apt/apt.conf.d/50-unattended-upgrades - regex not found
/etc/apt/apt.conf.d/50appstream - regex not found
/etc/apt/apt.conf.d/50command-not-found - regex not found
/etc/apt/apt.conf.d/50unattended-upgrades - regex not found
/etc/apt/apt.conf.d/70debconf - regex not found
/etc/apt/apt.conf.d/99needrestart - regex not found
/etc/apt/apt.conf.d/99update-notifier - regex not found
```

UBTU-24-300006 - Ubuntu 24.04 LTS library files must have mode 0755 or less permissive.

Info

If Ubuntu 24.04 LTS were to allow any user to make changes to software libraries, then those changes might be implemented without undergoing the appropriate testing and approvals that are part of a robust change management process. This requirement applies to operating systems with software libraries that are accessible and configurable, as in the case of interpreted languages.

Software libraries also include privileged programs that execute with escalated privileges.

Only qualified and authorized individuals must be allowed to obtain access to information system components for purposes of initiating changes, including upgrades and modifications.

Solution

Configure the systemwide shared library files contained in the directories `"/lib"`, `"/lib64"`, `"/usr/lib"`, and `"/usr/lib64"` to have mode 0755 or less permissive with the following command: `$ sudo find /lib /lib64 /usr/lib /usr/lib64 -type f -name '*.so*' -perm /022 -exec chmod go-w {} +`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.5
800-171R3	03.04.05
800-53	CM-5(6)
800-53R5	CM-5(6)
CAT	II
CCI	CCI-001499
CSF	PR.IP-1
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.4
ISO-27001-2022	A.8.9
ISO-27001-2022	A.8.19
ISO-27001-2022	A.8.31
ISO-27001-2022	A.8.32
ITSG-33	CM-5(6)
NESA	T3.2.3
NESA	T5.1.1
NESA	T5.6.1

NESA	T7.5.1
NESA	T7.5.3
NESA	T7.6.3
QCSC-V1	7.2
RULE-ID	SV-270696r1107306_rule
STIG-ID	UBTU-24-300006
VULN-ID	V-270696

Assets

strkbsubuntu9

No results found.

UBTU-24-300007 - Ubuntu 24.04 LTS library files must be owned by root.

Info

If Ubuntu 24.04 LTS were to allow any user to make changes to software libraries, then those changes might be implemented without undergoing the appropriate testing and approvals that are part of a robust change management process. This requirement applies to operating systems with software libraries that are accessible and configurable, as in the case of interpreted languages.

Software libraries also include privileged programs that execute with escalated privileges.

Only qualified and authorized individuals must be allowed to obtain access to information system components for purposes of initiating changes, including upgrades and modifications.

Solution

Configure the systemwide shared library files contained in the directories `"/lib"`, `"/lib64"`, `"/usr/lib"`, and `"/usr/lib64"` to be owned by root with the following command: `$ sudo find /lib /lib64 /usr/lib /usr/lib64 -type f -name '*.so*' ! -user root -exec chown root {} +`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.5
800-171R3	03.04.05
800-53	CM-5(6)
800-53R5	CM-5(6)
CAT	II
CCI	CCI-001499
CSF	PR.IP-1
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.4
ISO-27001-2022	A.8.9
ISO-27001-2022	A.8.19
ISO-27001-2022	A.8.31
ISO-27001-2022	A.8.32
ITSG-33	CM-5(6)
NESA	T3.2.3
NESA	T5.1.1
NESA	T5.6.1

NESA	T7.5.1
NESA	T7.5.3
NESA	T7.6.3
QCSC-V1	7.2
RULE-ID	SV-270697r1107308_rule
STIG-ID	UBTU-24-300007
VULN-ID	V-270697

Assets

strkbsubuntu9

No results found.

UBTU-24-300008 - Ubuntu 24.04 LTS library directories must be owned by root.

Info

If Ubuntu 24.04 LTS were to allow any user to make changes to software libraries, then those changes might be implemented without undergoing the appropriate testing and approvals that are part of a robust change management process. This requirement applies to operating systems with software libraries that are accessible and configurable, as in the case of interpreted languages.

Software libraries also include privileged programs that execute with escalated privileges.

Only qualified and authorized individuals must be allowed to obtain access to information system components for purposes of initiating changes, including upgrades and modifications.

Solution

Configure the library files and their respective parent directories to be protected from unauthorized access.

Run the following command: `$ sudo find /lib /lib64 /usr/lib /usr/lib64 ! -user root -type d -exec chown root '{}' \;`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.5
800-171R3	03.04.05
800-53	CM-5(6)
800-53R5	CM-5(6)
CAT	II
CCI	CCI-001499
CSF	PR.IP-1
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.4
ISO-27001-2022	A.8.9
ISO-27001-2022	A.8.19
ISO-27001-2022	A.8.31
ISO-27001-2022	A.8.32
ITSG-33	CM-5(6)
NESA	T3.2.3
NESA	T5.1.1
NESA	T5.6.1
NESA	T7.5.1

NESA	T7.5.3
NESA	T7.6.3
QCSC-V1	7.2
RULE-ID	SV-270698r1101751_rule
STIG-ID	UBTU-24-300008
VULN-ID	V-270698

Assets

strkbsubuntu9

No results found.

UBTU-24-300009 - Ubuntu 24.04 LTS library files must be group-owned by root or a system account.

Info

If Ubuntu 24.04 LTS were to allow any user to make changes to software libraries, then those changes might be implemented without undergoing the appropriate testing and approvals that are part of a robust change management process. This requirement applies to operating systems with software libraries that are accessible and configurable, as in the case of interpreted languages.

Software libraries also include privileged programs that execute with escalated privileges.

Only qualified and authorized individuals must be allowed to obtain access to information system components for purposes of initiating changes, including upgrades and modifications.

Solution

Configure the systemwide shared library files contained in the directories `"/lib"`, `"/lib64"`, `"/usr/lib"`, and `"/usr/lib64"` to be group owned by root with the following command: `$ sudo find /lib /lib64 /usr/lib /usr/lib64 -type f -name '*.so*' ! -group root -exec chown :root {} +`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.5
800-171R3	03.04.05
800-53	CM-5(6)
800-53R5	CM-5(6)
CAT	II
CCI	CCI-001499
CSF	PR.IP-1
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.4
ISO-27001-2022	A.8.9
ISO-27001-2022	A.8.19
ISO-27001-2022	A.8.31
ISO-27001-2022	A.8.32
ITSG-33	CM-5(6)
NESA	T3.2.3
NESA	T5.1.1
NESA	T5.6.1

NESA	T7.5.1
NESA	T7.5.3
NESA	T7.6.3
QCSC-V1	7.2
RULE-ID	SV-270699r1107310_rule
STIG-ID	UBTU-24-300009
VULN-ID	V-270699

Assets

strkbsubuntu9

No results found.

UBTU-24-300010 - Ubuntu 24.04 LTS library directories must be group-owned by root.

Info

If Ubuntu 24.04 LTS were to allow any user to make changes to software libraries, then those changes might be implemented without undergoing the appropriate testing and approvals that are part of a robust change management process. This requirement applies to operating systems with software libraries that are accessible and configurable, as in the case of interpreted languages.

Software libraries also include privileged programs that execute with escalated privileges.

Only qualified and authorized individuals must be allowed to obtain access to information system components for purposes of initiating changes, including upgrades and modifications.

Solution

Configure the system library directories to be protected from unauthorized access.

Run the following command: `$ sudo find /lib /usr/lib /lib64 ! -group root -type d -exec chgrp root '{}' \;`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.5
800-171R3	03.04.05
800-53	CM-5(6)
800-53R5	CM-5(6)
CAT	II
CCI	CCI-001499
CSF	PR.IP-1
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.4
ISO-27001-2022	A.8.9
ISO-27001-2022	A.8.19
ISO-27001-2022	A.8.31
ISO-27001-2022	A.8.32
ITSG-33	CM-5(6)
NESA	T3.2.3
NESA	T5.1.1
NESA	T5.6.1
NESA	T7.5.1

NESA	T7.5.3
NESA	T7.6.3
QCSC-V1	7.2
RULE-ID	SV-270700r1066589_rule
STIG-ID	UBTU-24-300010
VULN-ID	V-270700

Assets

strkbsubuntu9

No results found.

UBTU-24-300011 - Ubuntu 24.04 LTS must have system commands set to a mode of 0755 or less permissive.

Info

If Ubuntu 24.04 LTS were to allow any user to make changes to software libraries, then those changes might be implemented without undergoing the appropriate testing and approvals that are part of a robust change management process. This requirement applies to Ubuntu 24.04 LTS with software libraries that are accessible and configurable, as in the case of interpreted languages.

Software libraries also include privileged programs that execute with escalated privileges.

Only qualified and authorized individuals must be allowed to obtain access to information system components for purposes of initiating changes, including upgrades and modifications.

Solution

Configure the system commands to be protected from unauthorized access.

Run the following command: `$ sudo find /bin /sbin /usr/bin /usr/sbin /usr/local/bin /usr/local/sbin -perm /022 -type f -exec chmod 755 '{}' \;`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.5
800-171R3	03.04.05
800-53	CM-5(6)
800-53R5	CM-5(6)
CAT	II
CCI	CCI-001499
CSF	PR.IP-1
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.4
ISO-27001-2022	A.8.9
ISO-27001-2022	A.8.19
ISO-27001-2022	A.8.31
ISO-27001-2022	A.8.32
ITSG-33	CM-5(6)
NESA	T3.2.3
NESA	T5.1.1
NESA	T5.6.1

NESA	T7.5.1
NESA	T7.5.3
NESA	T7.6.3
QCSC-V1	7.2
RULE-ID	SV-270701r1066592_rule
STIG-ID	UBTU-24-300011
VULN-ID	V-270701

Assets

strkbsubuntu9

No results found.

UBTU-24-300012 - Ubuntu 24.04 LTS must have system commands owned by root or a system account.

Info

If Ubuntu 24.04 LTS were to allow any user to make changes to software libraries, then those changes might be implemented without undergoing the appropriate testing and approvals that are part of a robust change management process. This requirement applies to Ubuntu 24.04 LTS with software libraries that are accessible and configurable, as in the case of interpreted languages.

Software libraries also include privileged programs that execute with escalated privileges.

Only qualified and authorized individuals must be allowed to obtain access to information system components for purposes of initiating changes, including upgrades and modifications.

Solution

Configure the system commands and their respective parent directories to be protected from unauthorized access.

Run the following command, replacing "[FILE]" with any system command file not owned by "root" or a required system account: `$ sudo chown root [FILE]`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.5
800-171R3	03.04.05
800-53	CM-5(6)
800-53R5	CM-5(6)
CAT	II
CCI	CCI-001499
CSF	PR.IP-1
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.4
ISO-27001-2022	A.8.9
ISO-27001-2022	A.8.19
ISO-27001-2022	A.8.31
ISO-27001-2022	A.8.32
ITSG-33	CM-5(6)
NESA	T3.2.3
NESA	T5.1.1

NESA	T5.6.1
NESA	T7.5.1
NESA	T7.5.3
NESA	T7.6.3
QCSC-V1	7.2
RULE-ID	SV-270702r1066595_rule
STIG-ID	UBTU-24-300012
VULN-ID	V-270702

Assets

strkbsubuntu9

No results found.

UBTU-24-300013 - Ubuntu 24.04 LTS must have system commands group-owned by root or a system account.

Info

If Ubuntu 24.04 LTS were to allow any user to make changes to software libraries, then those changes might be implemented without undergoing the appropriate testing and approvals that are part of a robust change management process. This requirement applies to Ubuntu 24.04 LTS with software libraries that are accessible and configurable, as in the case of interpreted languages.

Software libraries also include privileged programs that execute with escalated privileges.

Only qualified and authorized individuals must be allowed to obtain access to information system components for purposes of initiating changes, including upgrades and modifications.

Solution

Configure the system commands to be protected from unauthorized access.

Run the following command, replacing "[FILE]" with any system command file not group-owned by "root" or a required system account: `$ sudo chgrp [SYSTEMACCOUNT] [FILE]`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.5
800-171R3	03.04.05
800-53	CM-5(6)
800-53R5	CM-5(6)
CAT	II
CCI	CCI-001499
CSF	PR.IP-1
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.2
ISO-27001-2022	A.8.4
ISO-27001-2022	A.8.9
ISO-27001-2022	A.8.19
ISO-27001-2022	A.8.31
ISO-27001-2022	A.8.32
ITSG-33	CM-5(6)
NESA	T3.2.3
NESA	T5.1.1
NESA	T5.6.1

NESA	T7.5.1
NESA	T7.5.3
NESA	T7.6.3
QCSC-V1	7.2
RULE-ID	SV-270703r1066598_rule
STIG-ID	UBTU-24-300013
VULN-ID	V-270703

Assets

strkbsubuntu9

No results found.

UBTU-24-300014 - Ubuntu 24.04 LTS must prevent the use of dictionary words for passwords.

Info

Password-based authentication applies to passwords regardless of whether they are used in single-factor or multifactor authentication.

Long passwords or passphrases are preferable over shorter passwords.

Enforced composition rules provide marginal security benefits while decreasing usability.

However, organizations may choose to establish certain rules for password generation (e.g., minimum character length for long passwords) under certain circumstances and can enforce this requirement in IA-5(1)(h).

Account recovery can occur, for example, in situations when a password is forgotten.

Cryptographically protected passwords include salted one-way cryptographic hashes of passwords.

The list of commonly used, compromised, or expected passwords includes passwords obtained from previous breach corpuses, dictionary words, and repetitive or sequential characters.

The list includes context-specific words, such as the name of the service, username, and derivatives thereof.

Solution

Configure Ubuntu 24.04 LTS to prevent the use of dictionary words for passwords. Add or update the following line in the "/etc/security/pwquality.conf" file to include the "dictcheck=1" parameter: dictcheck=1

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.7
800-171R3	03.05.07b.
800-53	IA-5(1)(b)
800-53R5	IA-5(1)(b)
CAT	II
CCI	CCI-004061
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ISO/IEC-27001	A.9.4.3
ITSG-33	IA-5(1)(b)
NESA	T5.2.3
NIAV2	AM22d

QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270704r1066601_rule
STIG-ID	UBTU-24-300014
SWIFT-CSCV1	4.1
VULN-ID	V-270704

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/security/pwquality.conf - regex '^[\\s]*dictcheck' found - expect
'^[\\s]*dictcheck[\\s]*=[\\s]*1[\\s]*$' found in the following lines:
80: dictcheck = 1
```

UBTU-24-300016 - Ubuntu 24.04 LTS must be configured so that when passwords are changed or new passwords are established, pwquality must be used.

Info

Use of a complex password helps to increase the time and resources required to compromise the password. Password complexity, or strength, is a measure of the effectiveness of a password in resisting attempts at guessing and brute-force attacks.

"pwquality" enforces complex password construction configuration and has the ability to limit brute-force attacks on the system.

Solution

Configure Ubuntu 24.04 LTS to use "pwquality" to enforce password complexity rules. Add the following line to "/etc/security/pwquality.conf" (or modify the line to have the required value): enforcing = 1 Add the following line to "/etc/pam.d/common-password" (or modify the line to have the required value): password requisite pam_pwquality.so retry=3 Note: Ensure the value of "retry" is between "1" and "3".

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	II
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270705r1066604_rule
STIG-ID	UBTU-24-300016
SWIFT-CSCV1	2.3
VULN-ID	V-270705

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
PASSED - common-password retry
Compliant file(s):
    /etc/pam.d/common-password - regex '^[\\s]*password[\\s]+(required|requisite)
[\\s]+pam_pwquality.so' found - expect '^[\\s]*password[\\s]+(required|requisite)
[\\s]+pam_pwquality.so([\\s]*retry[\\s]*=[\\s]*[1-3]([\\s]+|$))' found in the following lines:
    25: password requisite    pam_pwquality.so retry=3
-----
```

```
PASSED - enforcing
Compliant file(s):
    /etc/security/pwquality.conf - regex '^[\\s]*enforcing[\\s]*' found - expect
'^[\\s]*enforcing[\\s]*=[\\s]*1[\\s]*$' found in the following lines:
    81: enforcing = 1
```


UBTU-24-300017 - Ubuntu 24.04 LTS must enforce a delay of at least four seconds between logon prompts following a failed logon attempt.

Info

Limiting the number of logon attempts over a certain time interval reduces the chances that an unauthorized user may gain access to an account. The delay option is set in microseconds.

Solution

Configure Ubuntu 24.04 LTS to enforce a delay of at least four seconds between logon prompts following a failed logon attempt. Edit the file "/etc/pam.d/common-auth" and set the parameter "pam_faildelay" to a value of "4000000" or greater: `auth required pam_faildelay.so delay=4000000`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	III
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270706r1068361_rule
STIG-ID	UBTU-24-300017
SWIFT-CSCV1	2.3
VULN-ID	V-270706

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/pam.d/common-auth - regex '^[\s]*auth[\s]*required[\s]*pam_faildelay\.so' found -
expect '^[\s]*auth[\s]*required[\s]*pam_faildelay\.so+([\s]*delay=([4-9][0-9]{6,}|[1-9][0-9]{7,}))' found in the following lines:
17: auth required pam_faildelay.so delay=4000000
```

UBTU-24-300019 - Ubuntu 24.04 LTS must restrict privilege elevation to authorized personnel.

Info

If the "sudoers" file is not configured correctly, any user defined on the system can initiate privileged actions on the target system.

Solution

Configure the operating system to restrict privilege elevation to authorized personnel. Remove the following entries from the /etc/sudoers file or any configuration file under /etc/sudoers.d/: ALL ALL=(ALL) ALL ALL ALL=(ALL:ALL) ALL

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171R3	03.05.01b.
800-53	IA-11
800-53R5	SC-11b.
CAT	II
CCI	CCI-002038
CCI	CCI-004895
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(d)
QCSC-V1	13.2
RULE-ID	SV-274869r1107312_rule
STIG-ID	UBTU-24-300019
VULN-ID	V-274869

Assets

strkbsubuntu9

The following file(s) do not contain "[\s]*ALL[\s]+":
/etc/sudoers
/etc/sudoers.d/90-cloud-init-users
/etc/sudoers.d/README

UBTU-24-300020 - Ubuntu 24.04 LTS must require users to provide a password for privilege escalation.

Info

Without reauthentication, users may access resources or perform tasks for which they do not have authorization. When operating systems provide the capability to escalate a functional capability, it is critical that the user reauthenticate. Satisfies: SRG-OS-000373-GPOS-00156, SRG-OS-000373-GPOS-00157, SRG-OS-000373-GPOS-00158

Solution

Configure the operating system to not allow users to execute privileged actions without authenticating with a password. Remove any occurrence of "NOPASSWD" found in "/etc/sudoers" file or files in the "/etc/sudoers.d" directory. \$ sudo find /etc/sudoers /etc/sudoers.d -type f -exec sed -i '/NOPASSWD/ s/^/# /g' {} \;

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	II
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-274868r1107313_rule
STIG-ID	UBTU-24-300020
SWIFT-CSCV1	2.3
VULN-ID	V-274868

Assets

strkbsubuntu9

The following file(s) do not contain "(?i)^[^#\n\r]*nopasswd\b":
/etc/sudoers
/etc/sudoers.d/90-cloud-init-users

/etc/sudoers.d/README

UBTU-24-300021 - Ubuntu 24.04 LTS must require users to reauthenticate for privilege escalation or when changing roles.

Info

Without reauthentication, users may access resources or perform tasks for which they do not have authorization. When operating systems provide the capability to escalate a functional capability, it is critical the user reauthenticate.

Solution

Remove any occurrence of "!authenticate" found in "/etc/sudoers" file or files in the "/etc/sudoers.d" directory.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	II
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270707r1101786_rule
STIG-ID	UBTU-24-300021
SWIFT-CSCV1	2.3
VULN-ID	V-270707

Assets

strkbsubuntu9

The following file(s) do not contain "(?i)^[^#\n\r]*!authenticate\b":
/etc/sudoers
/etc/sudoers.d/90-cloud-init-users
/etc/sudoers.d/README

UBTU-24-300022 - Ubuntu 24.04 LTS must be configured so that remote X connections are disabled, unless to fulfill documented and validated mission requirements.

Info

The security risk of using X11 forwarding is that the client's X11 display server may be exposed to attack when the SSH client requests forwarding.

A system administrator (SA) must protect clients that may expose themselves to attack by unwittingly requesting X11 forwarding, which can warrant a "no" setting. X11 forwarding must be enabled with caution.

Users with the ability to bypass file permissions on the remote host (for the user's X11 authorization database) can access the local X11 display through the forwarded connection.

An attacker may then be able to perform activities such as keystroke monitoring if the ForwardX11Trusted option is also enabled. If X11 services are not required for the system's intended function, they must be disabled or restricted as appropriate to the system's needs.

Solution

Edit the "/etc/ssh/sshd_config" file to uncomment or add the line for the "X11Forwarding" keyword and set its value to "no" (this file may be named differently or be in a different location if using a version of SSH that is provided by a third-party vendor): X11Forwarding no
Restart the SSH daemon for the changes to take effect: \$ sudo systemctl restart sshd.service

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	I
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270708r1066613_rule
STIG-ID	UBTU-24-300022
SWIFT-CSCV1	2.3
VULN-ID	V-270708

Assets

strkbsubuntu9

Compliant file(s):

```
/etc/ssh/sshd_config - regex '(?i)^\s*X11Forwarding\s+' found - expect '(?i)^\s*X11Forwarding\s+?"no"?\s*$' found in the following lines:
99: X11Forwarding no
/etc/ssh/sshd_config.d/50-cloud-init.conf - regex '(?i)^\s*X11Forwarding\s+' found - expect '(?i)^\s*X11Forwarding\s+?"no"?\s*$' found in the following lines:
2: X11Forwarding no
/etc/ssh/sshd_config.d/50-cloudimg-settings.conf - regex '(?i)^\s*X11Forwarding\s+' found - expect '(?i)^\s*X11Forwarding\s+?"no"?\s*$' found in the following lines:
5: X11Forwarding no
/etc/ssh/sshd_config.d/60-cloudimg-settings.conf - regex '(?i)^\s*X11Forwarding\s+' found - expect '(?i)^\s*X11Forwarding\s+?"no"?\s*$' found in the following lines:
2: X11Forwarding no
```

UBTU-24-300023 - Ubuntu 24.04 LTS SSH daemon must prevent remote hosts from connecting to the proxy display.

Info

When X11 forwarding is enabled, there may be additional exposure to the server and client displays if the sshd proxy display is configured to listen on the wildcard address.

By default, sshd binds the forwarding server to the loopback address and sets the hostname part of the DISPLAY environment variable to localhost.

This prevents remote hosts from connecting to the proxy display.

Solution

Configure the SSH daemon to prevent remote hosts from connecting to the proxy display. Edit the "/etc/ssh/sshd_config" file to uncomment or add the line for the "X11UseLocalhost" keyword and set its value to "yes" (this file may be named differently or be in a different location if using a version of SSH that is provided by a third-party vendor): X11UseLocalhost yes Restart the SSH daemon for the changes to take effect: \$ sudo systemctl restart sshd.service

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	II
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270709r1066616_rule
STIG-ID	UBTU-24-300023
SWIFT-CSCV1	2.3
VULN-ID	V-270709

Assets

strkbsubuntu9

Compliant file(s):
/etc/ssh/sshd_config - regex '(?i)^\s*X11UseLocalhost\s+' found - expect '(?
i)^\s*X11UseLocalhost\s+"?yes"?[\s]*\$' found in the following lines:
101: X11UseLocalhost yes
/etc/ssh/sshd_config.d/50-cloud-init.conf - regex not found
/etc/ssh/sshd_config.d/50-cloudimg-settings.conf - regex not found
/etc/ssh/sshd_config.d/60-cloudimg-settings.conf - regex not found

UBTU-24-300025 - Ubuntu 24.04 LTS must disable the x86 Ctrl-Alt-Delete key sequence if a graphical user interface is installed.

Info

A locally logged-on user who presses Ctrl-Alt-Delete, when at the console, can reboot the system. If accidentally pressed, as could happen in the case of a mixed OS environment, this can create the risk of short-term loss of availability of systems due to unintentional reboot. In the graphical environment, risk of unintentional reboot from the Ctrl-Alt-Delete sequence is reduced because the user will be prompted before any action is taken.

Solution

Configure Ubuntu 24.04 LTS to disable the Ctrl-Alt-Delete sequence when using a graphical user interface. Create or edit a file named `/etc/dconf/db/local.d/00-mediakeys` with the following contents: `[org/gnome/settings-daemon/plugins/media-keys] logout=@as []` Update the dconf settings: `$ sudo dconf update`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	I
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270711r1184069_rule
STIG-ID	UBTU-24-300025
SWIFT-CSCV1	2.3
VULN-ID	V-270711

Assets

strkbsubuntu9

UBTU-24-300026 - Ubuntu 24.04 LTS must disable the x86 Ctrl-Alt-Delete key sequence.

Info

A locally logged-on user who presses Ctrl-Alt-Delete, when at the console, can reboot the system. If accidentally pressed, as could happen in the case of a mixed OS environment, this can create the risk of short-term loss of availability of systems due to unintentional reboot.

Solution

Configure the system to disable the Ctrl-Alt-Delete sequence for the command line with the following commands: `$ sudo systemctl disable ctrl-alt-del.target [...]` `$ sudo systemctl mask ctrl-alt-del.target` Created symlink `/etc/systemd/system/ctrl-alt-del.target ? /dev/null`. Reload the daemon to take effect: `$ sudo systemctl daemon-reload`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	I
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270712r1068363_rule
STIG-ID	UBTU-24-300026
SWIFT-CSCV1	2.3
VULN-ID	V-270712

Assets

strkbsubuntu9

The command `'/bin/systemctl status ctrl-alt-del.target'` returned :

```
* ctrl-alt-del.target
  Loaded: masked (Reason: Unit ctrl-alt-del.target is masked.)
```

Active: inactive (dead)

UBTU-24-300027 - Ubuntu 24.04 LTS must not have accounts configured with blank or null passwords.

Info

If an account has an empty password, anyone could log on and run commands with the privileges of that account.

Accounts with empty passwords must never be used in operational environments.

Solution

Configure all accounts on the system to have a password or lock the account with the following commands:

Perform a password reset: `$ sudo passwd [username]` Lock an account: `$ sudo passwd -l [username]`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	I
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270713r1066628_rule
STIG-ID	UBTU-24-300027
SWIFT-CSCV1	2.3
VULN-ID	V-270713

Assets

strkbsubuntu9

```
The command '/bin/awk -F: '!'$2 {print $1}' /etc/shadow | /bin/awk '{print} END {if (NR == 0)
print "No accounts have blank or null passwords"; else print "Blank or null passwords found on
system"}' returned :
```

No accounts have blank or null passwords

UBTU-24-300028 - Ubuntu 24.04 LTS must not allow accounts configured in Pluggable Authentication Modules (PAM) with blank or null passwords.

Info

If an account has an empty password, anyone could log on and run commands with the privileges of that account.

Accounts with empty passwords must never be used in operational environments.

Solution

If an account is configured for password authentication but does not have an assigned password, it is possible to log on to the account without authenticating. Remove any instances of the "nullok" option in "/etc/pam.d/common-password" to prevent logons with empty passwords.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	I
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270714r1134808_rule
STIG-ID	UBTU-24-300028
SWIFT-CSCV1	2.3
VULN-ID	V-270714

Assets

strkbsubuntu9

The following file(s) do not contain "`^[\\s]*[\\^#\\n\\r]+nullok`":
/etc/pam.d/common-auth
/etc/pam.d/common-password

UBTU-24-300029 - Ubuntu 24.04 LTS must generate audit records for all events that affect the systemd journal files.

Info

Once an attacker establishes access to a system, the attacker often attempts to create a persistent method of reestablishing access. One way to accomplish this is for the attacker to modify system level binaries and their operation. Auditing the systemd journal files provides logging that can be used for forensic purposes. To address access requirements, many operating systems may be integrated with enterprise-level authentication/access/auditing mechanisms that meet or exceed access control policy requirements.

Solution

Configure Ubuntu 24.04 LTS to generate audit records for events that affect `/var/log/journal`. Add or update the following rule to `/etc/audit/rules.d/stig.rules`: `-w /var/log/journal -p wa -k systemd_journal` To reload the rules file, issue the following command: `$ sudo augenrules --load`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	II
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270715r1066634_rule
STIG-ID	UBTU-24-300029
SWIFT-CSCV1	2.3
VULN-ID	V-270715

Assets

strkbsubuntu9

The command `'/sbin/auditctl -l | /bin/grep 'journal''` returned :

```
-w /var/log/journal -p wa -k systemd_journal
```


UBTU-24-300030 - Ubuntu 24.04 LTS default filesystem permissions must be defined in such a way that all authenticated users can read and modify only their own files.

Info

Setting the most restrictive default permissions ensures that when new accounts are created, they do not have unnecessary access.

Solution

Configure the system to define the default permissions for all authenticated users in such a way that the user can read and modify only their own files. Edit the "UMASK" parameter in the "/etc/login.defs" file to match the example below: UMASK 077

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	II
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270716r1066637_rule
STIG-ID	UBTU-24-300030
SWIFT-CSCV1	2.3
VULN-ID	V-270716

Assets

strkbsubuntu9

Compliant file(s):
/etc/login.defs - regex '^[\s]*UMASK[\s]' found - expect '^[\s]*UMASK[\s]+077[\s]*\$' found
in the following lines:
151: UMASK 077

UBTU-24-300031 - Ubuntu 24.04 LTS must not allow unattended or automatic login via SSH.

Info

Failure to restrict system access to authenticated users negatively impacts Ubuntu 24.04 LTS security.

Solution

Configure Ubuntu 24.04 LTS to allow the SSH daemon to not allow unattended or automatic login to the system. Add or edit the following lines in the "/etc/ssh/sshd_config" file: PermitEmptyPasswords no PermitUserEnvironment no Restart the SSH daemon for the changes to take effect: \$ sudo systemctl restart sshd.service

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	I
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-270717r1067177_rule
STIG-ID	UBTU-24-300031
SWIFT-CSCV1	2.3
VULN-ID	V-270717

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
PASSED - PermitEmptyPasswords
Compliant file(s):
/etc/ssh/sshd_config - regex '(?i)^[\\s]*PermitEmptyPasswords[\\s]+' found - expect '(?i)^[\\s]*PermitEmptyPasswords[\\s]+?"no"?[\\s]*$' found in the following lines:
```

```
67: PermitEmptyPasswords no
/etc/ssh/sshd_config.d/50-cloud-init.conf - regex not found
/etc/ssh/sshd_config.d/50-cloudimg-settings.conf - regex not found
/etc/ssh/sshd_config.d/60-cloudimg-settings.conf - regex not found

-----
PASSED - PermitUserEnvironment
Compliant file(s):
/etc/ssh/sshd_config - regex '(?i)^\s*PermitUserEnvironment\s+' found - expect '(?
i)^\s*PermitUserEnvironment\s+"?no"?[ \]*$' found in the following lines:
106: PermitUserEnvironment no
/etc/ssh/sshd_config.d/50-cloud-init.conf - regex not found
/etc/ssh/sshd_config.d/50-cloudimg-settings.conf - regex not found
/etc/ssh/sshd_config.d/60-cloudimg-settings.conf - regex not found
```

UBTU-24-300039 - Ubuntu 24.04 LTS must disable automatic mounting of Universal Serial Bus (USB) mass storage driver.

Info

Without authenticating devices, unidentified or unknown devices may be introduced, thereby facilitating malicious activity. Peripherals include, but are not limited to, devices such as flash drives, external storage, and printers. Satisfies: SRG-OS-000690-GPOS-00140, SRG-OS-000378-GPOS-00163

Solution

Configure Ubuntu 24.04 LTS to disable using the USB storage kernel module with the following command: \$ sudo su -c "echo install usb-storage /bin/false >> /etc/modprobe.d/DISASTIG.conf" Configure Ubuntu 24.04 LTS to disable the ability to use USB mass storage devices with the following command: \$ sudo su -c "echo blacklist usb-storage >> /etc/modprobe.d/DISASTIG.conf"

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171R3	03.05.02
800-53	IA-3
800-53R5	CM-7(9)(b)
800-53R5	IA-3
CAT	II
CCI	CCI-001958
CCI	CCI-003959
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ITSG-33	IA-3
ITSG-33	IA-3a.
NESA	T5.4.3
QCSC-V1	13.2
RULE-ID	SV-270718r1134811_rule
STIG-ID	UBTU-24-300039
TBA-FIISB	27.1

Assets**strkbsubuntu9**

All of the following must pass to satisfy this requirement:

PASSED - blacklist usb-storage

Compliant file(s):

/etc/modprobe.d/blacklist-ath_pci.conf - regex not found
/etc/modprobe.d/blacklist-firewire.conf - regex not found
/etc/modprobe.d/blacklist-framebuffer.conf - regex not found
/etc/modprobe.d/blacklist-nouveau.conf - regex not found
/etc/modprobe.d/blacklist-radeon-instinct.conf - regex not found
/etc/modprobe.d/blacklist-rare-network.conf - regex not found
/etc/modprobe.d/blacklist.conf - regex not found
/etc/modprobe.d/disable-algif_aead.conf - regex not found
/etc/modprobe.d/iwlwifi.conf - regex not found
/etc/modprobe.d/mdadm.conf - regex not found

/etc/modprobe.d/usb-storage.conf - regex '^[\s]*blacklist[\s]+usb-storage[\s]*\$' found -
expect '^[\s]*blacklist[\s]+usb-storage[\s]*\$' found in the following lines:
6: blacklist usb-storage

PASSED - install usb-storage

Compliant file(s):

/etc/modprobe.d/blacklist-ath_pci.conf - regex not found
/etc/modprobe.d/blacklist-firewire.conf - regex not found
/etc/modprobe.d/blacklist-framebuffer.conf - regex not found
/etc/modprobe.d/blacklist-nouveau.conf - regex not found
/etc/modprobe.d/blacklist-radeon-instinct.conf - regex not found
/etc/modprobe.d/blacklist-rare-network.conf - regex not found
/etc/modprobe.d/blacklist.conf - regex not found
/etc/modprobe.d/disable-algif_aead.conf - regex not found
/etc/modprobe.d/iwlwifi.conf - regex not found
/etc/modprobe.d/mdadm.conf - regex not found

/etc/modprobe.d/usb-storage.conf - regex '^[\s]*install[\s]+usb-storage[\s]+/bin/false[\s]*\$' found -
expect '^[\s]*install[\s]+usb-storage[\s]+/bin/false[\s]*\$' found in the following lines:

5: install usb-storage /bin/false

UBTU-24-400000 - Ubuntu 24.04 LTS must uniquely identify interactive users.

Info

To ensure accountability and prevent unauthenticated access, organizational users must be identified and authenticated to prevent potential misuse and compromise of the system. Organizational users include organizational employees or individuals the organization deems to have equivalent status of employees (e.g., contractors).

Organizational users (and processes acting on behalf of users) must be uniquely identified and authenticated to all accesses, except for the following: 1) Accesses explicitly identified and documented by the organization. Organizations document specific user actions that can be performed on the information system without identification or authentication; and 2) Accesses that occur through authorized use of group authenticators without individual authentication.

Organizations may require unique identification of individuals in group accounts (e.g., shared privilege accounts) or for detailed accountability of individual activity. Satisfies: SRG-OS-000104-GPOS-00051, SRG-OS-000121-GPOS-00062

Solution

Edit the file "/etc/passwd" and provide each interactive user account that has a duplicate UID with a unique UID.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.1
800-171R3	03.05.01a.
800-53	IA-2
800-53	IA-8
800-53R5	IA-2
800-53R5	IA-8
CAT	II
CCI	CCI-000764
CCI	CCI-000804
CN-L3	7.1.3.1(a)
CN-L3	7.1.3.1(e)
CN-L3	8.1.4.1(a)
CN-L3	8.1.4.2(a)
CN-L3	8.5.4.1(a)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)

HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ITSG-33	IA-2
ITSG-33	IA-2a.
ITSG-33	IA-8
ITSG-33	IA-8a.
NESA	T2.3.8
NESA	T4.3.1
NESA	T5.3.1
NESA	T5.4.2
NESA	T5.5.1
NESA	T5.5.2
NESA	T5.5.3
NIAV2	AM2
NIAV2	AM8
NIAV2	AM14b
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270720r1066649_rule
STIG-ID	UBTU-24-400000
SWIFT-CSCV1	2.8
TBA-FIISB	35.1
TBA-FIISB	36.1
VULN-ID	V-270720

Assets

strkbsubuntu9

No duplicate User IDs detected

UBTU-24-400110 - Ubuntu 24.04 LTS must prevent direct login to the root account.

Info

To ensure individual accountability and prevent unauthorized access, organizational users must be individually identified and authenticated. A group authenticator is a generic account used by multiple individuals. Use of a group authenticator alone does not uniquely identify individual users.

Examples of the group authenticator are the Unix OS "root" user account, the Windows "Administrator" account, the "sa" account, or a "helpdesk" account. For example, the Unix and Windows operating systems offer a "switch user" capability allowing users to authenticate with their individual credentials and, when needed, switch to the administrator role.

This method provides for unique individual authentication prior to using a group authenticator. Users (and any processes acting on behalf of users) need to be uniquely identified and authenticated for all accesses other than those accesses explicitly identified and documented by the organization, which outlines specific user actions that can be performed on Ubuntu 24.04 LTS without identification or authentication. Requiring individuals to be authenticated with an individual authenticator prior to using a group authenticator allows for traceability of actions, as well as adding an additional level of protection of the actions that can be taken with group account knowledge.

Solution

Configure Ubuntu 24.04 LTS to prevent direct logins to the root account by performing the following operations: `$ sudo passwd -l root`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.1
800-171R3	03.05.01a.
800-53	IA-2(5)
800-53R5	IA-2(5)
CAT	II
CCI	CCI-004045
CN-L3	7.1.3.1(a)
CN-L3	7.1.3.1(e)
CN-L3	8.1.4.1(a)
CN-L3	8.1.4.2(a)
CN-L3	8.5.4.1(a)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)

ISO-27001-2022	A.5.16
ITSG-33	IA-2(5)
ITSG-33	IA-2(5)(a)
ITSG-33	IA-2(5)(b)
NESA	T2.3.8
NESA	T5.3.1
NESA	T5.4.2
NESA	T5.5.1
NESA	T5.5.2
NESA	T5.5.3
NIAV2	AM2
NIAV2	AM8
NIAV2	AM14b
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270724r1066661_rule
STIG-ID	UBTU-24-400110
TBA-FIISB	35.1
TBA-FIISB	36.1
VULN-ID	V-270724

Assets

strkbsubuntu9

The command '/bin/passwd -S root' returned :

```
root L 2024-04-23 0 99999 7 -1
```

UBTU-24-400220 - Ubuntu 24.04 LTS must store only encrypted representations of passwords.

Info

Password complexity, or strength, is a measure of the effectiveness of a password in resisting attempts at guessing and brute-force attacks.

If the information system or application allows the user to consecutively reuse their password when that password has exceeded its defined lifetime, the end result is a password that is not changed per policy requirements.

Solution

Configure Ubuntu 24.04 LTS to store encrypted representations of passwords. Add or modify the "sha512" parameter value to the following line in "/etc/pam.d/common-password" file: password [success=1 default=ignore] pam_unix.so obscure sha512 shadow rounds=100000

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.2
800-171R3	03.05.07d.
800-53	IA-5(1)(d)
800-53R5	IA-5(1)(d)
CAT	II
CCI	CCI-004062
CN-L3	7.1.2.7(e)
CN-L3	7.1.3.1(b)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ISO/IEC-27001	A.9.4.3
ITSG-33	IA-5(1)(d)
NESA	T5.2.3
NIAV2	AM20

NIAV2	AM21
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270725r1101789_rule
STIG-ID	UBTU-24-400220
SWIFT-CSCV1	4.1
TBA-FIISB	26.2.2
VULN-ID	V-270725

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
PASSED - password encryption
Compliant file(s):
/etc/pam.d/common-password - regex '(?i)^\h*password\h+([\n\r]+)\h+pam_unix\.so\h+([\n\r]+)\h+)?(SHA512|YESCRYPT)\b' found - expect '(?i)^\h*password\h+([\n\r]+)\h+pam_unix\.so\h+([\n\r]+)\h+)?(SHA512|YESCRYPT)\b' found in the following lines:
    26: password [success=1 default=ignore] pam_unix.so obscure use_authok try_first_pass
    sha512 rounds=100000
-----
PASSED - password encryption rounds
Compliant file(s):
/etc/pam.d/common-password - regex '(?i)^\h*password\h+([\n\r]+)\h+pam_unix\.so\h+([\n\r]+)\h+)?rounds=[1-9]\d{5,}\b' found - expect '(?i)^\h*password\h+([\n\r]+)\h+pam_unix\.so\h+([\n\r]+)\h+)?rounds=[1-9]\d{5,}\b' found in the following lines:
    26: password [success=1 default=ignore] pam_unix.so obscure use_authok try_first_pass
    sha512 rounds=100000
```

UBTU-24-400260 - Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one uppercase character be used.

Info

Use of a complex password helps to increase the time and resources required to compromise the password. Password complexity, or strength, is a measure of the effectiveness of a password in resisting attempts at guessing and brute-force attacks. Password complexity is one factor of several that determines how long it takes to crack a password.

The more complex the password, the greater the number of possible combinations that need to be tested before the password is compromised. Satisfies: SRG-OS-000069-GPOS-00037, SRG-OS-000730-GPOS-00190

Solution

Configure Ubuntu 24.04 LTS to enforce password complexity by requiring that at least one uppercase character be used. Add or update the "/etc/security/pwquality.conf" file to contain the "ucredit" parameter: ucredit=-1

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.7
800-171R3	03.05.07a.
800-53	IA-5(1)(a)
800-53R5	IA-5(1)(g)
800-53R5	IA-5(1)(h)
CAT	II
CCI	CCI-004065
CCI	CCI-004066
CN-L3	7.1.2.7(e)
CN-L3	7.1.3.1(b)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ISO/IEC-27001	A.9.4.3
ITSG-33	IA-5(1)(a)

NESA	T5.2.3
NIAV2	AM19a
NIAV2	AM19b
NIAV2	AM19c
NIAV2	AM19d
NIAV2	AM22a
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270726r1066667_rule
STIG-ID	UBTU-24-400260
SWIFT-CSCV1	4.1
TBA-FIISB	26.2.1
TBA-FIISB	26.2.4
VULN-ID	V-270726

Assets

strkbsubuntu9

```
Compliant file(s):
  /etc/security/pwquality.conf - regex '^[\\s]*ucredit' found - expect
  '^[\\s]*ucredit[\\s]*=[\\s]*-1[\\s]*$' found in the following lines:
    82: ucredit = -1
```

UBTU-24-400270 - Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one lowercase character be used.

Info

Use of a complex password helps to increase the time and resources required to compromise the password. Password complexity, or strength, is a measure of the effectiveness of a password in resisting attempts at guessing and brute-force attacks. Password complexity is one factor of several that determines how long it takes to crack a password.

The more complex the password, the greater the number of possible combinations that need to be tested before the password is compromised. Satisfies: SRG-OS-000070-GPOS-00038, SRG-OS-000730-GPOS-00190

Solution

Configure Ubuntu 24.04 LTS to enforce password complexity by requiring that at least one lowercase character be used. Add or update the "/etc/security/pwquality.conf" file to contain the "lcredit" parameter: lcredit=-1

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.7
800-171R3	03.05.07a.
800-53	IA-5(1)(a)
800-53R5	IA-5(1)(g)
800-53R5	IA-5(1)(h)
CAT	II
CCI	CCI-004065
CCI	CCI-004066
CN-L3	7.1.2.7(e)
CN-L3	7.1.3.1(b)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ISO/IEC-27001	A.9.4.3
ITSG-33	IA-5(1)(a)

NESA	T5.2.3
NIAV2	AM19a
NIAV2	AM19b
NIAV2	AM19c
NIAV2	AM19d
NIAV2	AM22a
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270727r1066670_rule
STIG-ID	UBTU-24-400270
SWIFT-CSCV1	4.1
TBA-FIISB	26.2.1
TBA-FIISB	26.2.4
VULN-ID	V-270727

Assets

strkbsubuntu9

Compliant file(s):
 /etc/security/pwquality.conf - regex '^[\\s]*lcredit' found - expect
 '^[\\s]*lcredit[\\s]*=[\\s]*-1[\\s]*\$' found in the following lines:
 83: lcredit = -1

UBTU-24-400280 - Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one numeric character be used.

Info

Use of a complex password helps to increase the time and resources required to compromise the password. Password complexity, or strength, is a measure of the effectiveness of a password in resisting attempts at guessing and brute-force attacks. Password complexity is one factor of several that determines how long it takes to crack a password.

The more complex the password, the greater the number of possible combinations that need to be tested before the password is compromised. Satisfies: SRG-OS-000071-GPOS-00039, SRG-OS-000730-GPOS-00190

Solution

Configure Ubuntu 24.04 LTS to enforce password complexity by requiring that at least one numeric character be used. Add or update the "/etc/security/pwquality.conf" file to contain the "dcredit" parameter: dcredit=-1

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.7
800-171R3	03.05.07a.
800-53	IA-5(1)(a)
800-53R5	IA-5(1)(g)
800-53R5	IA-5(1)(h)
CAT	II
CCI	CCI-004065
CCI	CCI-004066
CN-L3	7.1.2.7(e)
CN-L3	7.1.3.1(b)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ISO/IEC-27001	A.9.4.3
ITSG-33	IA-5(1)(a)

NESA	T5.2.3
NIAV2	AM19a
NIAV2	AM19b
NIAV2	AM19c
NIAV2	AM19d
NIAV2	AM22a
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270728r1066673_rule
STIG-ID	UBTU-24-400280
SWIFT-CSCV1	4.1
TBA-FIISB	26.2.1
TBA-FIISB	26.2.4
VULN-ID	V-270728

Assets

strkbsubuntu9

Compliant file(s):
 /etc/security/pwquality.conf - regex '^[\\s]*dcredit' found - expect
 '^[\\s]*dcredit[\\s]*=[\\s]*-1[\\s]*\$' found in the following lines:
 84: dcredit = -1

UBTU-24-400290 - Ubuntu 24.04 LTS must require the change of at least eight characters when passwords are changed.

Info

If Ubuntu 24.04 LTS allows the user to consecutively reuse extensive portions of passwords, this increases the chances of password compromise by increasing the window of opportunity for attempts at guessing and brute-force attacks. The number of changed characters refers to the number of changes required with respect to the total number of positions in the current password.

In other words, characters may be the same within the two passwords; however, the positions of the like characters must be different. If the password length is an odd number, then number of changed characters must be rounded up.

For example, a password length of 15 characters must require the change of at least eight characters. Satisfies: SRG-OS-000072-GPOS-00040, SRG-OS-000730-GPOS-00190

Solution

Configure Ubuntu 24.04 LTS to require the change of at least eight characters when passwords are changed. Add or update the "/etc/security/pwquality.conf" file to include the "difok=8" parameter: difok=8

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.7
800-171R3	03.05.07a.
800-53	IA-5(1)(a)
800-53R5	IA-5(1)(g)
800-53R5	IA-5(1)(h)
CAT	II
CCI	CCI-004065
CCI	CCI-004066
CN-L3	7.1.2.7(e)
CN-L3	7.1.3.1(b)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17

ISO/IEC-27001	A.9.4.3
ITSG-33	IA-5(1)(a)
NESA	T5.2.3
NIAV2	AM19a
NIAV2	AM19b
NIAV2	AM19c
NIAV2	AM19d
NIAV2	AM22a
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270729r1066676_rule
STIG-ID	UBTU-24-400290
SWIFT-CSCV1	4.1
TBA-FIISB	26.2.1
TBA-FIISB	26.2.4
VULN-ID	V-270729

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/security/pwquality.conf - regex '^[\\s]*difok' found - expect
'^[\\s]*difok[\\s]*=[\\s]*([8-9]|[1-9][0-9]+)[\\s]*$' found in the following lines:
85: difok = 8
```

UBTU-24-400300 - Ubuntu 24.04 LTS must enforce 24 hours/1 day as the minimum password lifetime. Passwords for new users must have a 24 hours/1 day minimum password lifetime restriction.

Info

Enforcing a minimum password lifetime helps to prevent repeated password changes to defeat the password reuse or history enforcement requirement.

If users are allowed to immediately and continually change their password, then the password could be repeatedly changed in a short period of time to defeat the organization's policy regarding password reuse.

Satisfies: SRG-OS-000075-GPOS-00043, SRG-OS-000730-GPOS-00190

Solution

Configure Ubuntu 24.04 LTS to enforce a 24 hours/1 day minimum password lifetime. Add or modify the following line in the "/etc/login.defs" file: PASS_MIN_DAYS 1

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.7
800-171R3	03.05.07a.
800-53	IA-5(1)(a)
800-53R5	IA-5(1)(g)
800-53R5	IA-5(1)(h)
CAT	II
CCI	CCI-004065
CCI	CCI-004066
CN-L3	7.1.2.7(e)
CN-L3	7.1.3.1(b)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ISO/IEC-27001	A.9.4.3
ITSG-33	IA-5(1)(a)

NESA	T5.2.3
NIAV2	AM19a
NIAV2	AM19b
NIAV2	AM19c
NIAV2	AM19d
NIAV2	AM22a
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270730r1066679_rule
STIG-ID	UBTU-24-400300
SWIFT-CSCV1	4.1
TBA-FIISB	26.2.1
TBA-FIISB	26.2.4
VULN-ID	V-270730

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/login.defs - regex '^[\s]*PASS_MIN_DAYS[\s]+' found - expect
'^[\s]*PASS_MIN_DAYS[\s]+([1-9]|[1-9][0-9]+)' found in the following lines:
166: PASS_MIN_DAYS 1
```

UBTU-24-400310 - Ubuntu 24.04 LTS must enforce a 60-day maximum password lifetime restriction. Passwords for new users must have a 60-day maximum password lifetime restriction.

Info

Any password, no matter how complex, can eventually be cracked.

Therefore, passwords need to be changed periodically.

If Ubuntu 24.04 LTS does not limit the lifetime of passwords and force users to change their passwords, there is the risk that Ubuntu 24.04 LTS passwords could be compromised. Satisfies: SRG-OS-000076-GPOS-00044, SRG-OS-000730-GPOS-00190

Solution

Configure Ubuntu 24.04 LTS to enforce a 60-day maximum password lifetime. Add or modify the following line in the "/etc/login.defs" file: PASS_MAX_DAYS 60

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.7
800-171R3	03.05.07a.
800-53	IA-5(1)(a)
800-53R5	IA-5(1)(g)
800-53R5	IA-5(1)(h)
CAT	II
CCI	CCI-004065
CCI	CCI-004066
CN-L3	7.1.2.7(e)
CN-L3	7.1.3.1(b)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ISO/IEC-27001	A.9.4.3
ITSG-33	IA-5(1)(a)

NESA	T5.2.3
NIAV2	AM19a
NIAV2	AM19b
NIAV2	AM19c
NIAV2	AM19d
NIAV2	AM22a
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270731r1066682_rule
STIG-ID	UBTU-24-400310
SWIFT-CSCV1	4.1
TBA-FIISB	26.2.1
TBA-FIISB	26.2.4
VULN-ID	V-270731

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/login.defs - regex '^[\\s]*PASS_MAX_DAYS[\\s]+' found - expect
'^[\\s]*PASS_MAX_DAYS[\\s]+([1-9]|[1-5][0-9]|60)[\\s]*$' found in the following lines:
165: PASS_MAX_DAYS 60
```

UBTU-24-400320 - Ubuntu 24.04 LTS must enforce a minimum 15-character password length.

Info

The shorter the password, the lower the number of possible combinations that need to be tested before the password is compromised. Password complexity, or strength, is a measure of the effectiveness of a password in resisting attempts at guessing and brute-force attacks.

Password length is one factor of several that helps to determine strength and how long it takes to crack a password.

Use of more characters in a password helps to exponentially increase the time and/or resources required to compromise the password. Satisfies: SRG-OS-000078-GPOS-00046, SRG-OS-000730-GPOS-00190

Solution

Configure Ubuntu 24.04 LTS to enforce a minimum 15-character password length. Add or modify the "minlen" parameter value to the "/etc/security/pwquality.conf" file: minlen=15

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.7
800-171R3	03.05.07a.
800-53	IA-5(1)(a)
800-53R5	IA-5(1)(g)
800-53R5	IA-5(1)(h)
CAT	II
CCI	CCI-004065
CCI	CCI-004066
CN-L3	7.1.2.7(e)
CN-L3	7.1.3.1(b)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ISO/IEC-27001	A.9.4.3
ITSG-33	IA-5(1)(a)

NESA	T5.2.3
NIAV2	AM19a
NIAV2	AM19b
NIAV2	AM19c
NIAV2	AM19d
NIAV2	AM22a
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270732r1066685_rule
STIG-ID	UBTU-24-400320
SWIFT-CSCV1	4.1
TBA-FIISB	26.2.1
TBA-FIISB	26.2.4
VULN-ID	V-270732

Assets

strkbsubuntu9

```
Compliant file(s):
    /etc/security/pwquality.conf - regex '^[\\s]*minlen' found - expect
    '^[\\s]*minlen[\\s]*=[\\s]*(1[5-9]|[2-9][0-9]+)[\\s]*$' found in the following lines:
        11: minlen = 15
```

UBTU-24-400330 - Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one special character be used.

Info

Use of a complex password helps to increase the time and resources required to compromise the password. Password complexity or strength is a measure of the effectiveness of a password in resisting attempts at guessing and brute-force attacks. Password complexity is one factor in determining how long it takes to crack a password.

The more complex the password, the greater the number of possible combinations that need to be tested before the password is compromised. Special characters are those characters that are not alphanumeric.

Examples include: ~ ! @ # \$ % ^ *. Satisfies: SRG-OS-000266-GPOS-00101, SRG-OS-000730-GPOS-00190

Solution

Configure Ubuntu 24.04 LTS to enforce password complexity by requiring that at least one special character be used. Add or update the following line in the "/etc/security/pwquality.conf" file to include the "ocredit=-1" parameter: ocredit=-1

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.5.7
800-171R3	03.05.07a.
800-53	IA-5(1)(a)
800-53R5	IA-5(1)(g)
800-53R5	IA-5(1)(h)
CAT	II
CCI	CCI-004065
CCI	CCI-004066
CN-L3	7.1.2.7(e)
CN-L3	7.1.3.1(b)
CSF	PR.AC-1
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(a)(2)(i)
HIPAA	164.312(d)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.17
ISO/IEC-27001	A.9.4.3

ITSG-33	IA-5(1)(a)
NESA	T5.2.3
NIAV2	AM19a
NIAV2	AM19b
NIAV2	AM19c
NIAV2	AM19d
NIAV2	AM22a
QCSC-V1	5.2.2
QCSC-V1	13.2
RULE-ID	SV-270733r1066688_rule
STIG-ID	UBTU-24-400330
SWIFT-CSCV1	4.1
TBA-FIISB	26.2.1
TBA-FIISB	26.2.4
VULN-ID	V-270733

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/security/pwquality.conf - regex '^[\\s]*ocredit' found - expect
'^[\\s]*ocredit[\\s]*=[\\s]*-1[\\s]*$' found in the following lines:
30: ocredit = -1
```

UBTU-24-400400 - Ubuntu 24.04 LTS must encrypt all stored passwords with a FIPS 140-3 approved cryptographic hashing algorithm.

Info

Passwords need to be protected at all times, and encryption is the standard method for protecting passwords. If passwords are not encrypted, they can be plainly read (i.e., clear text) and easily compromised.

Solution

Configure Ubuntu 24.04 LTS to encrypt all stored passwords. Edit/modify the following line in the "/etc/login.defs" file and set "ENCRYPT_METHOD" to SHA512: ENCRYPT_METHOD SHA512

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	IA-7
800-53R5	IA-7
CAT	II
CCI	CCI-000803
CSF2.0	PR.AA-01
CSF2.0	PR.AA-03
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(d)
ITSG-33	IA-7
ITSG-33	IA-7a.
NESA	M5.2.1
NESA	M5.2.6
NESA	M5.3.1
NESA	T7.4.1
QCSC-V1	13.2
RULE-ID	SV-270739r1067124_rule
STIG-ID	UBTU-24-400400
VULN-ID	V-270739

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/login.defs - regex '^[\s]*ENCRYPT_METHOD' found - expect
'^[\s]*ENCRYPT_METHOD[\s]+(SHA512|YESCRYPT)[\s]*$' found in the following lines:
295: ENCRYPT_METHOD SHA512
```

UBTU-24-500010 - Ubuntu 24.04 LTS must generate audit records for privileged activities, nonlocal maintenance, diagnostic sessions, and other system-level access.

Info

If events associated with nonlocal administrative access or diagnostic sessions are not logged, a major tool for assessing and investigating attacks would not be available. This requirement addresses auditing-related issues associated with maintenance tools used specifically for diagnostic and repair actions on organizational information systems. Nonlocal maintenance and diagnostic activities are those activities conducted by individuals communicating through a network, either an external network (e.g., the internet) or an internal network.

Local maintenance and diagnostic activities are those activities carried out by individuals physically present at the information system or information system component and not communicating across a network connection. This requirement applies to hardware/software diagnostic test equipment or tools.

This requirement does not cover hardware/software components that may support information system maintenance, yet are a part of the system, for example, the software implementing "ping," "ls," "ipconfig," or the hardware and software implementing the monitoring port of an Ethernet switch. Satisfies: SRG-OS-000392-GPOS-00172, SRG-OS-000471-GPOS-00215

Solution

Configure Ubuntu 24.04 LTS to audit activities performed during nonlocal maintenance and diagnostic sessions. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -w /var/log/sudo.log -p wa -k maintenance To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171	3.7.5
800-171R3	03.03.03a.
800-171R3	03.07.05
800-53	AU-12c.
800-53	MA-4(1)(a)
800-53R5	AU-12c.
800-53R5	MA-4(1)(a)
CAT	II
CCI	CCI-000172
CCI	CCI-002884
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)

CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.MA-2
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO-27001-2022	A.8.16
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
ITSG-33	MA-4(1)
NESA	T2.3.4
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	5.2.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270740r1066709_rule
STIG-ID	UBTU-24-500010

SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
TBA-FIISB	45.2.3
VULN-ID	V-270740

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep 'sudo.log'' returned :

```
-w /var/log/sudo.log -p wa -k maintenance
```

UBTU-24-500050 - Ubuntu 24.04 LTS must use strong authenticators in establishing nonlocal maintenance and diagnostic sessions.

Info

Nonlocal maintenance and diagnostic activities are those activities conducted by individuals communicating through a network, either an external network (e.g., the internet) or an internal network.

Local maintenance and diagnostic activities are those activities carried out by individuals physically present at the information system or information system component and not communicating across a network connection.

Typically, strong authentication requires authenticators that are resistant to replay attacks and employ multifactor authentication.

Strong authenticators include, for example, PKI where certificates are stored on a token protected by a password, passphrase, or biometric.

Solution

Configure Ubuntu 24.04 LTS to use strong authentication when establishing nonlocal maintenance and diagnostic sessions. Add or modify the following line to `/etc/ssh/sshd_config`: `UsePAM yes`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.7.5
800-171R3	03.07.05b.
800-53	MA-4c.
800-53R5	MA-4c.
CAT	II
CCI	CCI-000877
CSF	PR.MA-2
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	MA-4c.
NESA	T2.3.4
NESA	T5.4.4
QCSC-V1	5.2.2
RULE-ID	SV-270741r1066712_rule
STIG-ID	UBTU-24-500050
TBA-FIISB	45.2.3
VULN-ID	V-270741

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/ssh/sshd_config - regex '(?i)^\s*UsePAM\s+' found - expect '(?i)^\s*UsePAM\s+\"?
yes\"?\s*$' found in the following lines:
94: UsePAM yes
```



```
/etc/ssh/sshd_config.d/50-cloud-init.conf - regex not found  
/etc/ssh/sshd_config.d/50-cloudimg-settings.conf - regex not found  
/etc/ssh/sshd_config.d/60-cloudimg-settings.conf - regex not found
```

UBTU-24-600000 - Ubuntu 24.04 LTS must immediately terminate all network connections associated with SSH traffic after a period of inactivity.

Info

Automatic session termination addresses the termination of user-initiated logical sessions in contrast to the termination of network connections that are associated with communications sessions (i.e., network disconnect).

A logical session (for local, network, and remote access) is initiated whenever a user (or process acting on behalf of a user) accesses an organizational information system.

Such user sessions can be terminated (and thus terminate user access) without terminating network sessions. Session termination terminates all processes associated with a user's logical session except those processes that are specifically created by the user (i.e., session owner) to continue after the session is terminated.

Conditions or trigger events requiring automatic session termination can include, for example, organization-defined periods of user inactivity, targeted responses to certain types of incidents, and time-of-day restrictions on information system use. This capability is typically reserved for specific Ubuntu 24.04 LTS functionality where the system owner, data owner, or organization requires additional assurance.

Solution

Configure Ubuntu 24.04 LTS to automatically terminate inactive SSH sessions after a period of inactivity. Modify or append the following line in the "/etc/ssh/sshd_config" file, replacing "[Count]" with a value of 1: ClientAliveCountMax 1 Restart the SSH daemon for the changes to take effect: \$ sudo systemctl restart ssh.service

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.13.9
800-171R3	03.13.09
800-53	SC-10
800-53R5	SC-10
CAT	II
CCI	CCI-001133
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.20
ITSG-33	SC-10
ITSG-33	SC-10a.
NESA	T2.3.8
NESA	T4.5.1
NESA	T5.5.1
RULE-ID	SV-270742r1066715_rule
STIG-ID	UBTU-24-600000
SWIFT-CSCV1	2.6

Assets**strkbsubuntu9**

Compliant file(s):

```
/etc/ssh/sshd_config - regex '(?i)^\s*ClientAliveCountMax\s+' found - expect '(?  
i)^\s*ClientAliveCountMax\s+"?1"?[\s]*$' found in the following lines:  
109: ClientAliveCountMax 1  
/etc/ssh/sshd_config.d/50-cloud-init.conf - regex not found  
/etc/ssh/sshd_config.d/50-cloudimg-settings.conf - regex not found  
/etc/ssh/sshd_config.d/60-cloudimg-settings.conf - regex not found
```

UBTU-24-600010 - Ubuntu 24.04 LTS must immediately terminate all network connections associated with SSH traffic at the end of the session or after 10 minutes of inactivity.

Info

Terminating an idle session within a short time period reduces the window of opportunity for unauthorized personnel to take control of a management session enabled on the console or console port that has been left unattended.

In addition, quickly terminating an idle session will also free up resources committed by the managed network element. Terminating network connections associated with communications sessions includes, for example, de-allocating associated TCP/IP address/port pairs at Ubuntu 24.04 LTS level, and de-allocating networking assignments at the application level if multiple application sessions are using a single operating system-level network connection.

This does not mean that Ubuntu 24.04 LTS terminates all sessions or network access; it only ends the inactive session and releases the resources associated with that session.

Solution

Configure Ubuntu 24.04 LTS to automatically terminate all network connections associated with SSH traffic at the end of a session or after a 10-minute period of inactivity. In the file `/etc/ssh/sshd_config` set `ClientAliveInterval` to a value of "600" or less: `ClientAliveInterval 600` Restart the SSH daemon for the changes to take effect: `$ sudo systemctl restart sshd.service`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.13.9
800-171R3	03.13.09
800-53	SC-10
800-53R5	SC-10
CAT	II
CCI	CCI-001133
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.20
ITSG-33	SC-10
ITSG-33	SC-10a.
NESA	T2.3.8
NESA	T4.5.1
NESA	T5.5.1
RULE-ID	SV-270743r1066718_rule
STIG-ID	UBTU-24-600010
SWIFT-CSCV1	2.6
VULN-ID	V-270743

Assets

strkbsubuntu9

```
Compliant file(s):
/etc/ssh/sshd_config - regex '(?i)^\s*ClientAliveInterval\s+' found - expect '(?i)^\s*ClientAliveInterval\s+"?([1-9]|[1-9][0-9]|[1-5][0-9]{2}|600)"?\s*$' found in the following lines:
    108: ClientAliveInterval 600
/etc/ssh/sshd_config.d/50-cloud-init.conf - regex not found
/etc/ssh/sshd_config.d/50-cloudimg-settings.conf - regex '(?i)^\s*ClientAliveInterval\s+' found - expect '(?i)^\s*ClientAliveInterval\s+"?([1-9]|[1-9][0-9]|[1-5][0-9]{2}|600)"?\s*$' found in the following lines:
    4: ClientAliveInterval 120
/etc/ssh/sshd_config.d/60-cloudimg-settings.conf - regex not found
```

UBTU-24-600140 - Ubuntu 24.04 LTS must restrict access to the kernel message buffer.

Info

Restricting access to the kernel message buffer limits access only to root. This prevents attackers from gaining additional system information as a nonprivileged user.

Solution

Configure Ubuntu 24.04 LTS to restrict access to the kernel message buffer. Set the system to the required kernel parameter by adding or modifying the following line in `/etc/sysctl.conf` or a config file in the `/etc/sysctl.d/` directory: `kernel.dmesg_restrict = 1` Remove any configurations that conflict with the above from the following locations: `/run/sysctl.d/ /etc/sysctl.d/ /usr/local/lib/sysctl.d/ /usr/lib/sysctl.d/ /lib/sysctl.d/ /etc/sysctl.conf` Reload settings from all system configuration files with the following command: `$ sudo sysctl --system`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.13.4
800-171R3	03.13.04
800-53	SC-4
800-53R5	SC-4
CAT	III
CCI	CCI-001090
CSF2.0	PR.DS-01
CSF2.0	PR.DS-02
CSF2.0	PR.DS-10
CSF2.0	PR.IR-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SC-4
ITSG-33	SC-4a.
RULE-ID	SV-270749r1137695_rule
STIG-ID	UBTU-24-600140
VULN-ID	V-270749

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
PASSED - sysctl
The command '/sbin/sysctl kernel.dmesg_restrict' returned :
kernel.dmesg_restrict = 1
```

```

-----
PASSED - kernel.dmesg_restrict conf files
Compliant file(s):
/etc/sysctl.conf - regex not found
/etc/sysctl.d/10-bufferbloat.conf - regex not found
/etc/sysctl.d/10-console-messages.conf - regex not found
/etc/sysctl.d/10-ipv6-privacy.conf - regex not found
/etc/sysctl.d/10-kernel-hardening.conf - regex not found
/etc/sysctl.d/10-magic-sysrq.conf - regex not found
/etc/sysctl.d/10-map-count.conf - regex not found
/etc/sysctl.d/10-network-security.conf - regex not found
/etc/sysctl.d/10-ptrace.conf - regex not found
/etc/sysctl.d/10-zero-page.conf - regex not found
/etc/sysctl.d/99-cloudimg-ipv6.conf - regex not found
/etc/sysctl.d/99-sysctl.conf - regex not found
/etc/sysctl.d/README.sysctl - regex not found
/etc/sysctl.d/stig_kernel.conf - regex '^[\s]*kernel.dmesg_restrict[\s]*=' found - expect
'^[\s]*kernel.dmesg_restrict[\s]*=[\s]*1[\s]*$' found in the following lines:
    3: kernel.dmesg_restrict=1
/lib/sysctl.d/10-apparmor.conf - regex not found
/lib/sysctl.d/50-pid-max.conf - regex not found
/lib/sysctl.d/99-protect-links.conf - regex not found
/usr/lib/sysctl.d/10-apparmor.conf - regex not found
/usr/lib/sysctl.d/50-pid-max.conf - regex not found
/usr/lib/sysctl.d/99-protect-links.conf - regex not found

```

UBTU-24-600150 - Ubuntu 24.04 LTS must set a sticky bit on all public directories to prevent unauthorized and unintended information transferred via shared system resources.

Info

Preventing unauthorized information transfers mitigates the risk of information, including encrypted representations of information, produced by the actions of prior users/roles (or the actions of processes acting on behalf of prior users/roles) from being available to any current users/roles (or current processes) that obtain access to shared system resources (e.g., registers, main memory, hard disks) after those resources have been released back to information systems.

The control of information in shared resources is also commonly referred to as object reuse and residual information protection. This requirement generally applies to the design of an information technology product, but it can also apply to the configuration of particular information system components that are, or use, such products.

This can be verified by acceptance/validation processes in DOD or other government agencies. There may be shared resources with configurable protections (e.g., files in storage) that may be assessed on specific information system components.

Solution

Configure all public directories to have the sticky bit set to prevent unauthorized and unintended information transferred via shared system resources. Set the sticky bit on all public directories using the following command, replacing "[Public Directory]" with any directory path missing the sticky bit: `$ sudo chmod +t [Public Directory]`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.13.4
800-171R3	03.13.04
800-53	SC-4
800-53R5	SC-4
CAT	II
CCI	CCI-001090
CSF2.0	PR.DS-01
CSF2.0	PR.DS-02
CSF2.0	PR.DS-10
CSF2.0	PR.IR-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SC-4
ITSG-33	SC-4a.
RULE-ID	SV-270750r1137695_rule
STIG-ID	UBTU-24-600150
VULN-ID	V-270750

Assets

strkbsubuntu9

No issues found.

UBTU-24-600180 - Ubuntu 24.04 LTS must synchronize internal information system clocks to the authoritative time source when the time difference is greater than one second.

Info

Inaccurate time stamps make it more difficult to correlate events and can lead to an inaccurate analysis. Determining the correct time a particular event occurred on a system is critical when conducting forensic analysis and investigating system events. Synchronizing internal information system clocks provides uniformity of time stamps for information systems with multiple system clocks and systems connected over a network. Organizations must consider setting time periods for different types of systems (e.g., financial, legal, or mission-critical systems). Organizations must also consider endpoints that may not have regular access to the authoritative time server (e.g., mobile, teleworking, and tactical endpoints). This requirement is related to the comparison done every 24 hours in SRG-OS-000355 because a comparison must be done to determine the time difference.

Solution

Configure chrony to synchronize the internal system clocks to the authoritative source when the time difference is greater than one second by doing the following: Edit the "/etc/chrony/chrony.conf" file and add: makestep 1 -1 Restart the chrony service: \$ sudo systemctl restart chrony.service

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.7
800-171R3	03.03.07
800-53	AU-8(1)
800-53R5	SC-45(1)(b)
CAT	III
CCI	CCI-004926
CN-L3	8.1.4.3(b)
CSF	PR.PT-1
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.17
ISO/IEC-27001	A.12.4.4
ITSG-33	AU-8(1)
NESA	T3.6.7
NIAV2	NS44
NIAV2	NS45
NIAV2	NS46
NIAV2	NS47

PCI-DSSV3.2.1	10.4
PCI-DSSV3.2.1	10.4.1
PCI-DSSV3.2.1	10.4.3
PCI-DSSV4.0	10.6
PCI-DSSV4.0	10.6.1
PCI-DSSV4.0	10.6.2
PCI-DSSV4.0	10.6.3
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270752r1068365_rule
STIG-ID	UBTU-24-600180
TBA-FIISB	37.4
VULN-ID	V-270752

Assets

strkbsubuntu9

```
Compliant file(s):
    /etc/chrony/chrony.conf - regex '(?i)^\s*makestep\s+1\s+' found - expect '(?i)^\s*makestep\s+1\s+-1\s*$' found in the following lines:
        58: makestep 1 -1
```

UBTU-24-600190 - Ubuntu 24.04 LTS must be configured to use TCP syncookies.

Info

Denial of service (DoS) occurs when a resource is not available for legitimate users, resulting in the organization not being able to accomplish its mission or causing it to operate at degraded capacity. Managing excess capacity ensures sufficient capacity is available to counter flooding attacks. Employing increased capacity and service redundancy may reduce the susceptibility to some DoS attacks. Managing excess capacity may include, for example, establishing selected usage priorities, quotas, or partitioning.

Solution

Configure Ubuntu 24.04 LTS to use TCP syncookies with the following command: `$ sudo sysctl -w net.ipv4.tcp_syncookies=1` If "1" is not the system's default value, add or update the following line in `/etc/sysctl.conf`: `net.ipv4.tcp_syncookies = 1`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SC-5(2)
800-53R5	SC-5(2)
CAT	II
CCI	CCI-001095
CSF	PR.DS-4
CSF2.0	DE.CM-01
CSF2.0	PR.IR-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.6
ITSG-33	SC-5(2)
NESA	T3.3.1
NIAV2	GS8e
NIAV2	GS10c
QCSC-V1	8.2.1
RULE-ID	SV-270753r1066748_rule
STIG-ID	UBTU-24-600190
SWIFT-CSCV1	6.4
VULN-ID	V-270753

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - 1.54 UBTU-22-253010

Compliant file(s):

- /etc/sysctl.conf - regex not found
- /etc/sysctl.d/10-bufferbloat.conf - regex not found
- /etc/sysctl.d/10-console-messages.conf - regex not found
- /etc/sysctl.d/10-ipv6-privacy.conf - regex not found
- /etc/sysctl.d/10-kernel-hardening.conf - regex not found
- /etc/sysctl.d/10-magic-sysrq.conf - regex not found
- /etc/sysctl.d/10-map-count.conf - regex not found
- /etc/sysctl.d/10-network-security.conf - regex not found
- /etc/sysctl.d/10-ptrace.conf - regex not found
- /etc/sysctl.d/10-zero-page.conf - regex not found
- /etc/sysctl.d/99-cloudimg-ipv6.conf - regex not found
- /etc/sysctl.d/99-sysctl.conf - regex not found
- /etc/sysctl.d/stig_kernel.conf - regex '^[\s]*net.ipv4.tcp_syncookies[\s]*=' found - expect
'^[\s]*net.ipv4.tcp_syncookies[\s]*=[\s]*1[\s]*\$' found in the following lines:
 - 1: net.ipv4.tcp_syncookies=1
- /lib/sysctl.d/10-apparmor.conf - regex not found
- /lib/sysctl.d/50-pid-max.conf - regex not found
- /lib/sysctl.d/99-protect-links.conf - regex not found
- /usr/lib/sysctl.d/10-apparmor.conf - regex not found
- /usr/lib/sysctl.d/50-pid-max.conf - regex not found
- /usr/lib/sysctl.d/99-protect-links.conf - regex not found

PASSED - 1.54 UBTU-22-253010

The command '/sbin/sysctl net.ipv4.tcp_syncookies' returned :

net.ipv4.tcp_syncookies = 1

UBTU-24-600230 - Ubuntu 24.04 LTS must disable all wireless network adapters.

Info

Without protection of communications with wireless peripherals, confidentiality and integrity may be compromised because unprotected communications can be intercepted and either read, altered, or used to compromise Ubuntu 24.04 LTS. This requirement applies to wireless peripheral technologies (e.g., wireless mice, keyboards, displays, etc.) used with an operating system.

Wireless peripherals (e.g., Wi-Fi/Bluetooth/IR Keyboards, Mice, and Pointing Devices and Near Field Communications [NFC]) present a unique challenge by creating an open, unsecured port on a computer. Wireless peripherals must meet DOD requirements for wireless data transmission and be approved for use by the approving authority (AO).

Even though some wireless peripherals, such as mice and pointing devices, do not ordinarily carry information that need to be protected, modification of communications with these wireless peripherals may be used to compromise Ubuntu 24.04 LTS.

Communication paths outside the physical protection of a controlled boundary are exposed to the possibility of interception and modification. Protecting the confidentiality and integrity of communications with wireless peripherals can be accomplished by physical means (e.g., employing physical barriers to wireless radio frequencies) or by logical means (e.g., employing cryptographic techniques).

If physical means of protection are employed, then logical means (cryptography) do not have to be employed, and vice versa.

If the wireless peripheral is only passing telemetry data, encryption of the data may not be required.

Solution

List all the wireless interfaces with the following command: `$ ls -l -d /sys/class/net/*/wireless | xargs dirname | xargs basename` For each interface, configure the system to disable wireless network interfaces with the following command: `$ sudo ifdown <interface name>` For each interface listed, find their respective module with the following command: `$ basename $(readlink -f /sys/class/net/<interface name>/device/driver)` where `<interface name>` must be substituted by the actual interface name. Create a file in the `/etc/modprobe.d` directory and for each module, add the following line: `install <module name> /bin/true` For each module from the system, execute the following command to remove it: `$ sudo modprobe -r <module name>`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.13.8
800-171R3	03.13.08
800-53	SC-8
800-53R5	SC-8
CAT	II
CCI	CCI-002418
CN-L3	8.1.2.2(a)
CN-L3	8.1.2.2(b)
CN-L3	8.1.4.7(a)
CN-L3	8.1.4.8(a)
CN-L3	8.2.4.5(c)
CN-L3	8.2.4.5(d)
CN-L3	8.5.2.2
CSF	PR.DS-2

CSF	PR.DS-5
CSF2.0	PR.DS-02
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.a
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(e)(1)
HIPAA	164.312(e)(2)(i)
ISO-27001-2022	A.5.10
ISO-27001-2022	A.5.14
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.20
ITSG-33	SC-8
ITSG-33	SC-8a.
NESA	T4.3.1
NESA	T4.3.2
NESA	T4.5.1
NESA	T4.5.2
NESA	T7.3.3
NESA	T7.4.1
NIAV2	IE8
NIAV2	IE9
NIAV2	IE12
NIAV2	NS29
NIAV2	SS24
PCI-DSSV3.2.1	2.3
PCI-DSSV3.2.1	4.1
PCI-DSSV4.0	2.2.7
PCI-DSSV4.0	4.2.1
QCSC-V1	5.2.2
QCSC-V1	6.2

RULE-ID	SV-270755r1066754_rule
STIG-ID	UBTU-24-600230
VULN-ID	V-270755

Assets

strkbsubuntu9

The command '/bin/ls -L -d /sys/class/net/*/wireless 2>/dev/null | /bin/awk '{print} END {if (NR == 0) print "No wireless device files found"; else print "fail"}}' returned :

No wireless device files found

UBTU-24-700010 - Ubuntu 24.04 LTS must generate error messages that provide information necessary for corrective actions without revealing information that could be exploited by adversaries.

Info

Any operating system providing too much information in error messages risks compromising the data and security of the structure, and organizations must carefully consider the content and structure of error messages. The extent to which information systems are able to identify and handle error conditions is guided by organizational policy and operational requirements.

Information that could be exploited by adversaries includes, for example, erroneous logon attempts with passwords entered by mistake as the username, mission/business information that can be derived from (if not stated explicitly by) information recorded, and personal information, such as account numbers, Social Security numbers, and credit card numbers. The /var/log/btmp, /var/log/wtmp, and /var/log/lastlog files have group write and global read permissions to allow for the lastlog function to perform.

Limiting the permissions beyond this configuration will result in the failure of functions that rely on the lastlog database.

Additionally, Canonical recommends excluding the history.log and eipp.log.xz from consideration in this check.

Solution

Configure Ubuntu 24.04 LTS to set permissions of all log files under the /var/log directory to "640" or more restricted by using the following command: Note: The btmp, wtmp, lastlog, history, and eipp.log.xz files are excluded.

Refer to the Discussion for details. `$ sudo find /var/log -perm /137 ! -name '[bw]tmp' ! -name '*lastlog' -type f -exec chmod 640 '{}'`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11a.
800-53R5	SI-11a.
CAT	II
CCI	CCI-001312
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11b.
RULE-ID	SV-270756r1134814_rule
STIG-ID	UBTU-24-700010
VULN-ID	V-270756

Assets

strkbsubuntu9

The command `'/bin/find /var/log -perm /137 ! -name '[bw]tmp' ! -name '*lastlog' -type f -exec stat -c "%n %a" {} \; | /bin/awk '{print} END {if (NR == 0) print "pass"; else print "fail"}'` returned :

pass

UBTU-24-700030 - Ubuntu 24.04 LTS must be configured so that the "journalctl" command is not accessible by unauthorized users.

Info

Any operating system providing too much information in error messages risks compromising the data and security of the structure, and content of error messages must be carefully considered by the organization. Organizations carefully consider the structure/content of error messages.

The extent to which information systems are able to identify and handle error conditions is guided by organizational policy and operational requirements.

Information that could be exploited by adversaries includes, for example, erroneous logon attempts with passwords entered by mistake as the username, mission/business information that can be derived from (if not stated explicitly by) information recorded, and personal information, such as account numbers, social security numbers, and credit card numbers.

Solution

Configure journalctl to have a permission set of "740": `$ sudo chmod 740 /usr/bin/journalctl`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11a.
800-53R5	SI-11a.
CAT	II
CCI	CCI-001312
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11b.
RULE-ID	SV-270758r1066763_rule
STIG-ID	UBTU-24-700030
VULN-ID	V-270758

Assets

strkbsubuntu9

The file `/usr/bin/journalctl` with `fmode owner: root group: root mode: 0740 uid: 0 gid: 0 uneven permissions : FALSE` is compliant with the policy value

`/usr/bin/journalctl`

UBTU-24-700040 - Ubuntu 24.04 LTS must be configured so that the "journalctl" command is owned by "root".

Info

Only authorized personnel are to be made aware of errors and the details of the errors. Error messages are an indicator of an organization's operational state or can identify Ubuntu 24.04 LTS or platform. Additionally, Personally Identifiable Information (PII) and operational information must not be revealed through error messages to unauthorized personnel or their designated representatives. The structure and content of error messages must be carefully considered by the organization and development team. The extent to which the information system is able to identify and handle error conditions is guided by organizational policy and operational requirements.

Solution

Configure journalctl to be owned by "root": `$ sudo chown root /usr/bin/journalctl`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11b.
800-53R5	SI-11b.
CAT	II
CCI	CCI-001314
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11c.
RULE-ID	SV-270759r1068367_rule
STIG-ID	UBTU-24-700040
VULN-ID	V-270759

Assets

strkbsubuntu9

The file /usr/bin/journalctl with fmode owner: root group: root mode: 0740 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/usr/bin/journalctl

UBTU-24-700050 - Ubuntu 24.04 LTS must be configured so that the "journalctl" command is group-owned by "root".

Info

Only authorized personnel are to be made aware of errors and the details of the errors. Error messages are an indicator of an organization's operational state or can identify Ubuntu 24.04 LTS or platform. Additionally, Personally Identifiable Information (PII) and operational information must not be revealed through error messages to unauthorized personnel or their designated representatives. The structure and content of error messages must be carefully considered by the organization and development team. The extent to which the information system is able to identify and handle error conditions is guided by organizational policy and operational requirements.

Solution

Configure journalctl to be group-owned by "root": `$ sudo chown :root /usr/bin/journalctl`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11b.
800-53R5	SI-11b.
CAT	II
CCI	CCI-001314
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11c.
RULE-ID	SV-270760r1066769_rule
STIG-ID	UBTU-24-700050
VULN-ID	V-270760

Assets

strkbsubuntu9

The file /usr/bin/journalctl with fmode owner: root group: root mode: 0740 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/usr/bin/journalctl

UBTU-24-700060 - Ubuntu 24.04 LTS must configure the directories used by the system journal to be group-owned by "systemd-journal".

Info

Only authorized personnel are to be made aware of errors and the details of the errors. Error messages are an indicator of an organization's operational state or can identify Ubuntu 24.04 LTS or platform. Additionally, Personally Identifiable Information (PII) and operational information must not be revealed through error messages to unauthorized personnel or their designated representatives. The structure and content of error messages must be carefully considered by the organization and development team. The extent to which the information system is able to identify and handle error conditions is guided by organizational policy and operational requirements.

Solution

Configure the system to set the appropriate group-ownership to the directories used by the systemd journal. Create a drop-in file if it does not already exist with the following command: \$ sudo vi /etc/tmpfiles.d/zzz-systemd-stig.conf Add or modify the following lines in the "/usr/lib/tmpfiles.d/zzz-systemd-stig.conf" file: z /run/log/journal 0640 root systemd-journal - - z /var/log/journal 0640 root systemd-journal - - Note: Restart the system for these settings to take effect.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11b.
800-53R5	SI-11b.
CAT	II
CCI	CCI-001314
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11c.
RULE-ID	SV-270761r1184074_rule
STIG-ID	UBTU-24-700060
VULN-ID	V-270761

Assets

strkbsubuntu9

No results found.

UBTU-24-700070 - Ubuntu 24.04 LTS must configure the files used by the system journal to be group-owned by "systemd-journal".

Info

Only authorized personnel are to be made aware of errors and the details of the errors. Error messages are an indicator of an organization's operational state or can identify Ubuntu 24.04 LTS or platform. Additionally, Personally Identifiable Information (PII) and operational information must not be revealed through error messages to unauthorized personnel or their designated representatives. The structure and content of error messages must be carefully considered by the organization and development team. The extent to which the information system is able to identify and handle error conditions is guided by organizational policy and operational requirements.

Solution

Configure the system to set the appropriate group-ownership to the files used by the systemd journal. Create a drop-in file if it does not already exist with the following command: `$ sudo vi /etc/tmpfiles.d/zzz-systemd-stig.conf` Add or modify the following lines in the `"/usr/lib/tmpfiles.d/zzz-systemd-stig.conf"` file: `Z /run/log/journal/%m ~0640 root systemd-journal - - z /var/log/journal/%m 0640 root systemd-journal - - z /var/log/journal/%m/system.journal 0640 root systemd-journal - -` Note: Restart the system for these settings to take effect.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11b.
800-53R5	SI-11b.
CAT	II
CCI	CCI-001314
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11c.
RULE-ID	SV-270762r1184076_rule
STIG-ID	UBTU-24-700070
VULN-ID	V-270762

Assets

strkbsubuntu9

No results found.

UBTU-24-700080 - Ubuntu 24.04 LTS must configure the directories used by the system journal to be owned by "root".

Info

Only authorized personnel are to be made aware of errors and the details of the errors. Error messages are an indicator of an organization's operational state or can identify Ubuntu 24.04 LTS or platform. Additionally, Personally Identifiable Information (PII) and operational information must not be revealed through error messages to unauthorized personnel or their designated representatives. The structure and content of error messages must be carefully considered by the organization and development team. The extent to which the information system is able to identify and handle error conditions is guided by organizational policy and operational requirements.

Solution

Configure the system to set the appropriate ownership to the directories used by the systemd journal. Create a drop-in file if it does not already exist with the following command: `$ sudo vi /etc/tmpfiles.d/zzz-systemd-stig.conf` Add or modify the following lines in the `"/usr/lib/tmpfiles.d/zzz-systemd-stig.conf"` file: `z /run/log/journal 0640 root systemd-journal - - z /var/log/journal 0640 root systemd-journal - -` Note: Restart the system for these settings to take effect.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11b.
800-53R5	SI-11b.
CAT	II
CCI	CCI-001314
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11c.
RULE-ID	SV-270763r1184078_rule
STIG-ID	UBTU-24-700080
VULN-ID	V-270763

Assets

strkbsubuntu9

No results found.

UBTU-24-700090 - Ubuntu 24.04 LTS must configure the files used by the system journal to be owned by "root"

Info

Only authorized personnel are to be made aware of errors and the details of the errors. Error messages are an indicator of an organization's operational state or can identify Ubuntu 24.04 LTS or platform. Additionally, Personally Identifiable Information (PII) and operational information must not be revealed through error messages to unauthorized personnel or their designated representatives. The structure and content of error messages must be carefully considered by the organization and development team. The extent to which the information system is able to identify and handle error conditions is guided by organizational policy and operational requirements.

Solution

Configure the system to set the appropriate ownership to the files used by the systemd journal. Create a drop-in file if it does not already exist with the following command: `$ sudo vi /etc/tmpfiles.d/zzz-systemd-stig.conf` Add or modify the following lines in the `"/usr/lib/tmpfiles.d/zzz-systemd-stig.conf"` file: `Z /run/log/journal/%m ~0640 root systemd-journal - - z /var/log/journal/%m 0640 root systemd-journal - - z /var/log/journal/%m/system.journal 0640 root systemd-journal - -` Note: Restart the system for these settings to take effect.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References	
800-53	SI-11b.
800-53R5	SI-11b.
CAT	II
CCI	CCI-001314
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11c.
RULE-ID	SV-270764r1184080_rule
STIG-ID	UBTU-24-700090
VULN-ID	V-270764

Assets

strkbsubuntu9

No results found.

UBTU-24-700100 - Ubuntu 24.04 LTS must configure the /var/log directory to be group-owned by syslog.

Info

Only authorized personnel are to be made aware of errors and the details of the errors. Error messages are an indicator of an organization's operational state or can identify Ubuntu 24.04 LTS or platform. Additionally, Personally Identifiable Information (PII) and operational information must not be revealed through error messages to unauthorized personnel or their designated representatives. The structure and content of error messages must be carefully considered by the organization and development team. The extent to which the information system is able to identify and handle error conditions is guided by organizational policy and operational requirements.

Solution

Configure Ubuntu 24.04 LTS to have syslog group-own the /var/log directory with the following command: \$
sudo chgrp syslog /var/log

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11b.
800-53R5	SI-11b.
CAT	II
CCI	CCI-001314
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11c.
RULE-ID	SV-270765r1066784_rule
STIG-ID	UBTU-24-700100
VULN-ID	V-270765

Assets

strkbsubuntu9

The file /var/log with fmode owner: root group: syslog mode: 0755 uid: 0 gid: 102 uneven permissions : FALSE is compliant with the policy value

/var/log

UBTU-24-700110 - Ubuntu 24.04 LTS must configure the /var/log directory to be owned by root.

Info

Only authorized personnel are to be made aware of errors and the details of the errors. Error messages are an indicator of an organization's operational state or can identify Ubuntu 24.04 LTS or platform. Additionally, Personally Identifiable Information (PII) and operational information must not be revealed through error messages to unauthorized personnel or their designated representatives. The structure and content of error messages must be carefully considered by the organization and development team. The extent to which the information system is able to identify and handle error conditions is guided by organizational policy and operational requirements.

Solution

Configure Ubuntu 24.04 LTS to have root own the /var/log directory by running the following command: `$ sudo chown root /var/log`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11b.
800-53R5	SI-11b.
CAT	II
CCI	CCI-001314
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11c.
RULE-ID	SV-270766r1066787_rule
STIG-ID	UBTU-24-700110
VULN-ID	V-270766

Assets

strkbsubuntu9

The file /var/log with fmode owner: root group: syslog mode: 0755 uid: 0 gid: 102 uneven permissions : FALSE is compliant with the policy value

/var/log

UBTU-24-700120 - Ubuntu 24.04 LTS must configure the /var/log directory to have mode "0755" or less permissive.

Info

Only authorized personnel are to be made aware of errors and the details of the errors. Error messages are an indicator of an organization's operational state or can identify Ubuntu 24.04 LTS or platform. Additionally, personally identifiable information (PII) and operational information must not be revealed through error messages to unauthorized personnel or their designated representatives. The structure and content of error messages must be carefully considered by the organization and development team. The extent to which the information system is able to identify and handle error conditions is guided by organizational policy and operational requirements.

Solution

Configure Ubuntu 24.04 LTS to have permissions of "0755" for the /var/log directory by running the following command: `$ sudo chmod 0755 /var/log`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11b.
800-53R5	SI-11b.
CAT	II
CCI	CCI-001314
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11c.
RULE-ID	SV-270767r1066790_rule
STIG-ID	UBTU-24-700120
VULN-ID	V-270767

Assets

strkbsubuntu9

The file /var/log with fmode owner: root group: syslog mode: 0755 uid: 0 gid: 102 uneven permissions : FALSE is compliant with the policy value

/var/log

UBTU-24-700130 - Ubuntu 24.04 LTS must configure the /var/log/syslog file to be group-owned by adm.

Info

Only authorized personnel are to be made aware of errors and the details of the errors. Error messages are an indicator of an organization's operational state or can identify Ubuntu 24.04 LTS or platform. Additionally, Personally Identifiable Information (PII) and operational information must not be revealed through error messages to unauthorized personnel or their designated representatives. The structure and content of error messages must be carefully considered by the organization and development team. The extent to which the information system is able to identify and handle error conditions is guided by organizational policy and operational requirements.

Solution

Configure Ubuntu 24.04 LTS to have adm group-own the /var/log/syslog file by running the following command: `$ sudo chgrp adm /var/log/syslog`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11b.
800-53R5	SI-11b.
CAT	II
CCI	CCI-001314
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11c.
RULE-ID	SV-270768r1066793_rule
STIG-ID	UBTU-24-700130
VULN-ID	V-270768

Assets

strkbsubuntu9

The file /var/log/syslog with fmode owner: syslog group: adm mode: 0640 uid: 102 gid: 4 uneven permissions : FALSE is compliant with the policy value

/var/log/syslog

UBTU-24-700140 - Ubuntu 24.04 LTS must configure /var/log/syslog file to be owned by syslog.

Info

Only authorized personnel are to be made aware of errors and the details of the errors. Error messages are an indicator of an organization's operational state or can identify Ubuntu 24.04 LTS or platform. Additionally, Personally Identifiable Information (PII) and operational information must not be revealed through error messages to unauthorized personnel or their designated representatives. The structure and content of error messages must be carefully considered by the organization and development team. The extent to which the information system is able to identify and handle error conditions is guided by organizational policy and operational requirements.

Solution

Configure Ubuntu 24.04 LTS to have syslog own the /var/log/syslog file by running the following command: \$ sudo chown syslog /var/log/syslog

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11b.
800-53R5	SI-11b.
CAT	II
CCI	CCI-001314
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11c.
RULE-ID	SV-270769r1066796_rule
STIG-ID	UBTU-24-700140
VULN-ID	V-270769

Assets

strkbsubuntu9

The file /var/log/syslog with fmode owner: syslog group: adm mode: 0640 uid: 102 gid: 4 uneven permissions : FALSE is compliant with the policy value

/var/log/syslog

UBTU-24-700150 - Ubuntu 24.04 LTS must configure /var/log/syslog file with mode "0640" or less permissive.

Info

Only authorized personnel are to be made aware of errors and the details of the errors. Error messages are an indicator of an organization's operational state or can identify Ubuntu 24.04 LTS or platform. Additionally, Personally Identifiable Information (PII) and operational information must not be revealed through error messages to unauthorized personnel or their designated representatives. The structure and content of error messages must be carefully considered by the organization and development team. The extent to which the information system is able to identify and handle error conditions is guided by organizational policy and operational requirements.

Solution

Configure Ubuntu 24.04 LTS to have permissions of "0640" for the /var/log/syslog file by running the following command: `$ sudo chmod 0640 /var/log/syslog`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-11b.
800-53R5	SI-11b.
CAT	II
CCI	CCI-001314
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-11c.
RULE-ID	SV-270770r1066799_rule
STIG-ID	UBTU-24-700150
VULN-ID	V-270770

Assets

strkbsubuntu9

The file /var/log/syslog with fmode owner: syslog group: adm mode: 0640 uid: 102 gid: 4 uneven permissions : FALSE is compliant with the policy value

/var/log/syslog

UBTU-24-700300 - Ubuntu 24.04 LTS must implement nonexecutable data to protect its memory from unauthorized code execution.

Info

Some adversaries launch attacks with the intent of executing code in nonexecutable regions of memory or in memory locations that are prohibited. Security safeguards employed to protect memory include, for example, data execution prevention and address space layout randomization. Data execution prevention safeguards can either be hardware-enforced or software-enforced with hardware providing the greater strength of mechanism. Examples of attacks are buffer overflow attacks.

Solution

Configure Ubuntu 24.04 LTS to enable NX. If "nx" is not showing up in "/proc/cpuinfo", and the system's BIOS setup configuration permits toggling the No Execution bit, set it to "enable".

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-16
800-53R5	SI-16
CAT	II
CCI	CCI-002824
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-16
RULE-ID	SV-270771r1066802_rule
STIG-ID	UBTU-24-700300
VULN-ID	V-270771

Assets

strkbsubuntu9

One of the following must pass to satisfy this requirement:

PASSED - dmesg execute disable

The command '/bin/dmesg | /bin/grep -i "execute disable"' returned :

```
[ 0.000000] NX (Execute Disable) protection: active
```

PASSED - cpuinfo flags nx

Compliant file(s):

/proc/cpuinfo - regex '^[\s]*flags[\s]*:' found - expect '^[\s]*flags[\s]*:.*[\s]nx([\s]|\$)' found in the following lines:

```
20: flags      : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36
clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good
nopl tsc_reliable nonstop_tsc cpuid extd_apicid aperfmperf pni pclmulqdq ssse3 fma cx16 pcid
sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy
abm sse4a misalignsse 3dnowprefetch osvw topoext vmcall fsgsbase bmi1 avx2 smep bmi2 erms
invpcid avx512f avx512dq rdseed adx smap avx512ifma clflushopt clwb avx512cd sha_ni avx512bw
avx512vl xsaveopt xsavec xgetbv1 xsaves user_shstk avx512_bf16 clzero xsaveerptr rdpru arat npt
nrip_save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthreshold v_vmsave_vmload
```

avx512vbmi umip avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg avx512_vpopcntdq
rdpid fsrm

48: flags : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36
clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good
nopl tsc_reliable nonstop_tsc cpuid extd_apicid aperfmperf pni pclmulqdq ssse3 fma cx16 pcid
sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy
abm sse4a misalignsse 3dnowprefetch osvw topoext vmcall fsgsbase bmi1 avx2 smep bmi2 erms
invpcid avx512f avx512dq rdseed adx smap avx512ifma clflushopt clwb avx512cd sha_ni avx512bw
avx512vl xsaveopt xsavec xgetbv1 xsaves user_shstk avx512_bf16 clzero xsaveerptr rdpru arat npt
nrip_save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthreshold v_vmsave_vmload
avx512vbmi umip avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg avx512_vpopcntdq
rdpid fsrm

UBTU-24-700310 - Ubuntu 24.04 LTS must implement address space layout randomization to protect its memory from unauthorized code execution.

Info

Some adversaries launch attacks with the intent of executing code in nonexecutable regions of memory or in memory locations that are prohibited. Security safeguards employed to protect memory include, for example, data execution prevention and address space layout randomization. Data execution prevention safeguards can either be hardware-enforced or software-enforced with hardware providing the greater strength of mechanism. Examples of attacks are buffer overflow attacks.

Solution

Remove the "kernel.randomize_va_space" entry found in the "/etc/sysctl.conf" file or any file located in the "/etc/sysctl.d/" directory. After the line has been removed, the kernel settings from all system configuration files must be reloaded before any of the changes will take effect.

Run the following command to reload all of the kernel system configuration files: `$ sudo sysctl --system`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-16
800-53R5	SI-16
CAT	II
CCI	CCI-002824
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-16
RULE-ID	SV-270772r1066805_rule
STIG-ID	UBTU-24-700310
VULN-ID	V-270772

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - sysctl

The command '/sbin/sysctl kernel.randomize_va_space' returned :

kernel.randomize_va_space = 2

PASSED - invalid configuration

The following file(s) do not contain "[\s]*kernel\.randomize_va_space[\s]*=[\s]*[^\s][\s]*\$":

```
/etc/sysctl.conf
/etc/sysctl.d/10-bufferbloat.conf
/etc/sysctl.d/10-console-messages.conf
/etc/sysctl.d/10-ipv6-privacy.conf
/etc/sysctl.d/10-kernel-hardening.conf
/etc/sysctl.d/10-magic-sysrq.conf
/etc/sysctl.d/10-map-count.conf
/etc/sysctl.d/10-network-security.conf
```

```
/etc/sysctl.d/10-ptrace.conf  
/etc/sysctl.d/10-zeropage.conf  
/etc/sysctl.d/99-cloudimg-ipv6.conf  
/etc/sysctl.d/99-sysctl.conf  
/etc/sysctl.d/README.sysctl  
/etc/sysctl.d/stig_kernel.conf
```

UBTU-24-700320 - Ubuntu 24.04 LTS must be configured so that Advance Package Tool (APT) removes all software components after updated versions have been installed.

Info

Previous versions of software components that are not removed from the information system after updates have been installed may be exploited by adversaries.

Some information technology products may remove older versions of software automatically from the information system.

Solution

Configure APT to remove all software components after updated versions have been installed. Add or update the following options to the "/etc/apt/apt.conf.d/50unattended-upgrades" file: Unattended-Upgrade::Remove-Unused-Dependencies "true"; Unattended-Upgrade::Remove-Unused-Kernel-Packages "true";

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.14.1
800-171R3	03.14.01
800-53	SI-2(6)
800-53R5	SI-2(6)
CAT	II
CCI	CCI-002617
CN-L3	8.1.4.4(e)
CN-L3	8.1.10.5(a)
CN-L3	8.1.10.5(b)
CN-L3	8.5.4.1(b)
CN-L3	8.5.4.1(d)
CN-L3	8.5.4.1(e)
CSF	ID.RA-1
CSF	PR.IP-12
CSF2.0	ID.IM-01
CSF2.0	ID.IM-02
CSF2.0	ID.IM-03
CSF2.0	PR.PS-02
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.6.8

ISO-27001-2022	A.8.8
ISO-27001-2022	A.8.32
ITSG-33	SI-2
NESA	T7.6.2
NESA	T7.7.1
NIAV2	PR9
PCI-DSSV3.2.1	6.2
PCI-DSSV4.0	6.3
PCI-DSSV4.0	6.3.3
QCSC-V1	11.2
RULE-ID	SV-270773r1066808_rule
STIG-ID	UBTU-24-700320
SWIFT-CSCV1	2.2
VULN-ID	V-270773

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
PASSED - Remove-Unused-Kernel-Packages
Compliant file(s):
/etc/apt/apt.conf.d/01-vendor-ubuntu - regex not found
/etc/apt/apt.conf.d/01autoremove - regex not found
/etc/apt/apt.conf.d/10periodic - regex not found
/etc/apt/apt.conf.d/15update-stamp - regex not found
/etc/apt/apt.conf.d/20apt-esm-hook.conf - regex not found
/etc/apt/apt.conf.d/20archive - regex not found
/etc/apt/apt.conf.d/20auto-upgrades - regex not found
/etc/apt/apt.conf.d/20packagekit - regex not found
/etc/apt/apt.conf.d/20snapd.conf - regex not found
/etc/apt/apt.conf.d/50-unattended-upgrades - regex '^[\s]*Unattended-Upgrade::Remove-
Unused-Kernel-Packages[\s]+' found - expect '^[\s]*Unattended-Upgrade::Remove-Unused-Kernel-
Packages[\s]+["']*true["'][\s]*;[\s]*$' found in the following lines:
  1: Unattended-Upgrade::Remove-Unused-Kernel-Packages "true";
/etc/apt/apt.conf.d/50appstream - regex not found
/etc/apt/apt.conf.d/50command-not-found - regex not found
/etc/apt/apt.conf.d/50unattended-upgrades - regex not found
/etc/apt/apt.conf.d/70debconf - regex not found
/etc/apt/apt.conf.d/99needrestart - regex not found
/etc/apt/apt.conf.d/99update-notifier - regex not found
-----
```

```
-----
PASSED - Remove-Unused-Dependencies
Compliant file(s):
/etc/apt/apt.conf.d/01-vendor-ubuntu - regex not found
/etc/apt/apt.conf.d/01autoremove - regex not found
/etc/apt/apt.conf.d/10periodic - regex not found
/etc/apt/apt.conf.d/15update-stamp - regex not found
/etc/apt/apt.conf.d/20apt-esm-hook.conf - regex not found
/etc/apt/apt.conf.d/20archive - regex not found
/etc/apt/apt.conf.d/20auto-upgrades - regex not found
/etc/apt/apt.conf.d/20packagekit - regex not found
/etc/apt/apt.conf.d/20snapd.conf - regex not found
-----
```

```
/etc/apt/apt.conf.d/50-unattended-upgrades - regex '^[\\s]*Unattended-Upgrade::Remove-
Unused-Dependencies[\\s]+' found - expect '^[\\s]*Unattended-Upgrade::Remove-Unused-
Dependencies[\\s]+["']*true["']*[\\s]*;[\\s]*$' found in the following lines:
    2: Unattended-Upgrade::Remove-Unused-Dependencies "true";
/etc/apt/apt.conf.d/50appstream - regex not found
/etc/apt/apt.conf.d/50command-not-found - regex not found
/etc/apt/apt.conf.d/50unattended-upgrades - regex not found
/etc/apt/apt.conf.d/70debconf - regex not found
/etc/apt/apt.conf.d/99needrestart - regex not found
/etc/apt/apt.conf.d/99update-notifier - regex not found
```

UBTU-24-900040 - Ubuntu 24.04 LTS must be configured so that audit configuration files are not write-accessible by unauthorized users.

Info

Without the capability to restrict which roles and individuals can select which events are audited, unauthorized personnel may be able to prevent the auditing of critical events. Misconfigured audits may degrade the system's performance by overwhelming the audit log.

Misconfigured audits may also make it more difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one.

Solution

Configure `/etc/audit/audit.rules`, `/etc/audit/rules.d/*`, and `/etc/audit/auditd.conf` files to have a mode of "0640" by using the following command: `$ sudo chmod -R 0640 /etc/audit/audit*.{rules,conf} /etc/audit/rules.d/*`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03
800-53	AU-12b.
800-53R5	AU-12b.
CAT	II
CCI	CCI-000171
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ITSG-33	AU-12b.
PCI-DSSV3.2.1	10.1

QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270775r1068369_rule
STIG-ID	UBTU-24-900040
SWIFT-CSCV1	6.4
VULN-ID	V-270775

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - auditd.conf

The file /etc/audit/auditd.conf with fmode owner: root group: root mode: 0640 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/etc/audit/auditd.conf

PASSED - rules.d/*

The file /etc/audit/rules.d/audit.rules with fmode owner: root group: root mode: 0640 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

The file /etc/audit/rules.d/stig.rules with fmode owner: root group: root mode: 0600 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/etc/audit/rules.d/audit.rules, /etc/audit/rules.d/stig.rules

PASSED - audit.rules

The file /etc/audit/audit.rules with fmode owner: root group: root mode: 0640 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/etc/audit/audit.rules

UBTU-24-900050 - Ubuntu 24.04 LTS must permit only authorized accounts to own the audit configuration files.

Info

Without the capability to restrict which roles and individuals can select which events are audited, unauthorized personnel may be able to prevent the auditing of critical events. Misconfigured audits may degrade the system's performance by overwhelming the audit log.

Misconfigured audits may also make it more difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one.

Solution

Configure `/etc/audit/audit.rules`, `/etc/audit/rules.d/*`, and `/etc/audit/auditd.conf` files to be owned by "root" user by using the following command: `$ sudo chown root /etc/audit/audit*.{rules,conf} /etc/audit/rules.d/*`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03
800-53	AU-12b.
800-53R5	AU-12b.
CAT	II
CCI	CCI-000171
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ITSG-33	AU-12b.
PCI-DSSV3.2.1	10.1

QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270776r1066817_rule
STIG-ID	UBTU-24-900050
SWIFT-CSCV1	6.4
VULN-ID	V-270776

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - audit.rules

The file /etc/audit/audit.rules with fmode owner: root group: root mode: 0640 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/etc/audit/audit.rules

PASSED - rules.d/*

The file /etc/audit/rules.d/audit.rules with fmode owner: root group: root mode: 0640 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

The file /etc/audit/rules.d/stig.rules with fmode owner: root group: root mode: 0600 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/etc/audit/rules.d/audit.rules, /etc/audit/rules.d/stig.rules

PASSED - auditd.conf

The file /etc/audit/auditd.conf with fmode owner: root group: root mode: 0640 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/etc/audit/auditd.conf

UBTU-24-90060 - Ubuntu 24.04 LTS must permit only authorized groups to own the audit configuration files.

Info

Without the capability to restrict which roles and individuals can select which events are audited, unauthorized personnel may be able to prevent the auditing of critical events. Misconfigured audits may degrade the system's performance by overwhelming the audit log.

Misconfigured audits may also make it more difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one.

Solution

Configure `/etc/audit/audit.rules`, `/etc/audit/rules.d/*`, and `/etc/audit/auditd.conf` files to be owned by "root" group by using the following command: `$ sudo chown :root /etc/audit/audit*.{rules,conf} /etc/audit/rules.d/*`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03
800-53	AU-12b.
800-53R5	AU-12b.
CAT	II
CCI	CCI-000171
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ITSG-33	AU-12b.
PCI-DSSV3.2.1	10.1

QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270777r1066820_rule
STIG-ID	UBTU-24-900060
SWIFT-CSCV1	6.4
VULN-ID	V-270777

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - audit.rules

The file /etc/audit/audit.rules with fmode owner: root group: root mode: 0640 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/etc/audit/audit.rules

PASSED - auditd.conf

The file /etc/audit/auditd.conf with fmode owner: root group: root mode: 0640 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/etc/audit/auditd.conf

PASSED - rules.d/*

The file /etc/audit/rules.d/audit.rules with fmode owner: root group: root mode: 0640 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

The file /etc/audit/rules.d/stig.rules with fmode owner: root group: root mode: 0600 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/etc/audit/rules.d/audit.rules, /etc/audit/rules.d/stig.rules

UBTU-24-900070 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the su command.

Info

Without generating audit records that are specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure Ubuntu 24.04 LTS to generate audit records when successful/unsuccessful attempts to use the "su" command occur. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/bin/su -F perm=x -F audit>=1000 -F audit!=1 -k privileged-priv_change To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270778r1066823_rule
STIG-ID	UBTU-24-900070
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270778

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/bin/su'' returned :

```
-a always,exit -S all -F path=/bin/su -F perm=x -F auid>=1000 -F auid!=-1 -F key=privileged-priv_change
-a always,exit -S all -F path=/usr/bin/sudo -F perm=x -F auid>=1000 -F auid!=-1 -F key=priv_cmd
-a always,exit -S all -F path=/usr/bin/sudoedit -F perm=x -F auid>=1000 -F auid!=-1 -F key=priv_cmd
```

UBTU-24-90080 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chfn command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful uses of the "chfn" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/chfn -F perm=x -F auid>=1000 -F auid!=-1 -k privileged-chfn To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270779r1066826_rule
STIG-ID	UBTU-24-900080
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270779

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/usr/bin/chfn'' returned :

```
-a always,exit -S all -F path=/usr/bin/chfn -F perm=x -F auid>=1000 -F auid!=-1 -F key=privileged-chfn
```

UBTU-24-90090 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the mount command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "mount" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/mount -F perm=x -F auid>=1000 -F auid!=1 -k privileged-mount To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270780r1066829_rule
STIG-ID	UBTU-24-900090
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270780

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/usr/bin/mount'' returned :

```
-a always,exit -S all -F path=/usr/bin/mount -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-mount
```

UBTU-24-900100 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the umount command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "umount" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/umount -F perm=x -F auid>=1000 -F auid!=--1 -k privileged-umount To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270781r1066832_rule
STIG-ID	UBTU-24-900100
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270781

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/usr/bin/umount'' returned :

```
-a always,exit -S all -F path=/usr/bin/umount -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-umount
```

UBTU-24-900110 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the ssh-agent command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "ssh-agent" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/ssh-agent -F perm=x -F auid>=1000 -F auid!=-1 -k privileged-ssh To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270782r1066835_rule
STIG-ID	UBTU-24-900110
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270782

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/usr/bin/ssh-agent'' returned :

```
-a always,exit -S all -F path=/usr/bin/ssh-agent -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-ssh
```

UBTU-24-900120 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the ssh-keysign command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "ssh-keysign" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/lib/openssh/ssh-keysign -F perm=x -F auid>=1000 -F auid!=-1 -k privileged-ssh To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270783r1066838_rule
STIG-ID	UBTU-24-900120
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270783

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep 'ssh-keysign'' returned :

```
-a always,exit -S all -F path=/usr/lib/openssh/ssh-keysign -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-ssh
```

UBTU-24-900130 - Ubuntu 24.04 LTS must generate audit records for any use of the setattr, fsetattr, lsetattr, removexattr, fremovexattr, and lremovexattr system calls.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter). The system call rules are loaded into a matching engine that intercepts each syscall that all programs on the system makes.

Therefore, it is very important to only use syscall rules when absolutely necessary since these affect performance.

The more rules, the bigger the performance hit.

The performance is helped, though, by combining syscalls into one rule whenever possible. Satisfies: SRG-OS-000064-GPOS-00033, SRG-OS-000462-GPOS-00206

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "setattr", "fsetattr", "lsetattr", "removexattr", "fremovexattr", and "lremovexattr" system calls. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F arch=b32 -S setattr,fsetattr,lsetattr,removexattr,fremovexattr,lremovexattr -F auid>=1000 -F auid!=1 -k perm_mod -a always,exit -F arch=b32 -S setattr,fsetattr,lsetattr,removexattr,fremovexattr,lremovexattr -F auid=0 -k perm_mod -a always,exit -F arch=b64 -S setattr,fsetattr,lsetattr,removexattr,fremovexattr,lremovexattr -F auid>=1000 -F auid!=1 -k perm_mod -a always,exit -F arch=b64 -S setattr,fsetattr,lsetattr,removexattr,fremovexattr,lremovexattr -F auid=0 -k perm_mod Note: For 32-bit architectures, only the 32-bit specific entries are required. To reload the rules file, issue the following command:
\$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7

CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270784r1068371_rule
STIG-ID	UBTU-24-900130
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270784

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - 32 bit fremovexattr auid>=1000

The command '/sbin/auditctl -l | /bin/grep 'xattr'' returned :

```
-a always,exit -F arch=b32 -S setxattr,lsetxattr,fsetxattr,removexattr,lremovexattr,fremovexattr -
F auid>=1000 -F auid!=-1 -F key=perm_mod
-a always,exit -F arch=b32 -S setxattr,lsetxattr,fsetxattr,removexattr,lremovexattr,fremovexattr -
F auid=0 -F key=perm_mod
```



```
-a always,exit -F arch=b32 -S setxattr,lsetxattr,fsetxattr,removexattr,lremovexattr,fremovexattr -  
F auid=0 -F key=perm_mod  
-a always,exit -F arch=b64 -S setxattr,lsetxattr,fsetxattr,removexattr,lremovexattr,fremovexattr -  
F auid>=1000 -F auid!=-1 -F key=perm_mod  
-a always,exit -F arch=b64 -S setxattr,lsetxattr,fsetxattr,removexattr,lremovexattr,fremovexattr -  
F auid=0 -F key=perm_mod
```

UBTU-24-900140 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chown, fchown, fchownat, and lchown system calls.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter). The system call rules are loaded into a matching engine that intercepts each syscall that all programs on the system makes.

Therefore, it is very important to only use syscall rules when absolutely necessary since these affect performance.

The more rules, the bigger the performance hit.

The performance is helped, though, by combining syscalls into one rule whenever possible. Satisfies: SRG-OS-000064-GPOS-00033, SRG-OS-000462-GPOS-00206

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "chown", "fchown", "fchownat", and "lchown" system calls. Add or update the following rules in the "/etc/audit/rules.d/stig.rules": -a always,exit -F arch=b32 -S chown,fchown,fchownat,lchown -F auid>=1000 -F auid!=1 -k perm_chng -a always,exit -F arch=b64 -S chown,fchown,fchownat,lchown -F auid>=1000 -F auid!=1 -k perm_chng Note: For 32-bit architectures, only the 32-bit specific entries are required. To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01

CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270785r1068373_rule
STIG-ID	UBTU-24-900140
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270785

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - fchownat b64

The command '/sbin/auditctl -l | /bin/grep 'chown'' returned :

```
-a always,exit -F arch=b32 -S lchown,fchown,chown,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng
```

```
-a always,exit -F arch=b64 -S chown,fchown,lchown,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng
```

PASSED - chown b32

The command '/sbin/auditctl -l | /bin/grep 'chown'' returned :


```

-a always,exit -F arch=b32 -S lchown,fchown,chmod,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng
-a always,exit -F arch=b64 -S chmod,fchown,lchown,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng

-----
PASSED - fchownat b32
The command '/sbin/auditctl -l | /bin/grep 'chown'' returned :

-a always,exit -F arch=b32 -S lchown,fchown,chmod,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng
-a always,exit -F arch=b64 -S chmod,fchown,lchown,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng

-----
PASSED - fchown b32
The command '/sbin/auditctl -l | /bin/grep 'chown'' returned :

-a always,exit -F arch=b32 -S lchown,fchown,chmod,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng
-a always,exit -F arch=b64 -S chmod,fchown,lchown,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng

-----
PASSED - lchown b64
The command '/sbin/auditctl -l | /bin/grep 'chown'' returned :

-a always,exit -F arch=b32 -S lchown,fchown,chmod,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng
-a always,exit -F arch=b64 -S chmod,fchown,lchown,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng

-----
PASSED - fchown b64
The command '/sbin/auditctl -l | /bin/grep 'chown'' returned :

-a always,exit -F arch=b32 -S lchown,fchown,chmod,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng
-a always,exit -F arch=b64 -S chmod,fchown,lchown,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng

-----
PASSED - chmod b64
The command '/sbin/auditctl -l | /bin/grep 'chown'' returned :

-a always,exit -F arch=b32 -S lchown,fchown,chmod,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng
-a always,exit -F arch=b64 -S chmod,fchown,lchown,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng

-----
PASSED - lchown b32
The command '/sbin/auditctl -l | /bin/grep 'chown'' returned :

-a always,exit -F arch=b32 -S lchown,fchown,chmod,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng
-a always,exit -F arch=b64 -S chmod,fchown,lchown,fchownat -F auid>=1000 -F auid!=-1 -F
key=perm_chng

```

UBTU-24-900150 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chmod, fchmod, and fchmodat system calls.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter). The system call rules are loaded into a matching engine that intercepts each syscall that all programs on the system makes.

Therefore, it is very important to only use syscall rules when absolutely necessary since these affect performance.

The more rules, the bigger the performance hit.

The performance is helped, though, by combining syscalls into one rule whenever possible. Satisfies: SRG-OS-000064-GPOS-00033, SRG-OS-000462-GPOS-00206

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "chmod", "fchmod", and "fchmodat" system calls. Add or update the following rules in the "/etc/audit/rules.d/stig.rules":
-a always,exit -F arch=b32 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=1 -k perm_chng -a always,exit -F arch=b64 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=1 -k perm_chng Note: For 32-bit architectures, only the 32-bit specific entries are required. To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01

CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270786r1068375_rule
STIG-ID	UBTU-24-900150
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270786

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - chmod b64

The command '/sbin/auditctl -l | /bin/grep chmod' returned :

```
-a always,exit -F arch=b32 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1 -F key=perm_chng
-a always,exit -F arch=b64 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1 -F key=perm_chng
```

PASSED - fchmod b32

The command '/sbin/auditctl -l | /bin/grep chmod' returned :

```
-a always,exit -F arch=b32 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1 -F key=perm_chng
-a always,exit -F arch=b64 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1 -F key=perm_chng
```

```
-----
PASSED - fchmod b64
The command '/sbin/auditctl -l | /bin/grep chmod' returned :

-a always,exit -F arch=b32 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1 -F key=perm_chng
-a always,exit -F arch=b64 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1 -F key=perm_chng

-----
PASSED - fchmodat b64
The command '/sbin/auditctl -l | /bin/grep chmod' returned :

-a always,exit -F arch=b32 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1 -F key=perm_chng
-a always,exit -F arch=b64 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1 -F key=perm_chng

-----
PASSED - fchmodat b32
The command '/sbin/auditctl -l | /bin/grep chmod' returned :

-a always,exit -F arch=b32 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1 -F key=perm_chng
-a always,exit -F arch=b64 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1 -F key=perm_chng

-----
PASSED - chmod b32
The command '/sbin/auditctl -l | /bin/grep chmod' returned :

-a always,exit -F arch=b32 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1 -F key=perm_chng
-a always,exit -F arch=b64 -S chmod,fchmod,fchmodat -F auid>=1000 -F auid!=-1 -F key=perm_chng
```

UBTU-24-900160 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the creat, open, openat, open_by_handle_at, truncate, and ftruncate system calls.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter). The system call rules are loaded into a matching engine that intercepts each syscall that all programs on the system makes.

Therefore, it is very important to only use syscall rules when absolutely necessary since these affect performance.

The more rules, the bigger the performance hit.

The performance is helped, though, by combining syscalls into one rule whenever possible. Satisfies: SRG-OS-000064-GPOS-00033, SRG-OS-000474-GPOS-00219

Solution

Configure the audit system to generate an audit event for any unsuccessful use of the "creat", "open", "openat", "open_by_handle_at", "truncate", and "ftruncate" system calls. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F arch=b32 -S creat,open,openat,open_by_handle_at,truncate,ftruncate -F exit=-EPERM -F auid>=1000 -F auid!=-1 -k perm_access -a always,exit -F arch=b32 -S creat,open,openat,open_by_handle_at,truncate,ftruncate -F exit=-EACCES -F auid>=1000 -F auid!=-1 -k perm_access -a always,exit -F arch=b64 -S creat,open,openat,open_by_handle_at,truncate,ftruncate -F exit=-EPERM -F auid>=1000 -F auid!=-1 -k perm_access -a always,exit -F arch=b64 -S creat,open,openat,open_by_handle_at,truncate,ftruncate -F exit=-EACCES -F auid>=1000 -F auid!=-1 -k perm_access Note: For 32-bit architectures, only the 32-bit specific entries are required. To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7

CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270787r1068378_rule
STIG-ID	UBTU-24-900160
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270787

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - 64 bit open EACCES

The command '/sbin/auditctl -l | /bin/grep -E 'open|truncate|creat'' returned :

```
-a always,exit -S all -F path=/usr/lib/openssh/ssh-keysign -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-ssh
-a always,exit -F arch=b32 -S open,creat,truncate,ftruncate,openat,open_by_handle_at -F exit=-
EPERM -F auid>=1000 -F auid!=-1 -F key=perm_access
```



```
-a always,exit -F arch=b32 -S open,creat,truncate,ftruncate,openat,open_by_handle_at -F exit=-EACCES -F auid>=1000 -F auid!=-1 -F key=perm_access
-a always,exit -F arch=b64 -S open,truncate,ftruncate,creat,openat,open_by_handle_at -F exit=-EPERM -F auid>=1000 -F auid!=-1 -F key=perm_access
-a always,exit -F arch=b64 -S open,truncate,ftruncate,creat,openat,open_by_handle_at -F exit=-EACCES -F auid>=1000 -F auid!=-1 -F key=perm_access
```

```
-----
PASSED - 32 bit open_by_handle_at EACCES
The command '/sbin/auditctl -l | /bin/grep -E 'open|truncate|creat'' returned :
```

```
-a always,exit -S all -F path=/usr/lib/openssh/ssh-keysign -F perm=x -F auid>=1000 -F auid!=-1 -F key=privileged-ssh
-a always,exit -F arch=b32 -S open,creat,truncate,ftruncate,openat,open_by_handle_at -F exit=-EPERM -F auid>=1000 -F auid!=-1 -F key=perm_access
-a always,exit -F arch=b32 -S open,creat,truncate,ftruncate,openat,open_by_handle_at -F exit=-EACCES -F auid>=1000 -F auid!=-1 -F key=perm_access
-a always,exit -F arch=b64 -S open,truncate,ftruncate,creat,openat,open_by_handle_at -F exit=-EPERM -F auid>=1000 -F auid!=-1 -F key=perm_access
-a always,exit -F arch=b64 -S open,truncate,ftruncate,creat,openat,open_by_handle_at -F exit=-EACCES -F auid>=1000 -F auid!=-1 -F key=perm_access
```

```
-----
PASSED - 32 bit creat EACCES
The command '/sbin/auditctl -l | /bin/grep -E 'open|truncate|creat'' returned :
```

```
-a always,exit -S all -F path=/usr/lib/openssh/ssh-keysign -F perm=x -F auid>=1000 -F auid!=-1 -F key=privileged-ssh
-a always,exit -F arch=b32 -S open,creat,truncate,ftruncate,openat,open_by_handle_at -F exit=-EPERM -F auid>=1000 -F auid!=-1 -F key=perm_access
-a always,exit -F arch=b32 -S open,creat,truncate,ftruncate,openat,open_by_handle_at -F exit=-EACCES -F auid>=1000 -F auid!=-1 -F key=perm_access
-a always,exit -F arch=b64 -S open,truncate,ftruncate,creat,openat,open_by_handle_at -F exit=-EPERM -F auid>=1000 -F auid!=-1 -F key=perm_access
-a always,exit -F arch=b64 -S open,truncate,ftruncate,creat,openat,open_by_handle_at -F exit=-EACCES -F auid>=1000 -F auid!=-1 -F key=perm_access
```

```
-----
PASSED - 32 bit open EACCES
The command '/sbin/auditctl -l | /bin/grep -E 'open|truncate|creat'' returned :
```

```
-a always,exit -S all -F path=/usr/lib/openssh/ssh-keysign -F perm=x -F auid>=1000 -F auid!=-1 -F key=privileged-ssh
-a always,exit -F arch=b32 -S open,creat,truncate,ftruncate,openat,open_by_handle_at -F exit=-EPERM -F auid>=1000 -F auid!=-1 -F key=perm_access
-a always,exit -F arch=b32 -S open,creat,truncate,ftruncate,openat,open_by_handle_at -F exit=-EACCES -F auid>=1000 -F auid!=-1 -F key=perm_access
-a always,exit -F arch=b64 -S open,truncate,ftruncate,creat,openat,open_by_handle_at -F exit=-EPERM -F auid>=1000 -F auid!=-1 -F key=perm_access
-a always,exit -F arch=b64 -S open,truncate,ftruncate,creat,openat,open_by_handle_at -F exit=-EACCES -F auid>=1000 -F auid!=-1 -F key=perm_access
```

UBTU-24-900170 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the sudo command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "sudo" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/sudo -F perm=x -F auid>=1000 -F auid!=-1 -k priv_cmd To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270788r1066853_rule
STIG-ID	UBTU-24-900170
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270788

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/usr/bin/sudo'' returned :

```
-a always,exit -S all -F path=/usr/bin/sudo -F perm=x -F auid>=1000 -F auid!=-1 -F key=priv_cmd
-a always,exit -S all -F path=/usr/bin/sudoedit -F perm=x -F auid>=1000 -F auid!=-1 -F
key=priv_cmd
```

UBTU-24-900180 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the sudoedit command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "sudoedit" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules": -a always,exit -F path=/usr/bin/sudoedit -F perm=x -F auid>=1000 -F auid!=1 -k priv_cmd To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270789r1066856_rule
STIG-ID	UBTU-24-900180
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270789

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/usr/bin/sudoedit'' returned :

```
-a always,exit -S all -F path=/usr/bin/sudoedit -F perm=x -F auid>=1000 -F auid!=-1 -F
key=priv_cmd
```

UBTU-24-900190 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chsh command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "chsh" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/chsh -F perm=x -F auid>=1000 -F auid!=-1 -k priv_cmd To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270790r1068380_rule
STIG-ID	UBTU-24-900190
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270790

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/usr/bin/chsh'' returned :

```
-a always,exit -S all -F path=/usr/bin/chsh -F perm=x -F auid>=1000 -F auid!=-1 -F key=priv_cmd
```

UBTU-24-900200 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the newgrp command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "newgrp" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/newgrp -F perm=x -F auid>=1000 -F auid!=-1 -k priv_cmd To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270791r1066862_rule
STIG-ID	UBTU-24-900200
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270791

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/usr/bin/newgrp'' returned :

```
-a always,exit -S all -F path=/usr/bin/newgrp -F perm=x -F auid>=1000 -F auid!=-1 -F key=priv_cmd
```

UBTU-24-900210 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chcon command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "chcon" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/chcon -F perm=x -F auid>=1000 -F auid!=--1 -k perm_chng To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270792r1066865_rule
STIG-ID	UBTU-24-900210
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270792

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/usr/bin/chcon'' returned :

```
-a always,exit -S all -F path=/usr/bin/chcon -F perm=x -F auid>=1000 -F auid!=-1 -F key=perm_chng
```

UBTU-24-900220 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the `apparmor_parser` command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the `"apparmor_parser"` command. Add or update the following rules in the `"/etc/audit/rules.d/stig.rules"` file: `-a always,exit -F path=/sbin/apparmor_parser -F perm=x -F auid>=1000 -F auid!=1 -k perm_chng` To reload the rules file, issue the following command: `$ sudo augenrules --load`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270793r1066868_rule
STIG-ID	UBTU-24-900220
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270793

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/sbin/apparmor_parser'' returned :

```
-a always,exit -S all -F path=/sbin/apparmor_parser -F perm=x -F auid>=1000 -F auid!=-1 -F
key=perm_chng
```

UBTU-24-900230 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the setfacl command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "setfacl" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/setfacl -F perm=x -F auid>=1000 -F auid!=1 -k perm_chng To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270794r1066871_rule
STIG-ID	UBTU-24-900230
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270794

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/usr/bin/setfacl'' returned :

```
-a always,exit -S all -F path=/usr/bin/setfacl -F perm=x -F auid>=1000 -F auid!=-1 -F
key=perm_chng
```

UBTU-24-900240 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chacl command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "chacl" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/chacl -F perm=x -F auid>=1000 -F auid!=1 -k perm_chng To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270795r1066874_rule
STIG-ID	UBTU-24-900240
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270795

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/usr/bin/chacl'' returned :

```
-a always,exit -S all -F path=/usr/bin/chacl -F perm=x -F auid>=1000 -F auid!=-1 -F key=perm_chng
```

UBTU-24-900250 - Ubuntu 24.04 LTS must generate audit records for the use and modification of faillog file.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter). Satisfies: SRG-OS-000064-GPOS-00033, SRG-OS-000470-GPOS-00214, SRG-OS-000473-GPOS-00218

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful modifications to the "faillog" file. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -w /var/log/faillog -p wa -k logins To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270796r1066877_rule
STIG-ID	UBTU-24-900250
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270796

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep 'faillog'' returned :

```
-w /var/log/faillog -p wa -k logins
```

UBTU-24-900260 - Ubuntu 24.04 LTS must generate audit records for the use and modification of the lastlog file.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter). Satisfies: SRG-OS-000064-GPOS-00033, SRG-OS-000470-GPOS-00214, SRG-OS-000473-GPOS-00218

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful modifications to the "lastlog" file. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -w /var/log/lastlog -p wa -k logins To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270797r1066880_rule
STIG-ID	UBTU-24-900260
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270797

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep 'lastlog'' returned :

```
-w /var/log/lastlog -p wa -k logins
```

UBTU-24-900270 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the passwd command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful uses of the "passwd" command. Add or update the following rule in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/passwd -F perm=x -F auid>=1000 -F auid!=1 -k privileged-passwd To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270798r1068382_rule
STIG-ID	UBTU-24-900270
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270798

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep 'passwd'' returned :

```
-w /etc/passwd -p wa -k usergroup_modification
-w /etc/security/opasswd -p wa -k usergroup_modification
-a always,exit -S all -F path=/usr/bin/passwd -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-passwd
-a always,exit -S all -F path=/usr/bin/gpasswd -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-gpasswd
```

UBTU-24-900280 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the unix_update command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful uses of the "unix_update" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/sbin/unix_update -F perm=x -F auid>=1000 -F auid!=-1 -k privileged-unix-update To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270799r1066886_rule
STIG-ID	UBTU-24-900280
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270799

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep 'unix_update'' returned :

```
-a always,exit -S all -F path=/sbin/unix_update -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-unix-update
```

UBTU-24-900290 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the gpasswd command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful uses of the "gpasswd" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/gpasswd -F perm=x -F auid>=1000 -F auid!=1 -k privileged-gpasswd To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270800r1066889_rule
STIG-ID	UBTU-24-900290
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270800

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep 'gpasswd'' returned :

```
-a always,exit -S all -F path=/usr/bin/gpasswd -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-gpasswd
```

UBTU-24-900300 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chage command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful uses of the "chage" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/chage -F perm=x -F auid>=1000 -F auid!=-1 -k privileged-chage To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270801r1066892_rule
STIG-ID	UBTU-24-900300
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270801

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep 'chage'' returned :

```
-a always,exit -S all -F path=/usr/bin/chage -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-chage
```

UBTU-24-900310 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the usermod command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful uses of the "usermod" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/sbin/usermod -F perm=x -F auid>=1000 -F auid!=-1 -k privileged-usermod To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270802r1066895_rule
STIG-ID	UBTU-24-900310
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270802

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep 'usermod'' returned :

```
-a always,exit -S all -F path=/usr/sbin/usermod -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-usermod
```

UBTU-24-900320 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the crontab command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful uses of the "crontab" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/bin/crontab -F perm=x -F auid>=1000 -F auid!=-1 -k privileged-crontab To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270803r1066898_rule
STIG-ID	UBTU-24-900320
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270803

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep 'crontab'' returned :

```
-a always,exit -S all -F path=/usr/bin/crontab -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-crontab
```

UBTU-24-900330 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the pam_timestamp_check command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful uses of the "pam_timestamp_check" command. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F path=/usr/sbin/pam_timestamp_check -F perm=x -F auid>=1000 -F auid!=-1 -k privileged-pam_timestamp_check To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270804r1066901_rule
STIG-ID	UBTU-24-900330
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270804

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep 'pam_timestamp_check'' returned :

```
-a always,exit -S all -F path=/usr/sbin/pam_timestamp_check -F perm=x -F auid>=1000 -F auid!=-1 -F
key=privileged-pam_timestamp_check
```

UBTU-24-900340 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the init_module and finit_module syscalls.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter). The system call rules are loaded into a matching engine that intercepts each syscall that all programs on the system makes.

Therefore, it is very important to only use syscall rules when absolutely necessary since these affect performance.

The more rules, the bigger the performance hit.

The performance is helped, though, by combining syscalls into one rule whenever possible. Satisfies: SRG-OS-000064-GPOS-00033, SRG-OS-000471-GPOS-00216

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "init_module" and "finit_module" syscalls. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F arch=b32 -S init_module,finit_module -F auid>=1000 -F auid!=-1 -k module_chng -a always,exit -F arch=b64 -S init_module,finit_module -F auid>=1000 -F auid!=-1 -k module_chng Note: For 32-bit architectures, only the 32-bit specific entries are required. To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01

CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270805r1068384_rule
STIG-ID	UBTU-24-900340
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270805

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - 32-bit finit_module

The command '/sbin/auditctl -l | /bin/grep 'init_module'' returned :

```
-a always,exit -F arch=b32 -S init_module,finit_module -F auid>=1000 -F auid!=-1 -F
key=module_chng
```

```
-a always,exit -F arch=b64 -S init_module,finit_module -F auid>=1000 -F auid!=-1 -F
key=module_chng
```

PASSED - 64-bit finit_module

The command '/sbin/auditctl -l | /bin/grep 'init_module'' returned :

```
-a always,exit -F arch=b32 -S init_module,finit_module -F auid>=1000 -F auid!=-1 -F  
key=module_chng  
-a always,exit -F arch=b64 -S init_module,finit_module -F auid>=1000 -F auid!=-1 -F  
key=module_chng
```

PASSED - 64-bit init_module

The command '/sbin/auditctl -l | /bin/grep 'init_module'' returned :

```
-a always,exit -F arch=b32 -S init_module,finit_module -F auid>=1000 -F auid!=-1 -F  
key=module_chng  
-a always,exit -F arch=b64 -S init_module,finit_module -F auid>=1000 -F auid!=-1 -F  
key=module_chng
```

PASSED - 32-bit init_module

The command '/sbin/auditctl -l | /bin/grep 'init_module'' returned :

```
-a always,exit -F arch=b32 -S init_module,finit_module -F auid>=1000 -F auid!=-1 -F  
key=module_chng  
-a always,exit -F arch=b64 -S init_module,finit_module -F auid>=1000 -F auid!=-1 -F  
key=module_chng
```


UBTU-24-900350 - Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the delete_module syscall.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter). Satisfies: SRG-OS-000064-GPOS-00033, SRG-OS-000477-GPOS-00222

Solution

Configure the audit system to generate an audit event for any successful/unsuccessful use of the "delete_module" syscall. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F arch=b32 -S delete_module -F auid>=1000 -F auid!=1 -k module_chng -a always,exit -F arch=b64 -S delete_module -F auid>=1000 -F auid!=1 -k module_chng Note: For 32-bit architectures, only the 32-bit specific entries are required. To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04

DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270806r1068386_rule
STIG-ID	UBTU-24-900350
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270806

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - delete_module b32

The command '/sbin/auditctl -l | /bin/grep 'delete_module'' returned :

```
-a always,exit -F arch=b32 -S delete_module -F auid>=1000 -F auid!=-1 -F key=module_chng
-a always,exit -F arch=b64 -S delete_module -F auid>=1000 -F auid!=-1 -F key=module_chng
```

PASSED - delete_module b64

The command '/sbin/auditctl -l | /bin/grep 'delete_module'' returned :

```
-a always,exit -F arch=b32 -S delete_module -F auid>=1000 -F auid!=-1 -F key=module_chng
-a always,exit -F arch=b64 -S delete_module -F auid>=1000 -F auid!=-1 -F key=module_chng
```

UBTU-24-900510 - Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to modify the /etc/sudoers file occur.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure Ubuntu 24.04 LTS to generate audit records for all modifications that affect "/etc/sudoers". Add or update the following rule to "/etc/audit/rules.d/stig.rules": -w /etc/sudoers -p wa -k privilege_modification To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270807r1066910_rule
STIG-ID	UBTU-24-900510
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270807

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep 'sudoers '' returned :

```
-w /etc/sudoers -p wa -k privilege_modification
```

UBTU-24-900520 - Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to modify the /etc/sudoers.d directory occur.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure Ubuntu 24.04 LTS to generate audit records for all modifications that affect "/etc/sudoers.d" directory. Add or update the following rule to "/etc/audit/rules.d/stig.rules": -w /etc/sudoers.d -p wa -k privilege_modification To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270808r1067100_rule
STIG-ID	UBTU-24-900520
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270808

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep sudoers.d' returned :

```
-w /etc/sudoers.d -p wa -k privilege_modification
```

UBTU-24-900540 - Ubuntu 24.04 LTS must generate audit records for any successful/unsuccessful use of unlink, unlinkat, rename, renameat, and rmdir system calls.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter). The system call rules are loaded into a matching engine that intercepts each syscall that all programs on the system makes.

Therefore, it is very important to only use syscall rules when absolutely necessary since these affect performance.

The more rules, the bigger the performance hit.

The performance is helped, though, by combining syscalls into one rule whenever possible.

Solution

Configure the audit system to generate audit events for any successful/unsuccessful use of "unlink", "unlinkat", "rename", "renameat", and "rmdir" system calls. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -a always,exit -F arch=b64 -S unlink,unlinkat,rename,renameat,rmdir -F auid>=1000 -F auid!=1 -k delete -a always,exit -F arch=b32 -S unlink,unlinkat,rename,renameat,rmdir -F auid>=1000 -F auid!=1 -k delete
Note: For 32-bit architectures, only the 32-bit specific entries are required. To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03

CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270809r1068388_rule
STIG-ID	UBTU-24-900540
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270809

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - 64 bit unlinkat

The command '/sbin/auditctl -l | /bin/grep -E 'unlink|rename|rmdir'' returned :

```
-a always,exit -F arch=b64 -S rename,rmdir,unlink,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F key=delete
-a always,exit -F arch=b32 -S unlink,rename,rmdir,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F key=delete
```

PASSED - 32 bit renameat

The command '/sbin/auditctl -l | /bin/grep -E 'unlink|rename|rmdir'' returned :

```
-a always,exit -F arch=b64 -S rename,rmdir,unlink,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F key=delete
```



```

-a always,exit -F arch=b32 -S unlink,rename,rmdir,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete

-----
PASSED - 64 bit renameat
The command '/sbin/auditctl -l | /bin/grep -E 'unlink|rename|rmdir'' returned :

-a always,exit -F arch=b64 -S rename,rmdir,unlink,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete
-a always,exit -F arch=b32 -S unlink,rename,rmdir,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete

-----
PASSED - 32 bit rename
The command '/sbin/auditctl -l | /bin/grep -E 'unlink|rename|rmdir'' returned :

-a always,exit -F arch=b64 -S rename,rmdir,unlink,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete
-a always,exit -F arch=b32 -S unlink,rename,rmdir,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete

-----
PASSED - 32 bit unlinkat
The command '/sbin/auditctl -l | /bin/grep -E 'unlink|rename|rmdir'' returned :

-a always,exit -F arch=b64 -S rename,rmdir,unlink,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete
-a always,exit -F arch=b32 -S unlink,rename,rmdir,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete

-----
PASSED - 32 bit unlink
The command '/sbin/auditctl -l | /bin/grep -E 'unlink|rename|rmdir'' returned :

-a always,exit -F arch=b64 -S rename,rmdir,unlink,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete
-a always,exit -F arch=b32 -S unlink,rename,rmdir,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete

-----
PASSED - 64 bit rmdir
The command '/sbin/auditctl -l | /bin/grep -E 'unlink|rename|rmdir'' returned :

-a always,exit -F arch=b64 -S rename,rmdir,unlink,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete
-a always,exit -F arch=b32 -S unlink,rename,rmdir,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete

-----
PASSED - 64 bit unlink
The command '/sbin/auditctl -l | /bin/grep -E 'unlink|rename|rmdir'' returned :

-a always,exit -F arch=b64 -S rename,rmdir,unlink,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete
-a always,exit -F arch=b32 -S unlink,rename,rmdir,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete

-----
PASSED - 32 bit rmdir
The command '/sbin/auditctl -l | /bin/grep -E 'unlink|rename|rmdir'' returned :

-a always,exit -F arch=b64 -S rename,rmdir,unlink,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete
-a always,exit -F arch=b32 -S unlink,rename,rmdir,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete

-----
PASSED - 64 bit rename
The command '/sbin/auditctl -l | /bin/grep -E 'unlink|rename|rmdir'' returned :

-a always,exit -F arch=b64 -S rename,rmdir,unlink,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete
-a always,exit -F arch=b32 -S unlink,rename,rmdir,unlinkat,renameat -F auid>=1000 -F auid!=-1 -F
key=delete

```

UBTU-24-900590 - Ubuntu 24.04 LTS must generate audit records for the /var/log/wtmp file.

Info

Without generating audit records that are specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate audit events showing start and stop times for user access via the "/var/log/wtmp" file. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -w /var/log/wtmp -p wa -k logins To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b

HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270810r1066919_rule
STIG-ID	UBTU-24-900590
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270810

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/var/log/wtmp'' returned :

```
-w /var/log/wtmp -p wa -k logins
```

UBTU-24-900600 - Ubuntu 24.04 LTS must generate audit records for the /var/run/utmp file.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate audit events showing start and stop times for user access via the "/var/run/utmp" file. Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -w /var/run/utmp -p wa -k logins To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b

HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270811r1066922_rule
STIG-ID	UBTU-24-900600
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270811

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/var/run/utmp'' returned :

```
-w /var/run/utmp -p wa -k logins
```

UBTU-24-900610 - Ubuntu 24.04 LTS must generate audit records for the /var/log/btmp file.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure the audit system to generate audit events showing start and stop times for user access via the "/var/log/btmp file". Add or update the following rules in the "/etc/audit/rules.d/stig.rules" file: -w /var/log/btmp -p wa -k logins To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b

HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270812r1066925_rule
STIG-ID	UBTU-24-900610
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270812

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/var/log/btmp'' returned :

```
-w /var/log/btmp -p wa -k logins
```

UBTU-24-900730 - Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to use modprobe command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure Ubuntu 24.04 LTS to audit the execution of the module management program "modprobe". Add or update the following rule in the "/etc/audit/rules.d/stig.rules" file: -w /sbin/modprobe -p x -k modules To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270813r1066928_rule
STIG-ID	UBTU-24-900730
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270813

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/sbin/modprobe'' returned :

```
-w /sbin/modprobe -p x -k modules
```

UBTU-24-900740 - Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to use the kmod command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure Ubuntu 24.04 LTS to audit the execution of the module management program "kmod". Add or update the following rule in the "/etc/audit/rules.d/stig.rules" file: -w /bin/kmod -p x -k modules To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270814r1066931_rule
STIG-ID	UBTU-24-900740
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270814

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep '/bin/kmod'' returned :

```
-w /bin/kmod -p x -k modules
```

UBTU-24-900750 - Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to use the fdisk command.

Info

Without generating audit records specific to the security and mission needs of the organization, it would be difficult to establish, correlate, and investigate the events relating to an incident or identify those responsible for one. Audit records can be generated from various components within the information system (e.g., module or policy filter).

Solution

Configure Ubuntu 24.04 LTS to audit the execution of the partition management program "fdisk". Add or update the following rule in the "/etc/audit/rules.d/stig.rules" file: -w /usr/sbin/fdisk -p x -k fdisk To reload the rules file, issue the following command: \$ sudo augenrules --load

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.1
800-171	3.3.2
800-171R3	03.03.03a.
800-53	AU-12c.
800-53R5	AU-12c.
CAT	II
CCI	CCI-000172
CN-L3	7.1.3.3(a)
CN-L3	7.1.3.3(b)
CN-L3	7.1.3.3(c)
CN-L3	8.1.3.5(a)
CN-L3	8.1.3.5(b)
CN-L3	8.1.4.3(a)
CSF	DE.CM-1
CSF	DE.CM-3
CSF	DE.CM-7
CSF	PR.PT-1
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	DE.CM-09
CSF2.0	PR.PS-04
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG

GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.1
ITSG-33	AU-12c.
NESA	T3.6.2
NESA	T3.6.5
NESA	T3.6.6
NIAV2	SM8
PCI-DSSV3.2.1	10.1
QCSC-V1	3.2
QCSC-V1	6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270815r1066934_rule
STIG-ID	UBTU-24-900750
SWIFT-CSCV1	6.4
TBA-FIISB	45.1.1
VULN-ID	V-270815

Assets

strkbsubuntu9

The command '/sbin/auditctl -l | /bin/grep 'fdisk'' returned :

```
-w /usr/sbin/fdisk -p x -k fdisk
```

UBTU-24-900980 - Ubuntu 24.04 LTS must alert the system administrator (SA) and information system security officer (ISSO) (at a minimum) in the event of an audit processing failure.

Info

It is critical for the appropriate personnel to be aware if a system is at risk of failing to process audit logs as required. Without this notification, the security personnel may be unaware of an impending failure of the audit capability, and system operation may be adversely affected. Audit processing failures include software/hardware errors, failures in the audit capturing mechanisms, and audit storage capacity being reached or exceeded. This requirement applies to each audit data storage repository (i.e., distinct information system component where audit records are stored), the centralized audit storage capacity of organizations (i.e., all audit data storage repositories combined), or both.

Solution

Configure "auditd" service to notify the SA and ISSO in the event of an audit processing failure. Edit the following line in "/etc/audit/auditd.conf" to ensure administrators are notified via email for those situations: action_mail_acct = <administrator_account> Note: Change "administrator_account" to an account for security personnel. Restart the "auditd" service so the changes take effect: \$ sudo systemctl restart auditd.service

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.4
800-171R3	03.03.04a.
800-53	AU-5a.
800-53R5	AU-5a.
CAT	III
CCI	CCI-000139
CN-L3	7.1.3.3(e)
CSF	PR.PT-1
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ITSG-33	AU-5a.
NESA	T3.6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270819r1068390_rule
STIG-ID	UBTU-24-900980
VULN-ID	V-270819

Assets

strkbsubuntu9

Compliant file(s):

```
/etc/audit/auditd.conf - regex '^[\s]*action_mail_acct[\s]*=' found - expect  
'^[\s]*action_mail_acct[\s]*=[\s]*auditor([\s]|$)' found in the following lines:  
21: action_mail_acct = auditor
```

UBTU-24-901220 - Ubuntu 24.04 LTS must record time stamps for audit records that can be mapped to Coordinated Universal Time (UTC) or Greenwich Mean Time (GMT).

Info

If time stamps are not consistently applied and there is no common time reference, it is difficult to perform forensic analysis. Time stamps generated by Ubuntu 24.04 LTS include date and time. Time is commonly expressed in Coordinated Universal Time (UTC), a modern continuation of Greenwich Mean Time (GMT), or local time with an offset from UTC.

Solution

To configure the system time zone to use UTC or GMT, run the following command, replacing [ZONE] with UTC or GMT: `$ sudo timedatectl set-timezone [ZONE]`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.7
800-171R3	03.03.07b.
800-53	AU-8b.
800-53R5	AU-8b.
CAT	III
CCI	CCI-001890
CN-L3	8.1.4.3(b)
CSF	PR.PT-1
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.17
ITSG-33	AU-8
NESA	T3.6.7
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270820r1066949_rule
STIG-ID	UBTU-24-901220
TBA-FIISB	37.4
VULN-ID	V-270820

Assets

strkbsubuntu9

The command `'/bin/timedatectl status'` returned :

Local time: Sun 2026-08-23 18:22:25 UTC
Universal time: Sun 2026-08-23 18:22:25 UTC
RTC time: Sun 2026-08-23 18:22:24
Time zone: Etc/UTC (UTC, +0000)
System clock synchronized: yes
NTP service: active
RTC in local TZ: no

UBTU-24-901230 - Ubuntu 24.04 LTS must configure audit tools with a mode of "0755" or less permissive.

Info

Protecting audit information also includes identifying and protecting the tools used to view and manipulate log data.

Therefore, protecting audit tools is necessary to prevent unauthorized operation on audit information.

Operating systems providing tools to interface with audit information will leverage user permissions, roles identifying the user accessing the tools, and the corresponding user rights to make decisions regarding access to audit tools. Audit tools include, but are not limited to, vendor-provided and open source audit tools needed to successfully view and manipulate audit information system activity and records.

Audit tools include custom queries and report generators. Satisfies: SRG-OS-000257-GPOS-00098, SRG-OS-000256-GPOS-00097

Solution

Configure the audit tools on Ubuntu 24.04 LTS to be protected from unauthorized access by setting the correct permissive mode using the following command: `$ sudo chmod 0755 [audit_tool]` Replace "[audit_tool]" with the audit tool that does not have the correct permissions.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.8
800-171R3	03.03.08
800-53	AU-9
800-53R5	AU-9
800-53R5	AU-9a.
CAT	II
CCI	CCI-001493
CCI	CCI-001494
CN-L3	7.1.2.3(d)
CN-L3	7.1.3.3(f)
CN-L3	8.1.3.5(c)
CN-L3	8.1.4.3(c)
CSF	PR.PT-1
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.15

ISO/IEC-27001	A.12.4.2
ITSG-33	AU-9
NESA	M5.2.3
NESA	M5.5.2
NESA	T3.6.4
NESA	T8.2.9
NIAV2	SM5
NIAV2	SM6
PCI-DSSV3.2.1	10.5
PCI-DSSV3.2.1	10.5.2
PCI-DSSV4.0	10.3.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270821r1134818_rule
STIG-ID	UBTU-24-901230
VULN-ID	V-270821

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - auditctl

The file /sbin/auditctl with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/auditctl

PASSED - aureport

The file /sbin/aureport with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/aureport

PASSED - ausearch

The file /sbin/ausearch with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/ausearch

PASSED - autrace

The file /sbin/autrace with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/autrace

PASSED - auditd

The file /sbin/auditd with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/auditd

PASSED - augenrules

The file /sbin/augenrules with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven
permissions : FALSE is compliant with the policy value

/sbin/augenrules

UBTU-24-901240 - Ubuntu 24.04 LTS must configure audit tools to be owned by root.

Info

Protecting audit information also includes identifying and protecting the tools used to view and manipulate log data.

Therefore, protecting audit tools is necessary to prevent unauthorized operation on audit information.

Operating systems providing tools to interface with audit information will leverage user permissions, roles identifying the user accessing the tools, and the corresponding user rights to make decisions regarding access to audit tools. Audit tools include, but are not limited to, vendor-provided and open source audit tools needed to successfully view and manipulate audit information system activity and records.

Audit tools include custom queries and report generators. Satisfies: SRG-OS-000257-GPOS-00098, SRG-OS-000256-GPOS-00097

Solution

Configure the audit tools on Ubuntu 24.04 LTS to be protected from unauthorized access by setting the file owner as root using the following command: `$ sudo chown root [audit_tool]` Replace "[audit_tool]" with each audit tool not owned by root.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.8
800-171R3	03.03.08
800-53	AU-9
800-53R5	AU-9
800-53R5	AU-9a.
CAT	II
CCI	CCI-001493
CCI	CCI-001494
CN-L3	7.1.2.3(d)
CN-L3	7.1.3.3(f)
CN-L3	8.1.3.5(c)
CN-L3	8.1.4.3(c)
CSF	PR.PT-1
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.2

ITSG-33	AU-9
NESA	M5.2.3
NESA	M5.5.2
NESA	T3.6.4
NESA	T8.2.9
NIAV2	SM5
NIAV2	SM6
PCI-DSSV3.2.1	10.5
PCI-DSSV3.2.1	10.5.2
PCI-DSSV4.0	10.3.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270822r1134821_rule
STIG-ID	UBTU-24-901240
VULN-ID	V-270822

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - ausearch

The file /sbin/ausearch with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/ausearch

PASSED - auditd

The file /sbin/auditd with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/auditd

PASSED - aureport

The file /sbin/aureport with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/aureport

PASSED - auditctl

The file /sbin/auditctl with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/auditctl

PASSED - autrace

The file /sbin/autrace with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/autrace

PASSED - augenrules

The file /sbin/augenrules with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/augenrules

UBTU-24-901250 - Ubuntu 24.04 LTS must configure the audit tools to be group owned by root.

Info

Protecting audit information also includes identifying and protecting the tools used to view and manipulate log data.

Therefore, protecting audit tools is necessary to prevent unauthorized operation on audit information.

Operating systems providing tools to interface with audit information will leverage user permissions, roles identifying the user accessing the tools, and the corresponding user rights to make decisions regarding access to audit tools. Audit tools include, but are not limited to, vendor-provided and open source audit tools needed to successfully view and manipulate audit information system activity and records.

Audit tools include custom queries and report generators. Satisfies: SRG-OS-000257-GPOS-00098, SRG-OS-000256-GPOS-00097

Solution

Configure the audit tools on Ubuntu 24.04 LTS to be protected from unauthorized access by setting the file group as root using the following command: `$ sudo chown :root [audit_tool]` Replace "[audit_tool]" with each audit tool not group owned by root.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.8
800-171R3	03.03.08
800-53	AU-9
800-53R5	AU-9
800-53R5	AU-9a.
CAT	II
CCI	CCI-001493
CCI	CCI-001494
CN-L3	7.1.2.3(d)
CN-L3	7.1.3.3(f)
CN-L3	8.1.3.5(c)
CN-L3	8.1.4.3(c)
CSF	PR.PT-1
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.2

ITSG-33	AU-9
NESA	M5.2.3
NESA	M5.5.2
NESA	T3.6.4
NESA	T8.2.9
NIAV2	SM5
NIAV2	SM6
PCI-DSSV3.2.1	10.5
PCI-DSSV3.2.1	10.5.2
PCI-DSSV4.0	10.3.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270823r1134824_rule
STIG-ID	UBTU-24-901250
VULN-ID	V-270823

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - auditd

The file /sbin/auditd with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/auditd

PASSED - autrace

The file /sbin/autrace with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/autrace

PASSED - auditctl

The file /sbin/auditctl with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/auditctl

PASSED - augenrules

The file /sbin/augenrules with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/augenrules

PASSED - aureport

The file /sbin/aureport with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven permissions : FALSE is compliant with the policy value

/sbin/aureport

PASSED - ausearch

The file /sbin/ausearch with fmode owner: root group: root mode: 0755 uid: 0 gid: 0 uneven
permissions : FALSE is compliant with the policy value

/sbin/ausearch

UBTU-24-901260 - Ubuntu 24.04 LTS must have directories that contain system commands set to a mode of "0755" or less permissive.

Info

Protecting audit information also includes identifying and protecting the tools used to view and manipulate log data.

Therefore, protecting audit tools is necessary to prevent unauthorized operation on audit information.

Operating systems providing tools to interface with audit information will leverage user permissions and roles identifying the user accessing the tools and the corresponding rights the user has to make access decisions regarding the deletion of audit tools. Audit tools include, but are not limited to, vendor-provided and open source audit tools needed to successfully view and manipulate audit information system activity and records. Audit tools include custom queries and report generators.

Solution

Configure the system commands directories to be protected from unauthorized access.

Run the following command: `$ sudo find /bin /sbin /usr/bin /usr/sbin /usr/local/bin /usr/local/sbin -perm /022 -type d -exec chmod -R 755 '{}' \;`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.8
800-171R3	03.03.08
800-53	AU-9
800-53R5	AU-9
CAT	II
CCI	CCI-001495
CN-L3	7.1.2.3(d)
CN-L3	7.1.3.3(f)
CN-L3	8.1.3.5(c)
CN-L3	8.1.4.3(c)
CSF	PR.PT-1
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.2
ITSG-33	AU-9

NESA	M5.2.3
NESA	M5.5.2
NESA	T3.6.4
NESA	T8.2.9
NIAV2	SM5
NIAV2	SM6
PCI-DSSV3.2.1	10.5
PCI-DSSV3.2.1	10.5.2
PCI-DSSV4.0	10.3.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270824r1066961_rule
STIG-ID	UBTU-24-901260
VULN-ID	V-270824

Assets

strkbsubuntu9

No results found.

UBTU-24-901270 - Ubuntu 24.04 LTS must have directories that contain system commands owned by root.

Info

Protecting audit information also includes identifying and protecting the tools used to view and manipulate log data.

Therefore, protecting audit tools is necessary to prevent unauthorized operation on audit information.

Operating systems providing tools to interface with audit information will leverage user permissions and roles identifying the user accessing the tools and the corresponding rights the user has to make access decisions regarding the deletion of audit tools. Audit tools include, but are not limited to, vendor-provided and open source audit tools needed to successfully view and manipulate audit information system activity and records. Audit tools include custom queries and report generators.

Solution

Configure the system commands directories to be protected from unauthorized access.

Run the following command: `$ sudo find /bin /sbin /usr/bin /usr/sbin /usr/local/bin /usr/local/sbin ! -user root -type d -exec chown root '{}' \;`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.8
800-171R3	03.03.08
800-53	AU-9
800-53R5	AU-9
CAT	II
CCI	CCI-001495
CN-L3	7.1.2.3(d)
CN-L3	7.1.3.3(f)
CN-L3	8.1.3.5(c)
CN-L3	8.1.4.3(c)
CSF	PR.PT-1
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.2
ITSG-33	AU-9

NESA	M5.2.3
NESA	M5.5.2
NESA	T3.6.4
NESA	T8.2.9
NIAV2	SM5
NIAV2	SM6
PCI-DSSV3.2.1	10.5
PCI-DSSV3.2.1	10.5.2
PCI-DSSV4.0	10.3.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270825r1066964_rule
STIG-ID	UBTU-24-901270
VULN-ID	V-270825

Assets

strkbsubuntu9

No results found.

UBTU-24-901280 - Ubuntu 24.04 LTS must have directories that contain system commands group-owned by root.

Info

Protecting audit information also includes identifying and protecting the tools used to view and manipulate log data.

Therefore, protecting audit tools is necessary to prevent unauthorized operation on audit information.

Operating systems providing tools to interface with audit information will leverage user permissions and roles identifying the user accessing the tools and the corresponding rights the user has to make access decisions regarding the deletion of audit tools. Audit tools include, but are not limited to, vendor-provided and open source audit tools needed to successfully view and manipulate audit information system activity and records. Audit tools include custom queries and report generators.

Solution

Configure the system commands directories to be protected from unauthorized access.

Run the following command: `$ sudo find /bin /sbin /usr/bin /usr/sbin /usr/local/bin /usr/local/sbin ! -group root -type d -exec chgrp root '{}' \;`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.8
800-171R3	03.03.08
800-53	AU-9
800-53R5	AU-9
CAT	II
CCI	CCI-001495
CN-L3	7.1.2.3(d)
CN-L3	7.1.3.3(f)
CN-L3	8.1.3.5(c)
CN-L3	8.1.4.3(c)
CSF	PR.PT-1
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.2
ITSG-33	AU-9

NESA	M5.2.3
NESA	M5.5.2
NESA	T3.6.4
NESA	T8.2.9
NIAV2	SM5
NIAV2	SM6
PCI-DSSV3.2.1	10.5
PCI-DSSV3.2.1	10.5.2
PCI-DSSV4.0	10.3.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270826r1066967_rule
STIG-ID	UBTU-24-901280
VULN-ID	V-270826

Assets

strkbsubuntu9

No results found.

UBTU-24-901300 - Ubuntu 24.04 LTS must be configured so that audit log files are not read or write-accessible by unauthorized users.

Info

Unauthorized disclosure of audit records can reveal system and configuration data to attackers, thus compromising its confidentiality. Audit information includes all information (e.g., audit records, audit settings, audit reports) needed to successfully audit operating system activity. Satisfies: SRG-OS-000057-GPOS-00027, SRG-OS-000058-GPOS-00028

Solution

Configure the audit log files to have a mode of "0600" or less permissive. Determine where the audit logs are stored with the following command: `$ sudo grep -iw log_file /etc/audit/auditd.conf log_file = /var/log/audit/audit.log` Using the path of the directory containing the audit logs, configure the audit log files to have a mode of "0600" or less permissive by using the following command: `$ sudo chmod 0600 /var/log/audit/*`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.8
800-171R3	03.03.08
800-53	AU-9
800-53R5	AU-9a.
CAT	II
CCI	CCI-000162
CCI	CCI-000163
CN-L3	7.1.2.3(d)
CN-L3	7.1.3.3(f)
CN-L3	8.1.3.5(c)
CN-L3	8.1.4.3(c)
CSF	PR.PT-1
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.2
ITSG-33	AU-9
NESA	M5.2.3

NESA	M5.5.2
NESA	T3.6.4
NESA	T8.2.9
NIAV2	SM5
NIAV2	SM6
PCI-DSSV3.2.1	10.5
PCI-DSSV3.2.1	10.5.2
PCI-DSSV4.0	10.3.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270827r1066970_rule
STIG-ID	UBTU-24-901300
VULN-ID	V-270827

Assets

strkbsubuntu9

```
The command 'logfile=$(/bin/grep -iw ^\s*log_file /etc/audit/auditd.conf); if [ ! -z
"${logfile}" ]; then /bin/stat -c "%n %a" $(/bin/dirname ${logfile} | /bin/grep "/" )/* | /bin/
grep -Ev '0?[0-6]00' | /bin/awk '{print} END {if (NR == 0) print "All files have a mode of 0600 or
less"; else print "Failing files found"}'; else echo "log_file variable in /etc/audit/auditd.conf
not set"; fi' returned :
```

All files have a mode of 0600 or less

UBTU-24-901310 - Ubuntu 24.04 LTS must be configured to permit only authorized users ownership of the audit log files.

Info

Unauthorized disclosure of audit records can reveal system and configuration data to attackers, thus compromising its confidentiality. Audit information includes all information (e.g., audit records, audit settings, audit reports) needed to successfully audit operating system activity. Satisfies: SRG-OS-000057-GPOS-00027, SRG-OS-000058-GPOS-00028, SRG-OS-000059-GPOS-00029

Solution

Configure the audit log directory and its underlying files to be owned by "root" user. Determine where the audit logs are stored with the following command: `$ sudo grep -iw log_file /etc/audit/auditd.conf log_file = /var/log/audit/audit.log` Using the path of the directory containing the audit logs, configure the audit log files to be owned by "root" user by using the following command: `$ sudo chown root /var/log/audit/*`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.8
800-171R3	03.03.08
800-53	AU-9
800-53R5	AU-9a.
CAT	II
CCI	CCI-000162
CCI	CCI-000163
CCI	CCI-000164
CN-L3	7.1.2.3(d)
CN-L3	7.1.3.3(f)
CN-L3	8.1.3.5(c)
CN-L3	8.1.4.3(c)
CSF	PR.PT-1
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.2
ITSG-33	AU-9

NESA	M5.2.3
NESA	M5.5.2
NESA	T3.6.4
NESA	T8.2.9
NIAV2	SM5
NIAV2	SM6
PCI-DSSV3.2.1	10.5
PCI-DSSV3.2.1	10.5.2
PCI-DSSV4.0	10.3.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270828r1066973_rule
STIG-ID	UBTU-24-901310
VULN-ID	V-270828

Assets

strkbsubuntu9

```
The command 'logfile=$( /bin/grep -iw ^\s*log_file /etc/audit/auditd.conf); if [ ! -z
"${logfile}" ]; then /bin/stat -c "%n %U" $(/bin/dirname $(/bin/grep -iw log_file /etc/audit/
auditd.conf) | /bin/grep "/" )/* | /bin/grep -Ev 'root' | /bin/awk '{print} END {if (NR == 0) print
"All files owned by root"; else print "Failing files found"}'; else echo "log_file variable in /
etc/audit/auditd.conf not set"; fi' returned :
```

All files owned by root

UBTU-24-901350 - Ubuntu 24.04 LTS must permit only authorized groups ownership of the audit log files.

Info

Unauthorized disclosure of audit records can reveal system and configuration data to attackers, thus compromising its confidentiality. Audit information includes all information (e.g., audit records, audit settings, audit reports) needed to successfully audit operating system activity. Satisfies: SRG-OS-000057-GPOS-00027, SRG-OS-000058-GPOS-00028, SRG-OS-000059-GPOS-00029

Solution

Configure the audit log directory and its underlying files to be owned by "root" group. Set the "log_group" parameter of the audit configuration file to the "root" value so when a new log file is created, its group owner is properly set: `$ sudo sed -i '/^log_group/D' /etc/audit/auditd.conf` `$ sudo sed -i '/^log_file/a'log_group = root' /etc/audit/auditd.conf` Signal the audit daemon to reload the configuration file to update the group owners of existing files: `$ sudo systemctl kill auditd -s SIGHUP`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.8
800-171R3	03.03.08
800-53	AU-9
800-53R5	AU-9a.
CAT	II
CCI	CCI-000162
CCI	CCI-000163
CCI	CCI-000164
CN-L3	7.1.2.3(d)
CN-L3	7.1.3.3(f)
CN-L3	8.1.3.5(c)
CN-L3	8.1.4.3(c)
CSF	PR.PT-1
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.2

ITSG-33	AU-9
NESA	M5.2.3
NESA	M5.5.2
NESA	T3.6.4
NESA	T8.2.9
NIAV2	SM5
NIAV2	SM6
PCI-DSSV3.2.1	10.5
PCI-DSSV3.2.1	10.5.2
PCI-DSSV4.0	10.3.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270829r1066976_rule
STIG-ID	UBTU-24-901350
VULN-ID	V-270829

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - file group

The command 'logfile=\$(/bin/grep -iw ^\s*log_file /etc/audit/auditd.conf); if [! -z "\${logfile}"]; then /bin/stat -c "%n %G" \$(/bin/dirname \${logfile} | /bin/grep "/")/* | /bin/grep -Ev 'root' | /bin/awk '{print} END {if (NR == 0) print "All files group owned by root"; else print "Failing files found"}'; else echo "log_file variable in /etc/audit/auditd.conf not set"; fi' returned :

All files group owned by root

PASSED - log_group = root

Compliant file(s):

/etc/audit/auditd.conf - regex '^\s*log_group[\s]*=' found - expect
'^\s*log_group[\s]*=[\s]*root[\s]*\$' found in the following lines:
8: log_group = root

UBTU-24-901380 - Ubuntu 24.04 LTS must be configured so that the audit log directory is not write-accessible by unauthorized users.

Info

If audit information were to become compromised, then forensic analysis and discovery of the true source of potentially malicious system activity is impossible to achieve. To ensure the veracity of audit information, Ubuntu 24.04 LTS must protect audit information from unauthorized deletion.

This requirement can be achieved through multiple methods, which will depend upon system architecture and design. Audit information includes all information (e.g., audit records, audit settings, audit reports) needed to successfully audit information system activity.

Solution

Configure the audit log directory to have a mode of "0750" or less permissive. Determine where the audit logs are stored with the following command: `$ sudo grep -iw ^log_file /etc/audit/auditd.conf log_file = /var/log/audit/audit.log` Using the path of the directory containing the audit logs, configure the audit log directory to have a mode of "0750" or less permissive by using the following command: `$ sudo chmod -R 750 /var/log/audit`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.8
800-171R3	03.03.08
800-53	AU-9
800-53R5	AU-9a.
CAT	II
CCI	CCI-000164
CN-L3	7.1.2.3(d)
CN-L3	7.1.3.3(f)
CN-L3	8.1.3.5(c)
CN-L3	8.1.4.3(c)
CSF	PR.PT-1
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.2
ITSG-33	AU-9
NESA	M5.2.3

NESA	M5.5.2
NESA	T3.6.4
NESA	T8.2.9
NIAV2	SM5
NIAV2	SM6
PCI-DSSV3.2.1	10.5
PCI-DSSV3.2.1	10.5.2
PCI-DSSV4.0	10.3.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270830r1068397_rule
STIG-ID	UBTU-24-901380
VULN-ID	V-270830

Assets

strkbsubuntu9

```
The command 'logfile=$(/bin/grep -iw ^\s*log_file /etc/audit/auditd.conf); if [ ! -z
"${logfile}" ]; then /bin/stat -c "%n %a" $(/bin/dirname ${logfile} | /bin/grep "/" ) | /bin/grep
-Ev '0?[0-7][0-5]0' | /bin/awk '{print} END {if (NR == 0) print "Log directory set to 0750 or
less"; else print "Failing files found"}'; else echo "log_file variable in /etc/audit/auditd.conf
not set"; fi' returned :
```

Log directory set to 0750 or less

UBTU-24-909000 - Ubuntu 24.04 LTS audit system must protect auditing rules from unauthorized change.

Info

Unauthorized disclosure of audit records can reveal system and configuration data to attackers, thus compromising its confidentiality.

Audit information includes all information (e.g., audit records, audit settings, audit reports) needed to successfully audit Ubuntu 24.04 LTS system activity.

In immutable mode, unauthorized users cannot execute changes to the audit system to potentially hide malicious activity and then put the audit rules back.

A system reboot would be noticeable, and a system administrator could then investigate the unauthorized changes.

Solution

Configure the audit system to set the audit rules to be immutable by adding the following line to the end of "/etc/audit/rules.d/audit.rules": -e 2

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.8
800-171R3	03.03.08
800-53	AU-9
800-53R5	AU-9a.
CAT	II
CCI	CCI-000163
CN-L3	7.1.2.3(d)
CN-L3	7.1.3.3(f)
CN-L3	8.1.3.5(c)
CN-L3	8.1.4.3(c)
CSF	PR.PT-1
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.2
ITSG-33	AU-9
NESA	M5.2.3

NESA	M5.5.2
NESA	T3.6.4
NESA	T8.2.9
NIAV2	SM5
NIAV2	SM6
PCI-DSSV3.2.1	10.5
PCI-DSSV3.2.1	10.5.2
PCI-DSSV4.0	10.3.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270832r1068399_rule
STIG-ID	UBTU-24-909000
VULN-ID	V-270832

Assets

strkbsubuntu9

The command `'/bin/grep -v "^$" /etc/audit/audit.rules | /bin/tail -1'` returned :

-e 2

UBTU-24-909890 - Ubuntu 24.04 LTS must use cryptographic mechanisms to protect the integrity of audit tools.

Info

Protecting the integrity of the tools used for auditing purposes is a critical step toward ensuring the integrity of audit information.

Audit information includes all information (e.g., audit records, audit settings, and audit reports) needed to successfully audit information system activity. Audit tools include, but are not limited to, vendor-provided and open source audit tools needed to successfully view and manipulate audit information system activity and records.

Audit tools include custom queries and report generators. It is not uncommon for attackers to replace the audit tools or inject code into the existing tools with the purpose of providing the capability to hide or erase system activity from the audit logs. To address this risk, audit tools must be cryptographically signed to provide the capability to identify when the audit tools have been modified, manipulated, or replaced.

An example is a checksum hash of the file or files.

Solution

Add or update the following selection lines for "/etc/aide/aide.conf" to protect the integrity of the audit tools: #
Audit Tools /sbin/auditctl p+i+n+u+g+s+b+acl+xattrs+sha512 /sbin/auditd p+i+n+u+g+s+b+acl+xattrs+sha512 /
sbin/ausearch p+i+n+u+g+s+b+acl+xattrs+sha512 /sbin/aureport p+i+n+u+g+s+b+acl+xattrs+sha512 /sbin/
autrace p+i+n+u+g+s+b+acl+xattrs+sha512 /sbin/augenrules p+i+n+u+g+s+b+acl+xattrs+sha512

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.8
800-171R3	03.03.08
800-53	AU-9(3)
800-53R5	AU-9(3)
CAT	II
CCI	CCI-001496
CN-L3	8.1.4.3(d)
CSF	PR.PT-1
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.5.33
ISO-27001-2022	A.8.15
ISO/IEC-27001	A.12.4.2
ITSG-33	AU-9(3)
NESA	M5.2.3
NESA	M5.5.2

NIAV2	SM5
NIAV2	SM6
PCI-DSSV3.2.1	10.5
PCI-DSSV3.2.1	10.5.2
PCI-DSSV4.0	10.3.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270831r1135002_rule
STIG-ID	UBTU-24-909890
TBA-FIISB	37.3.3
VULN-ID	V-270831

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

PASSED - autrace

Compliant file(s):

/etc/aide/aide.conf - regex '^[\s]*/sbin/autrace[\s]+p+i+n+u+g\s+b+acl+xattrs\+sha512' found - expect '^[\s]*/sbin/autrace[\s]+p+i+n+u+g\s+b+acl+xattrs\+sha512' found in the following lines:

118: /sbin/autrace p+i+n+u+g+s+b+acl+xattrs+sha512

PASSED - auditctl

Compliant file(s):

/etc/aide/aide.conf - regex '^[\s]*/sbin/auditctl[\s]+p+i+n+u+g\s+b+acl+xattrs\+sha512' found - expect '^[\s]*/sbin/auditctl[\s]+p+i+n+u+g\s+b+acl+xattrs\+sha512' found in the following lines:

114: /sbin/auditctl p+i+n+u+g+s+b+acl+xattrs+sha512

PASSED - aureport

Compliant file(s):

/etc/aide/aide.conf - regex '^[\s]*/sbin/aureport[\s]+p+i+n+u+g\s+b+acl+xattrs\+sha512' found - expect '^[\s]*/sbin/aureport[\s]+p+i+n+u+g\s+b+acl+xattrs\+sha512' found in the following lines:

117: /sbin/aureport p+i+n+u+g+s+b+acl+xattrs+sha512

PASSED - augenrules

Compliant file(s):

/etc/aide/aide.conf - regex '^[\s]*/sbin/augenrules[\s]+p+i+n+u+g\s+b+acl+xattrs\+sha512' found - expect '^[\s]*/sbin/augenrules[\s]+p+i+n+u+g\s+b+acl+xattrs\+sha512' found in the following lines:

119: /sbin/augenrules p+i+n+u+g+s+b+acl+xattrs+sha512

PASSED - auditd

Compliant file(s):

/etc/aide/aide.conf - regex '^[\s]*/sbin/auditd[\s]+p+i+n+u+g\s+b+acl+xattrs\+sha512' found - expect '^[\s]*/sbin/auditd[\s]+p+i+n+u+g\s+b+acl+xattrs\+sha512' found in the following lines:

115: /sbin/auditd p+i+n+u+g+s+b+acl+xattrs+sha512

PASSED - ausearch

Compliant file(s):

```
/etc/aide/aide.conf - regex '^[\s]*/sbin/ausearch[\s]+p+i+n+u+g+s+b+acl+xattrs\
+sha512' found - expect '^[\s]*/sbin/ausearch[\s]+p+i+n+u+g+s+b+acl+xattrs+sha512' found
in the following lines:
    116: /sbin/ausearch p+i+n+u+g+s+b+acl+xattrs+sha512
```

Audits ERROR,INFO,WARNING

UBTU-24-100110 - Ubuntu 24.04 LTS must configure AIDE to perform file integrity checking on the file system if installed.

Info

Without verification, security functions may not operate correctly and the failure may go unnoticed. Security function is defined as the hardware, software, and/or firmware of the information system responsible for enforcing the system security policy and supporting the isolation of code and data on which the protection is based.

Security functionality includes, but is not limited to, establishing system accounts, configuring access authorizations (i.e., permissions, privileges), setting events to be audited, and setting intrusion detection parameters. This requirement applies to Ubuntu 24.04 LTS performing security function verification/testing and/or systems and environments that require this functionality. NOTE: Nessus has provided the target output to assist in reviewing the benchmark to ensure target compliance.

Solution

Initialize the AIDE package (this may take a few minutes): `$ sudo aideinit` Running `aide --init...` The new database will need to be renamed to be read by AIDE: `$ sudo cp -p /var/lib/aide/aide.db.new /var/lib/aide/aide.db` Perform a manual check: `$ sudo aide -c /etc/aide/aide.conf --check` Example output: ... Start timestamp: 2024-10-30 14:22:38 -0400 (AIDE 0.18.6) AIDE found differences between database and filesystem!! ... Done.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-6a.
800-53R5	SI-6a.
CAT	II
CCI	CCI-002696
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-6a.
RULE-ID	SV-270650r1155241_rule
STIG-ID	UBTU-24-100110
VULN-ID	V-270650

Assets

strkbsubuntu9

The command `'/bin/dpkg -s aide 2>&1 | /bin/grep -E '(Status:|not installed)''` returned :

Status: install ok installed

UBTU-24-100120 - Ubuntu 24.04 LTS must be configured so that the script which runs each 30 days or less to check file integrity is the default one.

Info

Without verification, security functions may not operate correctly and the failure may go unnoticed. Security function is defined as the hardware, software, and/or firmware of the information system responsible for enforcing the system security policy and supporting the isolation of code and data on which the protection is based.

Security functionality includes, but is not limited to, establishing system accounts, configuring access authorizations (i.e., permissions, privileges), setting events to be audited, and setting intrusion detection parameters. Notifications provided by information systems include, for example, electronic alerts to system administrators, messages to local computer consoles, and/or hardware indications, such as lights. This requirement applies to Ubuntu 24.04 LTS performing security function verification/testing and/or systems and environments that require this functionality. NOTE: Nessus has provided the target output to assist in reviewing the benchmark to ensure target compliance.

Solution

The cron file for AIDE is fairly complex as it creates the report.

This file is installed with the "aide-common" package, and the default can be restored by copying it from the package: Download the original package to the /tmp dir: `$ cd /tmp; apt download aide-common` Extract the aide script to its original place: `$ dpkg-deb --fsys-tarfile /tmp/aide-common_*.deb | sudo tar -x --wildcards ./usr/share/aide/config/cron.daily/dailyaidecheck* -C /` Copy it to the cron.daily directory: `$ sudo cp -f /usr/share/aide/config/cron.daily/aide /etc/cron.daily/dailyaidecheck*`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SI-6b.
800-53R5	SI-6b.
CAT	II
CCI	CCI-002699
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SI-6a.
RULE-ID	SV-270651r1068395_rule
STIG-ID	UBTU-24-100120
VULN-ID	V-270651

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

WARNING - Manual Review Required - sha256sum /etc/aide/aide.conf
The command '/bin/sha256sum /etc/aide/aide.conf' returned :

47980d946a87ba3d8630e5b03f79dc12ae242cc834028dc879af892f3e8509b1 /etc/aide/aide.conf

WARNING - Manual Review Required - cron aide
The command '/bin/grep -r aide /etc/cron* /etc/crontab' returned :

/etc/cron.daily/dailyaidecheck:SCRIPT="/usr/share/aide/bin/dailyaidecheck"


```
/etc/cron.daily/dailyaidecheck:          capsh --caps="cap_dac_read_search,cap_audit_write
+eip cap_setpcap,cap_setuid,cap_setgid+ep" --keep=1 --user=_aide --
addamb=cap_dac_read_search,cap_audit_write -- -c "${SCRIPT}" --crondaily"
```

WARNING - Manual Review Required - systemctl list-timers aide
The command '/bin/systemctl list-timers | grep aide' returned :

```
Mon 2026-08-24 02:22:45 UTC          8h -          -
dailyaidecheck.timer                dailyaidecheck.service
```

UBTU-24-200250 - Ubuntu 24.04 LTS must automatically remove or disable emergency accounts after 72 hours.

Info

Temporary accounts are privileged or nonprivileged accounts established during pressing circumstances, such as new software or hardware configuration or an incident response, where the need for prompt account activation requires bypassing normal account authorization procedures.

If any inactive temporary accounts are left enabled on the system and are not either manually removed or automatically expired within 72 hours, the security posture of the system will be degraded and exposed to exploitation by unauthorized users or insider threat actors. Temporary accounts are different from emergency accounts.

Emergency accounts, also known as "last resort" or "break glass" accounts, are local logon accounts enabled on the system for emergency use by authorized system administrators to manage a system when standard logon methods are failing or not available.

Emergency accounts are not subject to manual removal or scheduled expiration requirements. The automatic expiration of temporary accounts may be extended as needed by the circumstances but it must not be extended indefinitely.

A documented permanent account must be established for privileged users who need long-term maintenance accounts. Satisfies: SRG-OS-000002-GPOS-00002, SRG-OS-000123-GPOS-00064 NOTE: Nessus has provided the target output to assist in reviewing the benchmark to ensure target compliance.

Solution

Configure Ubuntu 24.04 LTS to expire temporary accounts after 72 hours with the following command: `$ sudo chage -E $(date -d +3days +%Y-%m-%d) <temporary_account_name>`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.1
800-171R3	03.01.01
800-53	AC-2(2)
800-53R5	AC-2(2)
CAT	II
CCI	CCI-000016
CCI	CCI-001682
CN-L3	7.1.3.2(e)
CSF	PR.AC-1
CSF	PR.AC-4
CSF2.0	DE.CM-01
CSF2.0	DE.CM-03
CSF2.0	PR.AA-01
CSF2.0	PR.AA-05
CSF2.0	PR.DS-10
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b

HIPAA	164.306(a)(1)
HIPAA	164.312(a)(1)
ISO-27001-2022	A.5.16
ISO-27001-2022	A.5.18
ISO-27001-2022	A.8.2
ISO/IEC-27001	A.9.2.1
ITSG-33	AC-2(2)
NIAV2	AM28
NIAV2	NS5j
NIAV2	SS14e
QCSC-V1	5.2.2
QCSC-V1	8.2.1
QCSC-V1	13.2
QCSC-V1	15.2
RULE-ID	SV-270682r1066535_rule
STIG-ID	UBTU-24-200250
VULN-ID	V-270682

Assets

strkbsubuntu9

The command 'UIDMIN=`/bin/awk '/^\s\*UID\_MIN/{print \$2}' /etc/login.defs`; /bin/awk -v var="\$UIDMIN" -F: '\$3>=var{print \$1}' /etc/passwd | while read i; do /bin/chage -l \$i | /bin/grep -E '(Account|Password) expires'; /bin/echo "User: \$i"; done | /bin/awk -F: 'BEGIN{printf("%-30s %-15s %s\n", "User", "Account", "Password")} /Account/{a=\$2} /Password/{p=\$2} /User/{printf("%-30s %-15s %s\n", \$2, a, p)}}' returned :

User	Account	Password
nobody	never	never
strkadmin24	never	never
vagrant	never	never

UBTU-24-200680 - Ubuntu 24.04 LTS must be configured to enforce the acknowledgement of the Standard Mandatory DOD Notice and Consent Banner for all SSH connections.

Info

The banner must be acknowledged by the user prior to allowing the user access to Ubuntu 24.04 LTS. This provides assurance that the user has seen the message and accepted the conditions for access. If the consent banner is not acknowledged by the user, DOD will not be in compliance with system use notifications required by law. Ubuntu 24.04 LTS must prevent further activity until the user executes a positive action to manifest agreement. NOTE: Nessus has not performed this check. Please review the benchmark to ensure target compliance.

Solution

Configure Ubuntu 24.04 LTS to prompt a user to acknowledge the Standard Mandatory DOD Notice and Consent Banner before granting access: `$ sudo vi /etc/profile.d/ssh_confirm.sh #!/bin/bash if [-n "$SSH_CLIENT"] || [-n "$SSH_TTY"]; then while true; do read -p " You are accessing a U.S. Government (USG) Information System (IS) that is provided for USG-authorized use only. By using this IS (which includes any device attached to this IS), you consent to the following conditions: -The USG routinely intercepts and monitors communications on this IS for purposes including, but not limited to, penetration testing, COMSEC monitoring, network operations and defense, personnel misconduct (PM), law enforcement (LE), and counterintelligence (CI) investigations. -At any time, the USG may inspect and seize data stored on this IS. -Communications using, or data stored on, this IS are not private, are subject to routine monitoring, interception, and search, and may be disclosed or used for any USG-authorized purpose. -This IS includes security measures (e.g., authentication and access controls) to protect USG interests--not for your personal benefit or privacy. -Notwithstanding the above, using this IS does not constitute consent to PM, LE or CI investigative searching or monitoring of the content of privileged communications, or work product, related to personal representation or services by attorneys, psychotherapists, or clergy, and their assistants. Such communications and work product are private and confidential. See User Agreement for details. Do you agree? [y/N] " yn case $yn in [Yy]*) break ;; [Nn]*) exit 1 ;; esac done fi`
Note: The "ssh_confirm.sh" script is provided as a supplemental file to this document.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.1.9
800-171R3	03.01.09
800-53	AC-8b.
800-53R5	AC-8b.
CAT	II
CCI	CCI-000050
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.5
ITSG-33	AC-8b.
NESA	M5.2.5
NESA	T5.5.1
NIAV2	AM10f
RULE-ID	SV-270694r1066571_rule

STIG-ID	UBTU-24-200680
TBA-FIISB	45.2.4
VULN-ID	V-270694

Assets

strkbsubuntu9

UBTU-24-300041 - Ubuntu 24.04 LTS must be configured to prohibit or restrict the use of functions, ports, protocols, and/or services, as defined in the Ports, Protocols, and Services Management Category Assurance List (PPSM CAL) and vulnerability assessments.

Info

To prevent unauthorized connection of devices, unauthorized transfer of information, or unauthorized tunneling (i.e., embedding of data types within data types), organizations must disable or restrict unused or unnecessary physical and logical ports/protocols on information systems. Operating systems are capable of providing a wide variety of functions and services.

Some of the functions and services provided by default may not be necessary to support essential organizational operations.

Additionally, it is sometimes convenient to provide multiple services from a single component (e.g., VPN and IPS); however, doing so increases risk over limiting the services provided by any one component. To support the requirements and principles of least functionality, Ubuntu 24.04 LTS must support the organizational requirements, providing only essential capabilities and limiting the use of ports, protocols, and/or services to only those required, authorized, and approved to conduct official business or to address authorized quality of life issues. NOTE: Nessus has provided the target output to assist in reviewing the benchmark to ensure target compliance.

Solution

Add all ports, protocols, or services allowed by the PPSM CLSA by using the following command: `$ sudo ufw allow <direction> <port/protocol/service>` Where the direction is "in" or "out" and the port is the one corresponding to the protocol or service allowed. To deny access to ports, protocols, or services, use: `$ sudo ufw deny <direction> <port/protocol/service>`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.6
800-171	3.4.7
800-171R3	03.04.06b.
800-53	CM-7b.
800-53R5	CM-7b.
CAT	II
CCI	CCI-000382
CN-L3	7.1.3.5(c)
CN-L3	7.1.3.7(d)
CN-L3	8.1.4.4(b)
CSF	PR.IP-1
CSF	PR.PT-3
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	CM-7a.

NIAV2	SS13b
NIAV2	SS14a
NIAV2	SS14c
PCI-DSSV3.2.1	2.2.2
PCI-DSSV4.0	2.2.4
QCSC-V1	3.2
RULE-ID	SV-270719r1067172_rule
STIG-ID	UBTU-24-300041
SWIFT-CSCV1	2.3
VULN-ID	V-270719

Assets

strkbsubuntu9

The command '/sbin/ufw show raw' returned :

IPV4 (raw):

Chain INPUT (policy DROP 194 packets, 14002 bytes)

pkts	bytes	target	prot	opt	in	out	source	destination
31991	35142924	ufw-before-logging-input	0	--	*	*	0.0.0.0/0	0.0.0.0/0
31991	35142924	ufw-before-input	0	--	*	*	0.0.0.0/0	0.0.0.0/0
194	14002	ufw-after-input	0	--	*	*	0.0.0.0/0	0.0.0.0/0
194	14002	ufw-after-logging-input	0	--	*	*	0.0.0.0/0	0.0.0.0/0
194	14002	ufw-reject-input	0	--	*	*	0.0.0.0/0	0.0.0.0/0
194	14002	ufw-track-input	0	--	*	*	0.0.0.0/0	0.0.0.0/0

Chain FORWARD (policy DROP 0 packets, 0 bytes)

pkts	bytes	target	prot	opt	in	out	source	destination
0	0	ufw-before-logging-forward	0	--	*	*	0.0.0.0/0	
0.0.0.0/0	0	ufw-before-forward	0	--	*	*	0.0.0.0/0	0.0.0.0/0
0	0	ufw-after-forward	0	--	*	*	0.0.0.0/0	0.0.0.0/0
0	0	ufw-after-logging-forward	0	--	*	*	0.0.0.0/0	
0.0.0.0/0	0	ufw-reject-forward	0	--	*	*	0.0.0.0/0	0.0.0.0/0
0	0	ufw-track-forward	0	--	*	*	0.0.0.0/0	0.0.0.0/0

Chain OUTPUT (policy ACCEPT 2 packets, 168 bytes)

pkts	bytes	target	prot	opt	in	out	source	destination
28870	4087236	ufw-before-logging-output	0	--	*	*	0.0.0.0/0	
0.0.0.0/0	28870	4087236	ufw-before-output	0	--	*	*	0.0.0.0/0
1320	118781	ufw-after-output	0	--	*	*	0.0.0.0/0	0.0.0.0/0
1320	118781	ufw-after-logging-output	0	--	*	*	0.0.0.0/0	0.0.0.0/0
1320	118781	ufw-reject-output	0	--	*	*	0.0.0.0/0	0.0.0.0/0
1320	118781	ufw-track-output	0	--	*	*	0.0.0.0/0	0.0.0.0/0

Chain ufw-after-forward (1 references)

```

pkts      bytes target      prot opt in      out      source      destination
Chain ufw-after-input (1 references)
pkts      bytes target      prot opt in      out      source      destination
0          0 ufw-skip-to-policy-input 17 -- *      *      0.0.0.0/0    0.0.0.0/0
udp dpt:137
0          0 ufw-skip-to-policy-input 17 -- *      *      0.0.0.0/0    0.0.0.0/0
udp dpt:138
0          0 ufw-skip-to-policy-input 6  -- *      *      0.0.0.0/0    0.0.0.0/0
tcp dpt:139
0          0 ufw-skip-to-policy-input 6  -- *      *      0.0.0.0/0    0.0.0.0/0
tcp dpt:445
0          0 ufw-skip-to-policy-input 17 -- *      *      0.0.0.0/0    0.0.0.0/0
udp dpt:67
0          0 ufw-skip-to-policy-input 17 -- *      *      0.0.0.0/0    0.0.0.0/0
udp dpt:68
0          0 ufw-skip-to-policy-input 0  -- *      *      0.0.0.0/0    0.0.0.0/0
ADDRTYPE match dst-type BROADCAST

Chain ufw-after-logging-forward (1 references)
pkts      bytes target      prot opt in      out      source      destination
0          0 LOG          0  -- *      *      0.0.0.0/0    0.0.0.0/0
limit: avg 3/min burst 10 LOG flags 0 level 4 prefix "[UFW BLOCK] "

Chain ufw-after-logging-input (1 references)
pkts      bytes target      prot opt in      out      source      destination
51         4158 LOG          0  -- *      *      0.0.0.0/0    0.0.0.0/0
limit: avg 3/min burst 10 LOG flags 0 level 4 prefix "[UFW BLOCK] "

Chain ufw-after-logging-output (1 references)
pkts      bytes target      prot opt in      out      source      destination

Chain ufw-after-output (1 references)
pkts      bytes target      prot opt in      out      source      destination

Chain ufw-before-forward (1 references)
pkts      bytes target      prot opt in      out      source      destination
0          0 ACCEPT       0  -- *      *      0.0.0.0/0    0.0.0.0/0
ctstate RELATED,ESTABLISHED
0          0 ACCEPT       1  -- *      *      0.0.0.0/0    0.0.0.0/0
icmp type 3
0          0 ACCEPT       1  -- *      *      0.0.0.0/0    0.0.0.0/0
icmp type 11
0          0 ACCEPT       1  -- *      *      0.0.0.0/0    0.0.0.0/0
icmp type 12
0          0 ACCEPT       1  -- *      *      0.0.0.0/0    0.0.0.0/0
icmp type 8
0          0 ufw-user-forward 0  -- *      *      0.0.0.0/0    0.0.0.0/0

Chain ufw-before-input (1 references)
pkts      bytes target      prot opt in      out      source      destination
2428      358174 ACCEPT       0  -- lo      *      0.0.0.0/0    0.0.0.0/0
29280     34765976 ACCEPT       0  -- *      *      0.0.0.0/0    0.0.0.0/0
ctstate RELATED,ESTABLISHED
10         520 ufw-logging-deny 0  -- *      *      0.0.0.0/0    0.0.0.0/0
ctstate INVALID
10         520 DROP        0  -- *      *      0.0.0.0/0    0.0.0.0/0
ctstate INVALID
0          0 ACCEPT       1  -- *      *      0.0.0.0/0    0.0.0.0/0
icmp type 3
0          0 ACCEPT       1  -- *      *      0.0.0.0/0    0.0.0.0/0
icmp type 11
0          0 ACCEPT       1  -- *      *      0.0.0.0/0    0.0.0.0/0
icmp type 12
2          64 ACCEPT       1  -- *      *      0.0.0.0/0    0.0.0.0/0
icmp type 8
0          0 ACCEPT       17 -- *      *      0.0.0.0/0    0.0.0.0/0      udp
spt:67 dpt:68
271       18190 ufw-not-local 0  -- *      *      0.0.0.0/0    0.0.0.0/0
0          0 ACCEPT       17 -- *      *      0.0.0.0/0    224.0.0.251      udp
dpt:5353
0          0 ACCEPT       17 -- *      *      0.0.0.0/0    239.255.255.250  udp
dpt:1900

```



```

271    18190 ufw-user-input 0    -- *      *      0.0.0.0/0      0.0.0.0/0

Chain ufw-before-logging-forward (1 references)
pkts    bytes target    prot opt in    out    source    destination
Chain ufw-before-logging-input (1 references)
pkts    bytes target    prot opt in    out    source    destination
Chain ufw-before-logging-output (1 references)
pkts    bytes target    prot opt in    out    source    destination
Chain ufw-before-output (1 references)
pkts    bytes target    prot opt in    out    source    destination
2428    358174 ACCEPT    0    -- *      lo    0.0.0.0/0    0.0.0.0/0
25122   3610281 ACCEPT    0    -- *      *    0.0.0.0/0    0.0.0.0/0
ctstate RELATED,ESTABLISHED
1320    118781 ufw-user-output 0    -- *      *    0.0.0.0/0    0.0.0.0/0

Chain ufw-logging-allow (0 references)
pkts    bytes target    prot opt in    out    source    destination
0        0 LOG      0    -- *      *    0.0.0.0/0    0.0.0.0/0
limit: avg 3/min burst 10 LOG flags 0 level 4 prefix "[UFW ALLOW] "

Chain ufw-logging-deny (2 references)
pkts    bytes target    prot opt in    out    source    destination
10       520 RETURN    0    -- *      *    0.0.0.0/0    0.0.0.0/0
ctstate INVALID limit: avg 3/min burst 10
0        0 LOG      0    -- *      *    0.0.0.0/0    0.0.0.0/0
limit: avg 3/min burst 10 LOG flags 0 level 4 prefix "[UFW BLOCK] "

Chain ufw-not-local (1 references)
pkts    bytes target    prot opt in    out    source    destination
271     18190 RETURN    0    -- *      *    0.0.0.0/0    0.0.0.0/0
ADDRTYPE match dst-type LOCAL
0        0 RETURN    0    -- *      *    0.0.0.0/0    0.0.0.0/0
ADDRTYPE match dst-type MULTICAST
0        0 RETURN    0    -- *      *    0.0.0.0/0    0.0.0.0/0
ADDRTYPE match dst-type BROADCAST
0        0 ufw-logging-deny 0    -- *      *    0.0.0.0/0    0.0.0.0/0
limit: avg 3/min burst 10
0        0 DROP      0    -- *      *    0.0.0.0/0    0.0.0.0/0

Chain ufw-reject-forward (1 references)
pkts    bytes target    prot opt in    out    source    destination
Chain ufw-reject-input (1 references)
pkts    bytes target    prot opt in    out    source    destination
Chain ufw-reject-output (1 references)
pkts    bytes target    prot opt in    out    source    destination
Chain ufw-skip-to-policy-forward (0 references)
pkts    bytes target    prot opt in    out    source    destination
0        0 DROP      0    -- *      *    0.0.0.0/0    0.0.0.0/0
Chain ufw-skip-to-policy-input (7 references)
pkts    bytes target    prot opt in    out    source    destination
0        0 DROP      0    -- *      *    0.0.0.0/0    0.0.0.0/0
Chain ufw-skip-to-policy-output (0 references)
pkts    bytes target    prot opt in    out    source    destination
0        0 ACCEPT    0    -- *      *    0.0.0.0/0    0.0.0.0/0
Chain ufw-track-forward (1 references)
pkts    bytes target    prot opt in    out    source    destination
Chain ufw-track-input (1 references)
pkts    bytes target    prot opt in    out    source    destination
Chain ufw-track-output (1 references)
pkts    bytes target    prot opt in    out    source    destination
670     40200 ACCEPT    6    -- *      *    0.0.0.0/0    0.0.0.0/0
ctstate NEW

```

```

        648      78413 ACCEPT      17  --  *      *      0.0.0.0/0      0.0.0.0/0
ctstate NEW

Chain ufw-user-forward (1 references)
  pkts      bytes target      prot opt in      out      source      destination
Chain ufw-user-input (1 references)
  pkts      bytes target      prot opt in      out      source      destination
    77      4188 ACCEPT      6    --  *      *      0.0.0.0/0      0.0.0.0/0      tcp
dpt:22

Chain ufw-user-limit (0 references)
  pkts      bytes target      prot opt in      out      source      destination
    0         0 LOG      0    --  *      *      0.0.0.0/0      0.0.0.0/0
limit: avg 3/min burst 5 LOG flags 0 level 4 prefix "[UFW LIMIT BLOCK] "
    0         0 REJECT      0    --  *      *      0.0.0.0/0      0.0.0.0/0
reject-with icmp-port-unreachable

Chain ufw-user-limit-accept (0 references)
  pkts      bytes target      prot opt in      out      source      destination
    0         0 ACCEPT      0    --  *      *      0.0.0.0/0      0.0.0.0/0

Chain ufw-user-logging-forward (0 references)
  pkts      bytes target      prot opt in      out      source      destination
Chain ufw-user-logging-input (0 references)
  pkts      bytes target      prot opt in      out      source      destination
Chain ufw-user-logging-output (0 references)
  pkts      bytes target      prot opt in      out      source      destination
Chain ufw-user-output (1 references)
  pkts      bytes target      prot opt in      out      source      destination
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts      bytes target      prot opt in      out      source      destination
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts      bytes target      prot opt in      out      source      destination
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts      bytes target      prot opt in      out      source      destination
Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts      bytes target      prot opt in      out      source      destination
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts      bytes target      prot opt in      out      source      destination
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts      bytes target      prot opt in      out      source      destination
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
  pkts      bytes target      prot opt in      out      source      destination
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts      bytes target      prot opt in      out      source      destination
Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts      bytes target      prot opt in      out      source      destination
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts      bytes target      prot opt in      out      source      destination
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts      bytes target      prot opt in      out      source      destination

IPV6:
Chain INPUT (policy DROP 0 packets, 0 bytes)
  pkts      bytes target      prot opt in      out      source      destination
    0         0 ufw6-before-logging-input 0  --  *      *      ::/0      ::/0
    0         0 ufw6-before-input 0  --  *      *      ::/0      ::/0
    0         0 ufw6-after-input 0  --  *      *      ::/0      ::/0

```

```

0      0 ufw6-after-logging-input 0  -- *      *      ::/0      ::/0
0      0 ufw6-reject-input 0  -- *      *      ::/0      ::/0
0      0 ufw6-track-input 0  -- *      *      ::/0      ::/0

Chain FORWARD (policy DROP 0 packets, 0 bytes)
pkts    bytes target    prot opt in    out    source    destination
0      0 ufw6-before-logging-forward 0  -- *      *      ::/0      ::/0
0      0 ufw6-before-forward 0  -- *      *      ::/0      ::/0
0      0 ufw6-after-forward 0  -- *      *      ::/0      ::/0
0      0 ufw6-after-logging-forward 0  -- *      *      ::/0      ::/0
0      0 ufw6-reject-forward 0  -- *      *      ::/0      ::/0
0      0 ufw6-track-forward 0  -- *      *      ::/0      ::/0

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts    bytes target    prot opt in    out    source    destination
2      112 ufw6-before-logging-output 0  -- *      *      ::/0      ::/0
2      112 ufw6-before-output 0  -- *      *      ::/0      ::/0
0      0 ufw6-after-output 0  -- *      *      ::/0      ::/0
0      0 ufw6-after-logging-output 0  -- *      *      ::/0      ::/0
0      0 ufw6-reject-output 0  -- *      *      ::/0      ::/0
0      0 ufw6-track-output 0  -- *      *      ::/0      ::/0

Chain ufw6-after-forward (1 references)
pkts    bytes target    prot opt in    out    source    destination

Chain ufw6-after-input (1 references)
pkts    bytes target    prot opt in    out    source    destination
0      0 ufw6-skip-to-policy-input 17  -- *      *      ::/0      ::/0
      udp dpt:137
0      0 ufw6-skip-to-policy-input 17  -- *      *      ::/0      ::/0
      udp dpt:138
0      0 ufw6-skip-to-policy-input 6  -- *      *      ::/0      ::/0
      tcp dpt:139
0      0 ufw6-skip-to-policy-input 6  -- *      *      ::/0      ::/0
      tcp dpt:445
0      0 ufw6-skip-to-policy-input 17  -- *      *      ::/0      ::/0
      udp dpt:546
0      0 ufw6-skip-to-policy-input 17  -- *      *      ::/0      ::/0
      udp dpt:547

Chain ufw6-after-logging-forward (1 references)
pkts    bytes target    prot opt in    out    source    destination
0      0 LOG      0  -- *      *      ::/0      ::/0
limit: avg 3/min burst 10 LOG flags 0 level 4 prefix "[UFW BLOCK] "

Chain ufw6-after-logging-input (1 references)
pkts    bytes target    prot opt in    out    source    destination
0      0 LOG      0  -- *      *      ::/0      ::/0
limit: avg 3/min burst 10 LOG flags 0 level 4 prefix "[UFW BLOCK] "

Chain ufw6-after-logging-output (1 references)
pkts    bytes target    prot opt in    out    source    destination

Chain ufw6-after-output (1 references)
pkts    bytes target    prot opt in    out    source    destination

Chain ufw6-before-forward (1 references)
pkts    bytes target    prot opt in    out    source    destination
0      0 DROP      0  -- *      *      ::/0      ::/0
type:0
rt

```

```

0 0 ACCEPT 0 -- * * ::/0 ::/0
ctstate RELATED,ESTABLISHED
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 1
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 2
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 3
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 4
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 128
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 129
0 0 ufw6-user-forward 0 -- * * ::/0 ::/0

```

Chain ufw6-before-input (1 references)

```

pkts bytes target prot opt in out source destination
0 0 ACCEPT 0 -- lo * * ::/0 ::/0
0 0 DROP 0 -- * * ::/0 ::/0
type:0
0 0 ACCEPT 0 -- * * ::/0 ::/0
ctstate RELATED,ESTABLISHED
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 129
0 0 ufw6-logging-deny 0 -- * * ::/0 ::/0
ctstate INVALID
0 0 DROP 0 -- * * ::/0 ::/0
ctstate INVALID
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 1
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 2
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 3
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 4
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 128
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 133 HL match HL == 255
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 134 HL match HL == 255
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 135 HL match HL == 255
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 136 HL match HL == 255
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 141 HL match HL == 255
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 142 HL match HL == 255
0 0 ACCEPT 58 -- * * fe80::/10 ::/0
ipv6-icmp type 130
0 0 ACCEPT 58 -- * * fe80::/10 ::/0
ipv6-icmp type 131
0 0 ACCEPT 58 -- * * fe80::/10 ::/0
ipv6-icmp type 132
0 0 ACCEPT 58 -- * * fe80::/10 ::/0
ipv6-icmp type 143
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 148 HL match HL == 255
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 149 HL match HL == 255
0 0 ACCEPT 58 -- * * fe80::/10 ::/0
ipv6-icmp type 151 HL match HL == 1
0 0 ACCEPT 58 -- * * fe80::/10 ::/0
ipv6-icmp type 152 HL match HL == 1
0 0 ACCEPT 58 -- * * fe80::/10 ::/0
ipv6-icmp type 153 HL match HL == 1
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 144
0 0 ACCEPT 58 -- * * ::/0 ::/0
ipv6-icmp type 145

```

0	0 ACCEPT	58	--	*	*	::/0	::/0	
ipv6-icmp	type 146							
0	0 ACCEPT	58	--	*	*	::/0	::/0	
ipv6-icmp	type 147							
0	0 ACCEPT	17	--	*	*	fe80::/10	fe80::/10	udp
spt:547 dpt:546								
0	0 ACCEPT	17	--	*	*	::/0	ff02::fb	udp
dpt:5353								
0	0 ACCEPT	17	--	*	*	::/0	ff02::f	udp
dpt:1900								
0	0 ufw6-user-input	0	--	*	*	::/0	::/0	

Chain ufw6-before-logging-forward (1 references)

pkts	bytes	target	prot	opt	in	out	source	destination
------	-------	--------	------	-----	----	-----	--------	-------------

Chain ufw6-before-logging-input (1 references)

pkts	bytes	target	prot	opt	in	out	source	destination
------	-------	--------	------	-----	----	-----	--------	-------------

Chain ufw6-before-logging-output (1 references)

pkts	bytes	target	prot	opt	in	out	source	destination
------	-------	--------	------	-----	----	-----	--------	-------------

Chain ufw6-before-output (1 references)

pkts	bytes	target	prot	opt	in	out	source	destination
0	0 ACCEPT	0	--	*		lo	::/0	::/0
0	0 DROP	0	--	*		*	::/0	::/0
type:0								rt
0	0 ACCEPT	0	--	*		*	::/0	::/0
ctstate	RELATED,ESTABLISHED							
0	0 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 1							
0	0 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 2							
0	0 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 3							
0	0 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 4							
0	0 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 128							
0	0 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 129							
2	112 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 133 HL match	HL == 255						
0	0 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 136 HL match	HL == 255						
0	0 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 135 HL match	HL == 255						
0	0 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 134 HL match	HL == 255						
0	0 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 141 HL match	HL == 255						
0	0 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 142 HL match	HL == 255						
0	0 ACCEPT	58	--	*		*	fe80::/10	::/0
ipv6-icmp	type 130							
0	0 ACCEPT	58	--	*		*	fe80::/10	::/0
ipv6-icmp	type 131							
0	0 ACCEPT	58	--	*		*	fe80::/10	::/0
ipv6-icmp	type 132							
0	0 ACCEPT	58	--	*		*	fe80::/10	::/0
ipv6-icmp	type 143							
0	0 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 148 HL match	HL == 255						
0	0 ACCEPT	58	--	*		*	::/0	::/0
ipv6-icmp	type 149 HL match	HL == 255						
0	0 ACCEPT	58	--	*		*	fe80::/10	::/0
ipv6-icmp	type 151 HL match	HL == 1						
0	0 ACCEPT	58	--	*		*	fe80::/10	::/0
ipv6-icmp	type 152 HL match	HL == 1						
0	0 ACCEPT	58	--	*		*	fe80::/10	::/0
ipv6-icmp	type 153 HL match	HL == 1						
0	0 ufw6-user-output	0	--	*	*	*	::/0	::/0

Chain ufw6-logging-allow (0 references)

```

    pkts      bytes target    prot opt in     out     source      destination
    0          0 LOG      0      -- *    *         ::/0        ::/0
limit: avg 3/min burst 10 LOG flags 0 level 4 prefix "[UFW ALLOW] "

Chain ufw6-logging-deny (1 references)
    pkts      bytes target    prot opt in     out     source      destination
    0          0 RETURN   0      -- *    *         ::/0        ::/0
ctstate INVALID limit: avg 3/min burst 10
    0          0 LOG      0      -- *    *         ::/0        ::/0
limit: avg 3/min burst 10 LOG flags 0 level 4 prefix "[UFW BLOCK] "

Chain ufw6-reject-forward (1 references)
    pkts      bytes target    prot opt in     out     source      destination

Chain ufw6-reject-input (1 references)
    pkts      bytes target    prot opt in     out     source      destination

Chain ufw6-reject-output (1 references)
    pkts      bytes target    prot opt in     out     source      destination

Chain ufw6-skip-to-policy-forward (0 references)
    pkts      bytes target    prot opt in     out     source      destination
    0          0 DROP     0      -- *    *         ::/0        ::/0

Chain ufw6-skip-to-policy-input (6 references)
    pkts      bytes target    prot opt in     out     source      destination
    0          0 DROP     0      -- *    *         ::/0        ::/0

Chain ufw6-skip-to-policy-output (0 references)
    pkts      bytes target    prot opt in     out     source      destination
    0          0 ACCEPT   0      -- *    *         ::/0        ::/0

Chain ufw6-track-forward (1 references)
    pkts      bytes target    prot opt in     out     source      destination

Chain ufw6-track-input (1 references)
    pkts      bytes target    prot opt in     out     source      destination

Chain ufw6-track-output (1 references)
    pkts      bytes target    prot opt in     out     source      destination
    0          0 ACCEPT   6      -- *    *         ::/0        ::/0
ctstate NEW
    0          0 ACCEPT   17     -- *    *         ::/0        ::/0
ctstate NEW

Chain ufw6-user-forward (1 references)
    pkts      bytes target    prot opt in     out     source      destination

Chain ufw6-user-input (1 references)
    pkts      bytes target    prot opt in     out     source      destination
    0          0 ACCEPT   6      -- *    *         ::/0        ::/0
dpt:22                                     tcp

Chain ufw6-user-limit (0 references)
    pkts      bytes target    prot opt in     out     source      destination
    0          0 LOG      0      -- *    *         ::/0        ::/0
limit: avg 3/min burst 5 LOG flags 0 level 4 prefix "[UFW LIMIT BLOCK] "
    0          0 REJECT   0      -- *    *         ::/0        ::/0
reject-with icmp6-port-unreachable

Chain ufw6-user-limit-accept (0 references)
    pkts      bytes target    prot opt in     out     source      destination
    0          0 ACCEPT   0      -- *    *         ::/0        ::/0

Chain ufw6-user-logging-forward (0 references)
    pkts      bytes target    prot opt in     out     source      destination

Chain ufw6-user-logging-input (0 references)
    pkts      bytes target    prot opt in     out     source      destination

Chain ufw6-user-logging-output (0 references)
    pkts      bytes target    prot opt in     out     source      destination

Chain ufw6-user-output (1 references)
    pkts      bytes target    prot opt in     out     source      destination

```

Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)						
pkts	bytes	target	prot	opt	in	out
					source	destination
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)						
pkts	bytes	target	prot	opt	in	out
					source	destination
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)						
pkts	bytes	target	prot	opt	in	out
					source	destination
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)						
pkts	bytes	target	prot	opt	in	out
					source	destination
Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)						
pkts	bytes	target	prot	opt	in	out
					source	destination
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)						
pkts	bytes	target	prot	opt	in	out
					source	destination
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)						
pkts	bytes	target	prot	opt	in	out
					source	destination

UBTU-24-600060 - Ubuntu 24.04 LTS must use DOD PKI-established certificate authorities (CAs) for verification of the establishment of protected sessions.

Info

Untrusted CAs can issue certificates, but they may be issued by organizations or individuals that seek to compromise DOD systems or by organizations with insufficient security controls.

If the CA used for verifying the certificate is not a DOD-approved CA, trust of this CA has not been established. The DOD will only accept PKI-certificates obtained from a DOD-approved internal or external certificate authority.

Reliance on CAs for the establishment of secure sessions includes, for example, the use of SSL/TLS certificates.

NOTE: Nessus has provided the target output to assist in reviewing the benchmark to ensure target compliance.

Solution

Configure Ubuntu 24.04 LTS to use of DOD PKI-established CAs for verification of the establishment of protected sessions. Edit the "/etc/ca-certificates.conf" file, adding the character "!" to the beginning of all uncommented lines that do not start with the "!" character with the following command: `$ sudo sed -i -E 's/^[^!#]+)/!\\1/' /etc/ca-certificates.conf` Add at least one CA to the "/usr/local/share/ca-certificates" directory in the PEM format. Update the "/etc/ssl/certs" directory with the following command: `$ sudo update-ca-certificates`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.13.15
800-171R3	03.13.15
800-53	SC-23(5)
800-53R5	SC-23(5)
CAT	II
CCI	CCI-002470
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SC-23
ITSG-33	SC-23a.
NESA	T4.5.1
QCSC-V1	5.2.1
RULE-ID	SV-270745r1066724_rule
STIG-ID	UBTU-24-600060
VULN-ID	V-270745

Assets

strkbsubuntu9

The command `'/bin/ls /etc/ssl/certs | /bin/grep -i DOD'` did not return any result

UBTU-24-60090 - Ubuntu 24.04 LTS handling data requiring "data at rest" protections must employ cryptographic mechanisms to prevent unauthorized disclosure and modification of the information at rest.

Info

Information at rest refers to the state of information when it is located on a secondary storage device (e.g., disk drive and tape drive, when used for backups) within an operating system. This requirement addresses protection of user-generated data, as well as operating system-specific configuration data.

Organizations may choose to employ different mechanisms to achieve confidentiality and integrity protections, as appropriate, in accordance with the security category and/or classification of the information. Satisfies: SRG-OS-000185-GPOS-00079, SRG-OS-000404-GPOS-00183, SRG-OS-000405-GPOS-00184, SRG-OS-000780-GPOS-00240 NOTE: Nessus has provided the target output to assist in reviewing the benchmark to ensure target compliance.

Solution

To encrypt an entire partition, dedicate a partition for encryption in the partition layout. Note: Encrypting a partition in an already-installed system is more difficult because it will need to be resized and existing partitions changed.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.13.16
800-171R3	03.13.08
800-53	SC-28
800-53	SC-28(1)
800-53R5	SC-28
800-53R5	SC-28(1)
800-53R5	SC-28(3)
CAT	II
CCI	CCI-001199
CCI	CCI-002475
CCI	CCI-002476
CCI	CCI-004910
CN-L3	8.1.4.7(b)
CN-L3	8.1.4.8(b)
CSF	PR.DS-1
CSF2.0	PR.DS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.a
GDPR	32.1.b
HIPAA	164.306(a)(1)

HIPAA	164.312(a)(2)(iv)
HIPAA	164.312(e)(2)(ii)
ISO-27001-2022	A.5.10
ISO-27001-2022	A.5.33
ITSG-33	SC-28
ITSG-33	SC-28a.
ITSG-33	SC-28(1)
PCI-DSSV3.2.1	3.4
PCI-DSSV4.0	3.3.2
PCI-DSSV4.0	3.5.1
QCSC-V1	5.2.2
QCSC-V1	6.2
RULE-ID	SV-270747r1066730_rule
STIG-ID	UBTU-24-600090
TBA-FIISB	28.1
VULN-ID	V-270747

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

WARNING - Manual Review Required - fdisk
The command '/usr/sbin/fdisk -l' returned :

Disk /dev/nvme0n1: 30 GiB, 32213303296 bytes, 62916608 sectors
Disk model: MSFT NVMe Accelerator v1.0
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 33554432 bytes
Disklabel type: gpt
Disk identifier: DB3F2CB1-1F56-44CC-A805-1AC69067E6A4

Device	Start	End	Sectors	Size	Type
/dev/nvme0n1p1	2099200	62916574	60817375	29G	Linux filesystem
/dev/nvme0n1p14	2048	10239	8192	4M	BIOS boot
/dev/nvme0n1p15	10240	227327	217088	106M	EFI System
/dev/nvme0n1p16	227328	2097152	1869825	913M	Linux extended boot

Partition table entries are not in disk order.

WARNING - Manual Review Required - crypttab
The command '/bin/cat /etc/crypttab' returned :

<target name> <source device> <key file> <options>

UBTU-24-600130 - Ubuntu 24.04 LTS must ensure only users who need access to security functions are part of sudo group.

Info

An isolation boundary provides access control and protects the integrity of the hardware, software, and firmware that perform security functions. Security functions are the hardware, software, and/or firmware of the information system responsible for enforcing the system security policy and supporting the isolation of code and data on which the protection is based.

Operating systems implement code separation (i.e., separation of security functions from nonsecurity functions) in a number of ways, including through the provision of security kernels via processor rings or processor modes.

For nonkernel code, security function isolation is often achieved through file system protections that serve to protect the code on disk and address space protections that protect executing code. Developers and implementers can increase the assurance in security functions by employing well-defined security policy models; structured, disciplined, and rigorous hardware and software development techniques; and sound system/security engineering principles.

Implementation may include isolation of memory space and libraries. Ubuntu 24.04 LTS restricts access to security functions through the use of access control mechanisms and by implementing least privilege capabilities. NOTE: Nessus has provided the target output to assist in reviewing the benchmark to ensure target compliance.

Solution

Configure the sudo group with only members requiring access to security functions. To remove a user from the sudo group, run: `$ sudo gpasswd -d <username> sudo`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SC-3
800-53R5	SC-3
CAT	I
CCI	CCI-001084
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SC-3
ITSG-33	SC-3a.
NESA	T3.4.1
NESA	T4.3.1
NESA	T4.3.2
RULE-ID	SV-270748r1066733_rule
STIG-ID	UBTU-24-600130
VULN-ID	V-270748

Assets

strkbsubuntu9

Non-compliant file(s):
 /etc/group - regex '^sudo:' found - expect '^Manual Review Required\$' not found in the following lines:

21: sudo:x:27:strkadmin24,vagrant

UBTU-24-600200 - Ubuntu 24.04 LTS must configure the uncomplicated firewall to rate-limit impacted network interfaces.

Info

Denial of service (DoS) occurs when a resource is not available for legitimate users, resulting in the organization not being able to accomplish its mission or causing it to operate at degraded capacity. This requirement addresses the configuration of Ubuntu 24.04 LTS to mitigate the impact of DoS attacks that have occurred or are ongoing on system availability.

For each system, known and potential DoS attacks must be identified and solutions for each type implemented. A variety of technologies exist to limit or, in some cases, eliminate the effects of DoS attacks (e.g., limiting processes or establishing memory partitions).

Employing increased capacity and bandwidth, combined with service redundancy, may reduce the susceptibility to some DoS attacks. NOTE: Nessus has provided the target output to assist in reviewing the benchmark to ensure target compliance.

Solution

Configure the application firewall to protect against or limit the effects of DoS attacks by ensuring Ubuntu 24.04 LTS is implementing rate-limiting measures on impacted network interfaces. To change the Status of ufw to "active" use the following command: `$ sudo ufw enable` Check all the services listening to the ports with the following command: `$ sudo ss -l46ut Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process` tcp LISTEN 0 128 [::]:ssh [::]:* For each service with a port listening to connections, run the following command, replacing "[service]" with the service that needs to be rate limited. `$ sudo ufw limit [service]` Rate-limiting can also be done on an interface.

An example of adding a rate-limit on the eth0 interface follows: `$ sudo ufw limit in on eth0`

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	SC-5
800-53R5	SC-5a.
CAT	II
CCI	CCI-002385
CSF	DE.CM-1
CSF	PR.DS-4
CSF2.0	DE.CM-01
CSF2.0	PR.IR-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ITSG-33	SC-5
ITSG-33	SC-5a.
NESA	T3.3.1
NIAV2	GS8e
NIAV2	GS10c
QCSC-V1	8.2.1

RULE-ID SV-270754r1066751_rule

STIG-ID UBTU-24-600200

VULN-ID V-270754

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

WARNING - Manual Review Required - ss -l46ut
The command '/bin/ss -l46ut' returned :

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
udp	UNCONN	0	0	:::1:323	:::*	
tcp	LISTEN	0	100	:::smtp	:::*	
tcp	LISTEN	0	4096	:::ssh	:::*	

WARNING - Manual Review Required - ufw status
The command '/sbin/ufw status verbose' returned :

Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To	Action	From
--	-----	----
22/tcp	ALLOW IN	Anywhere
22/tcp (v6)	ALLOW IN	Anywhere (v6)

UBTU-24-700400 - Ubuntu 24.04 LTS must be a vendor-supported release.

Info

An operating system release is considered "supported" if the vendor continues to provide security patches for the product.
With an unsupported release, it will not be possible to resolve security issues discovered in the system software. The support status of the OS depends on its subscription status. End Of Life dates for Ubuntu 24.04 releases are as follows: Standard Support: April 2029 Extended Security Maintenance (ESM): April 2036 ESM is available via an Ubuntu Pro subscription. NOTE: Nessus has provided the target output to assist in reviewing the benchmark to ensure target compliance.

Solution

Upgrade to a supported version of Ubuntu 24.04 LTS.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.4.2
800-171R3	03.04.02a.
800-53	CM-6b.
800-53R5	CM-6b.
CAT	I
CCI	CCI-000366
CN-L3	8.1.10.6(d)
CSF	PR.IP-1
CSF2.0	DE.CM-09
CSF2.0	PR.PS-01
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
ISO-27001-2022	A.8.9
ITSG-33	CM-6b.
NESA	T3.2.1
RULE-ID	SV-278917r1155246_rule
STIG-ID	UBTU-24-700400
SWIFT-CSCV1	2.3
VULN-ID	V-278917

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

WARNING - Manual Review Required - DISTRIB_DESCRIPTION

Non-compliant file(s):

/etc/lsb-release - regex '^[\\s]*DISTRIB_DESCRIPTION[\\s]*=' found - expect '^Manual Review Required\$' not found in the following lines:
4: DISTRIB_DESCRIPTION="Ubuntu 24.04.4 LTS"

WARNING - Manual Review Required - pro status
The command '/bin/pro status' returned :

WARNING: this output is intended to be human readable, and subject to change.
In scripts, prefer using machine readable data from the `pro api` command,
or use `pro status --format json`.

SERVICE	AVAILABLE	DESCRIPTION
anbox-cloud	yes	Scalable Android in the cloud
esm-apps	yes	Expanded Security Maintenance for Applications
esm-infra	yes	Expanded Security Maintenance for Infrastructure
fips-updates	yes	FIPS compliant crypto packages with stable security updates
landscape	yes	Management and administration tool for Ubuntu
livepatch	yes	Canonical Livepatch service
realtime-kernel	yes	Ubuntu kernel with PREEMPT_RT patches integrated
usg	yes	Security compliance and audit tools

For a list of all Ubuntu Pro services, run 'pro status --all'

This machine is not attached to an Ubuntu Pro subscription.
See <https://ubuntu.com/pro>

UBTU-24-900920 - Ubuntu 24.04 LTS must allocate audit record storage capacity to store at least one week's worth of audit records, when audit records are not immediately sent to a central audit record storage facility.

Info

To ensure operating systems have a sufficient storage capacity in which to write the audit logs, operating systems must be able to allocate audit record storage capacity. The task of allocating audit record storage capacity is usually performed during initial installation of Ubuntu 24.04 LTS. NOTE: Nessus has provided the target output to assist in reviewing the benchmark to ensure target compliance.

Solution

Allocate enough storage capacity for at least one week's worth of audit records when audit records are not immediately sent to a central audit record storage facility. If audit records are stored on a partition made specifically for audit records, use the "parted" program to resize the partition with sufficient space to contain one week's worth of audit records. If audit records are not stored on a partition made specifically for audit records, a new partition with sufficient amount of space will need be to be created. Set the auditd server to point to the mount point where the audit records must be located: `$ sudo sed -i -E 's@^(log_file\s*=\s*).*\@\\1<log mountpoint>/audit.log@' /etc/audit/auditd.conf` where <log mountpoint> is the aforementioned mount point.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	AU-4
800-53R5	AU-4
CAT	III
CCI	CCI-001849
CSF	PR.DS-4
CSF	PR.PT-1
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.6
ITSG-33	AU-4
NESA	T3.3.1
NESA	T3.6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270816r1066937_rule
STIG-ID	UBTU-24-900920
VULN-ID	V-270816

Assets

strkbsubuntu9

```
The command 'logfile=$(/bin/grep -iw log_file /etc/audit/auditd.conf); if [ ! -z "${logfile}" ];  
then /bin/df -h $(/bin/dirname $(/bin/grep -iw log_file /etc/audit/auditd.conf) | /bin/grep "/");  
else echo "log_file variable in /etc/audit/auditd.conf not set"; fi' returned :
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/root	29G	2.7G	26G	10%	/

UBTU-24-900950 - Ubuntu 24.04 LTS must have a crontab script running weekly to offload audit events of standalone systems.

Info

Information stored in one location is vulnerable to accidental or incidental deletion or alteration. Offloading is a common process in information systems with limited audit storage capacity. NOTE: Nessus has provided the target output to assist in reviewing the benchmark to ensure target compliance.

Solution

Create a script that offloads audit logs to external media and runs weekly. The script must be located in the "/etc/cron.weekly" directory.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-53	AU-4(1)
800-53R5	AU-4(1)
CAT	III
CCI	CCI-001851
CSF	PR.DS-4
CSF	PR.PT-1
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ISO-27001-2022	A.8.6
ITSG-33	AU-4
NESA	T3.3.1
NESA	T3.6.2
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270817r1066940_rule
STIG-ID	UBTU-24-900950
VULN-ID	V-270817

Assets

strkbsubuntu9

The command '/bin/ls -l /etc/cron.weekly' returned :

```
total 4
-rwxr-xr-x 1 root root 1055 Mar 29 2024 man-db
```

UBTU-24-900960 - Ubuntu 24.04 LTS must immediately notify the system administrator (SA) and information system security officer (ISSO) (at a minimum) when allocated audit record storage volume reaches 75 percent of the repository maximum audit record storage capacity.

Info

If security personnel are not notified immediately when storage volume reaches 75 percent utilization, they are unable to plan for audit record storage capacity expansion. NOTE: Nessus has provided the target output to assist in reviewing the benchmark to ensure target compliance.

Solution

Edit `/etc/audit/auditd.conf` and set the `"space_left_action"` parameter to `"exec"` or `"email"`. If the `"space_left_action"` parameter is set to `"email"`, set the `"action_mail_acct"` parameter to an email address for the SA and ISSO. If the `"space_left_action"` parameter is set to `"exec"`, ensure the command being executed notifies the SA and ISSO. Edit `/etc/audit/auditd.conf` and set the `"space_left"` parameter to be at least 25 percent of the repository maximum audit record storage capacity.

See Also

https://dl.dod.cyber.mil/wp-content/uploads/stigs/zip/U_CAN_Ubuntu_24-04_LTS_V1R5_STIG.zip

References

800-171	3.3.4
800-171R3	03.03.04
800-53	AU-5(1)
800-53R5	AU-5(1)
CAT	III
CCI	CCI-001855
CN-L3	7.1.3.7(e)
CSF	PR.PT-1
DISA_BENCHMARK	CAN_Ubuntu_24-04_STIG
GDPR	32.1.b
HIPAA	164.306(a)(1)
HIPAA	164.312(b)
ITSG-33	AU-5(1)
NESA	T3.3.1
QCSC-V1	8.2.1
QCSC-V1	13.2
RULE-ID	SV-270818r1066943_rule
STIG-ID	UBTU-24-900960
TBA-FIISB	37.3.2
VULN-ID	V-270818

Assets

strkbsubuntu9

All of the following must pass to satisfy this requirement:

```
-----
PASSED - space_left_action = email
Compliant file(s):
    /etc/audit/auditd.conf - regex '(?i)^\s*space_left_action\s*=\s*' found - expect '(?
i)^\s*space_left_action\s*=\s*email\s*$' found in the following lines:
    19: space_left_action = EMAIL

-----
WARNING - Manual Review Required - action_mail_acct
Non-compliant file(s):
    /etc/audit/auditd.conf - regex '(?i)^\s*action_mail_acct\s*=\s*' found - expect
    '^Manual Review Required$' not found in the following lines:
    21: action_mail_acct = auditor

-----
WARNING - Manual Review Required - space_left
Non-compliant file(s):
    /etc/audit/auditd.conf - regex '(?i)^\s*space_left\s*=\s*' found - expect '^Manual
Review Required$' not found in the following lines:
    18: space_left = 25%
```